Validated Dirichlet approximation and weighted arithmetic moments

GRH research working note

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Abstract

Adaptive outward dyadic arithmetic and theta-integral quadrature give a requested-error construction for three fixed completed Dirichlet functions. Fresh samples and executed rational readout training certify the supplied uniform 10^{-24} examples on the stated compact; no practical extreme-precision runtime bound is asserted. Separately, a fixed-twist sextic Hecke-family moment improves an original masked remainder estimate, while retaining a positive cutoff loss. We give the precise external analytic inputs and distinguish these results from zero-location and spectral determinant claims. A classical quadratic pivot improves one row sector; joint Gauss phases have an exact masked identity. An ordinary all-character conductor comparison improves the unsigned raw envelope to $H + DH^{1/6} + (DH)^{2/3}$, while an executed finite arithmetic Fourier readout certifies correlation and signed-observable errors without asserting a varying-family signed saving. A moving common-profile construction aligns the actual coefficients on nonprincipal sixth-power rows, and an exact joint example rules out variance-only centering; the fixed-profile signed target remains open. An exact conductor-12 character twist and gcd–Möbius inversion give a separate bound for the signed coprime slice, retaining the unresolved 7/6 power. A signed common-divisor zeta expansion and a complete-column exact laboratory retain every raw mask; weighted divisor certificates do not close the required asymptotic saving. An exact rough-prime support identity gives an explicit target-budget mask correction; the remaining coherent signed main term is still unestimated. No proof of GRH or complete Lean formalization is asserted.

1 A finite, validated approximation result

Let $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$ and $\chi_5 = (0, 1, -1, -1, 1)$, extended periodically, with parities $a = 1, 1, 0$. We approximate the completed functions

$$\Lambda_q(s) = (q/\pi)^{(s+a)/2}\Gamma((s + a)/2)L(s, \chi_q).$$

The primitive theta/Mellin identity is the explicit classical analytic input; no hypothesis about zeros is used. These three examples do not assert a uniform result for all conductors, general Hecke functions or Artin holomorphy.

Theorem 1 (Scope of the supplied finite certificate)**.** *For each $q = 3, 4, 5$, the supplied exact-rational calculation constructs a complex polynomial P_q of degree 28 satisfying*

$$\sup_{|s-1/2|\le 7/10} |P_q(s) - \Lambda_q(s)| < 5 \cdot 10^{-15}.$$

The computation includes enclosures of all 64 true Fourier-circle nodes, completed-function sample error, analytic truncation, Fourier aliasing, the exact closed-form 180-step gradient-descent iterate and coefficient midpoint rounding. The Python implementation and informal analytic proof have independent checks; this is not a kernel-checked Lean theorem.

Proof. Write $F(c+z)=\sum_{n\ge 0} a_n z^n$, with $c=1/2$, $R=7/10$, $\rho=6/5$, $r=12/5$. The independently recomputed rational theta majorants are strictly below $M_q=4,10,6$, respectively, on $|z|\le r$. Use true roots $\zeta_j=e^{2\pi ij/64}$ and rational labels y_j with proved error $|y_j-F(c+\rho\zeta_j)|\le\eta$. Use the real Euclidean gradient on complex coefficient coordinates for

$$\mathcal{L}(b)=\frac{1}{128}\sum_{j=0}^{63}\left|\sum_{n=0}^{28} b_n\zeta_j^n-y_j\right|^2.$$

Discrete Fourier orthogonality gives the least-squares optimum $b_n^*=64^{-1}\sum_j y_j\bar{\zeta}_j^n$, $0\le n\le 28$, and gradient $\nabla\mathcal{L}(b)=b-b^*$. For step $1/4$ and zero initialization, the exact 180-step iterate is $b_n^{180}=(1-(3/4)^{180})b_n^*$. The code evaluates this closed form; it does not claim an explicit180-update optimizer run.

Set $u=R/\rho=7/12$, $v=R/r=7/24$ and $w=(\rho/r)^{64}=2^{-64}$. Let $\widehat{b}_n$ be an outward rectangle enclosing b_n^* and ϵ_n the sum of the coordinate halfwidths of the rectangle for b_n^{180}. Choose each actual polynomial coefficient as that rectangle's exact rational midpoint c_n. The normalized basis is explicit:

$$P_q(s)=\sum_{n=0}^{28} c_n\left(\frac{s-1/2}{6/5}\right)^n.$$

Thus the exported coefficient with index n multiplies the normalized variable, rather than the raw power $(s-1/2)^n$. Its error on the compact disk is bounded by

$$E_{\text{opt}}=(3/4)^{180}\sum_{n=0}^{28}|\widehat{b}_n|_1u^n,\qquad E_{\text{arithmetic}}=\sum_{n=0}^{28}\epsilon_nu^n,$$

$$E_{\text{oracle}}=\eta\frac{1-u^{29}}{1-u},\qquad E_{\text{alias}}=M_q\frac{w}{1-w}\frac{1-v^{29}}{1-v},$$

$$E_{\text{Taylor}}=M_q\frac{v^{29}}{1-v}.$$

Here $|\widehat{b}_n|_1$ is the sum of the maximum absolute real and imaginary coordinates, a safe upper bound for every enclosed complex modulus. Cauchy's estimate $|a_n|\le M_qr^{-n}$ and the Fourier identity $b_n^*=a_n\rho^n+\sum_{\ell\ge1}a_{n+64\ell}\rho^{n+64\ell}$ for exact labels prove the alias and Taylor bounds. Label perturbations contribute at most η per Fourier coefficient. The remaining two terms follow directly from the exact iterate and coefficient rectangles. Each term is rounded upward to a 180-bit dyadic rational; the reported total is the exact sum of those reported terms. The certified totals below are strictly below the asserted threshold. Detailed sample enclosures are proved in the extended edition and encoded with exact integer endpoints. □

Conductor	Maximum sample radius	Uniform polynomial bound
3	3.026×10^{-17}	1.786×10^{-15}
4	7.206×10^{-17}	4.455×10^{-15}
5	1.404×10^{-16}	2.906×10^{-15}

The displayed values are rounded upward; exact rational records establish the strict inequalities.

2 What this does and does not certify

The final JSON records exact rational centers, interval endpoints, source hashes and all five error terms. Float values are display summaries only. A high-precision comparison is an independent

diagnostic, not the reason an interval contains its mathematical target. Arithmetic uses outward integer operations, and invalid float/order inputs are rejected, including through typed caches. Earlier pilots and failed validation probes are retained.

This is a fixed finite tolerance/configuration. Increasing accuracy needs new truncation and precision choices; the fixed $N = 11, T = 128$ interface is not an arbitrary-accuracy oracle. The broader approximation existence theorem is separate. The certificate says nothing about the location of zeros: P_q is not required to be real-rooted or a self-adjoint spectral determinant. The independent arithmetic determinant identity, positivity and original signed short-row/covariance estimates remain open. No GRH proof or complete new Lean4 formalization is claimed.

3 Validated primitive operations

Real intervals are closed dyadic intervals $[l2^{-224}, h2^{-224}]$ with integer endpoints; complex intervals are Cartesian products. Addition is exact at this scale. For multiplication take the minimum/maximum of all four integer endpoint products, divide by 2^{224} and round down/up. For a reciprocal whose interval omits zero, reverse the endpoint order and round $2^{448}/h$ down and $2^{448}/l$ up. Negative denominators and scalar division use the same endpoint rules. Each operation encloses all represented real values by monotonicity; induction proves inclusion for every expression built from them. Rectangular complex multiplication uses these real rules. The sum of coordinate halfwidths bounds the complex distance from its exact rational center. Binary floating arguments are never silently converted.

For $0 < x < 1$, the alternating arctangent sum and the endpoint obtained by adding the next signed term enclose $\arctan x$. Sixty terms at $1/5$ and $1/239$, together with Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, give the interval for π. The negative coefficient reverses the second interval. The identity follows from tangent addition and the range of the two angles; no decimal value of π is a trusted input.

For rational $x > 0$, range-reduce $x = 2^k v$, $1 \le v < 2$. With $z = (v-1)/(v+1) \le 1/3$,

$$\log v = 2\sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \qquad 0 \le \mathcal{R} \le \frac{2z^{2J+1}}{(2J+1)(1-z^2)}.$$

The geometric tail bounds each remaining denominator below by $2J+1$. Use $J = 92$, the analogous expression for $\log 2$, and sign-aware interval multiplication by k. This covers negative k as well.

For complex z, repeatedly halve until its interval ℓ^1 magnitude bound is $A \le 1/2$. The degree 60 Taylor polynomial for e^z then has remainder at most

$$\frac{A^{61}}{61!}\frac{1}{1 - A/62}.$$

Successive absolute terms have ratio at most $A/62$, so geometric comparison proves this bound. Add its symmetric interval to each coordinate, then square outward the required number of times. The identity $e^{2z} = (e^z)^2$ and interval inclusion prove the final enclosure. Caches are type-sensitive: otherwise a cached exact argument can accidentally bypass rejection of an equal-valued float. No implementation change is allowed during the recorded run; the producer verifies source bytes again before emitting its certificate.

4 Recomputing the compact growth constants

For the larger growth disk, $c = 1/2$, $r = 12/5$, take $k = 2$ when $a = 1$ and $k = 1$ when $a = 0$. Both real Mellin exponents on that disk are at most k. Define

$$Q_k(x) = \sum_{j=0}^{k} \frac{k!}{(k-j)!x^{j+1}}, \qquad U_{16}(x) = (1 + x/16)^{-16}, \qquad \delta = 3/q.$$

Integration by parts gives $\int_1^\infty t^k e^{-xt}\,dt = e^{-x}Q_k(x)$. The same Gaussian split and geometric-tail comparison used below, now with cutoff $n_0 = 8$, imply the exact rational majorant

$$\widehat{M}_q = 2\left[\sum_{n=1}^{7} n^a U_{16}(\delta n^2)Q_k(\delta n^2) + C_a Q_k(64\delta)\frac{U_{16}(32\delta)}{1 - U_{16}(8\delta)}\right], \quad C_0 = 1, \quad C_1 = (1 + \delta^{-1})/2.$$

Replacing the actual $d = \pi/q$ by the smaller positive δ increases the integral majorant. Exact fraction comparison gives $\widehat{M}_3 < 4$, $\widehat{M}_4 < 10$, $\widehat{M}_5 < 6$. The producer recomputes these fractions, asserts the strict comparisons and exports them; the constants in the readout proof are thus checked rather than supplied by a floating growth estimate.

5 The completed-function integral and its tails

Let $d = \pi/q$ and $\theta_q(t) = \sum_{n\geq 1} \chi_q(n)n^a e^{-dn^2 t}$. The standard primitive theta transformation, followed by splitting the Mellin integral at1, gives

$$\Lambda_q(s) = \int_1^\infty \theta_q(t)t^{(s+a)/2-1}\,dt + \int_1^\infty \theta_q(t)t^{(1-s+a)/2-1}\,dt.$$

The second root-number coefficient is1 for these three characters: their finite Gauss sums are $i\sqrt{3}$, $2i$ and $\sqrt{5}$, respectively. For the last, $2\cos(2\pi/5) - 2\cos(4\pi/5) = \sqrt{5}$; the first two are direct conjugate-root differences. Division by $i^a\sqrt{q}$ gives the claimed normalization. Exponential decay makes the two integrals entire. This classical theta identity, not an assumption of Artin holomorphy or desired positivity, is the analytic input of the implementation.

The actual circle nodes are enclosed by computing $e^{2\pi ij/64}$ with the validated primitives. All input boxes fit $D = [-1, 2] + i[-4/3, 4/3]$. Thus both exponents $p = (s+a)/2-1$ and $p' = (1-s+a)/2-1$ satisfy $|\Re p| \leq 2$, $|\Im p| \leq 2/3$. The domain guard precedes all term discards. Retain $1 \leq n \leq 11$ and integrate $1 \leq t \leq 128$.

Put $\delta = 3/q < d$, $U(x) = (1+x/128)^{-128}$ and $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. Since $(1+x/128)^{128} \leq e^x$, $e^{-x} \leq U(x) < 1$ for $x > 0$. Integration by parts gives $\int_1^\infty t^2 e^{-xt}\,dt = e^{-x}Q(x)$. With $C_0 = 1$, $C_1 = (1 + \delta^{-1})/2$ the omitted-n tail of both integrals is

$$2C_a Q(\delta 12^2)\frac{U(\delta 12^2/2)}{1 - U(12\delta)}.$$

Indeed $n^a e^{-\delta n^2/2} \leq C_a$, the function Q decreases, and $(12 + j)^2 \geq 12^2 + 24j$. These inequalities yield a geometric bound after splitting the exponent in halves. The retained-n time tail is at most

$$\sum_{n=1}^{11} 2n^a 128^3 U(128\delta n^2)Q(128\delta n^2).$$

It follows by substituting $t = 128u$ and majorizing $t^{\Re p}$ by t^2. The two tails are disjoint: omitted n use the full integral, whereas retained n contribute only $t > 128$.

6 High-order integration with a certified remainder

Split each dyadic interval $[2^j, 2^{j+1}]$, $j = 0, \dots, 6$, into four equal panels. For its midpoint m and halfwidth h, $h/m \le 1/9$. For $f(z) = e^{-\lambda z} z^p$, $\lambda = \pi n^2/q$, the principal logarithm is analytic on $|z - m| \le \rho = m/2$. On that circle,

$$|f(z)| \le M_\rho = 3 \max\{(3m/2)^2, (m/2)^{-2}, 1\} U(\delta n^2 m/2).$$

Here $\Re z \ge m/2$, $|\arg z| \le 1$ and $e^{|\Im p|} \le e^{2/3} \le 3$. The maximum handles both signs of $\Re p$. Cauchy's coefficient estimate and cancellation of odd powers on $[-h, h]$ therefore bound the error after degree $2K$ by

$$\frac{2hM_\rho (h/\rho)^{2K+2}}{(2K+3)(1-(h/\rho)^2)}.$$

Indeed integrate each remaining even term, replace its denominator by the first omitted denominator, then sum a geometric series. Here $K = 12$. Multiply by $2n^a$ for both Mellin terms and sum all 28 panels and retained n. Keeping this majorant even for later discarded terms only overcounts error.

To evaluate the polynomial stably, set $u = h/m$, $q_{-1} = 0$, $q_0 = 1$ and use

$$q_{j+1} = \frac{u(p - \lambda m - j) q_j - (\lambda m) u^2 q_{j-1}}{j+1}.$$

Multiplying $f' = (-\lambda + p/t) f$ by $t = m + x$ and matching coefficients proves this recurrence, with $q_j = f^{(j)}(m) h^j / (j! f(m))$. The integrated polynomial is $2he^{-\lambda m} m^p \sum_{j=0}^{K} q_{2j}/(2j+1)$; all its arithmetic is outward.

Fixed absolute exponential-rounding errors can be amplified by high-order coefficients in late panels. The code therefore first bounds a complete (n, panel) pair, including both powers, by

$$4hn^a (m+h)^2 U(\delta n^2 (m-h)).$$

For real $t \ge 1$ its phase has modulus1 and $t^{\Re p} \le (m+h)^2$, proving the bound. If it is below 2^{-90}, skip that pair entirely and add its bound to the final radius. This is controlled truncation, not numerical underflow. The final sample radius is the interval-sum halfwidth plus the omitted-n, time, Cauchy and discard bounds. All are recorded separately and checked as nonnegative. No high-precision special-function routine is trusted here.

7 Reproduction and remaining interfaces

Run the preserved source on the existing Linux Python environment with no new packages: `python3` on `examples/analytic_oracle/theta_oracle.py`, with argument `-output NEW`. It refuses to overwrite an existing experiment directory. The final-v5 experiment contains 192 samples, three rational degree 28 polynomials, copied source and SHA-256 hashes. Independent validation checks rational endpoint arithmetic, transcendental enclosures, 1160 exact coefficient identities, high- precision completed-function comparisons and the exported component sums.

The fixed-parameter oracle solves a concrete evaluation/certification gap. It does not identify a positive arithmetic operator, force real-rooted polynomials, prove the signed original-family short-row bound or supply complete Lean4 verification. Surface/order and neural architectures may use these certificates as independent targets, but a small learned loss is not a substitute for the remaining arithmetic and categorical theorems.

8 Requested-error readouts from preserved certified labels

The accepted final-v5 sample certificate is preserved unchanged. For each $q = 3, 4, 5$, it supplies $m = 64$ rational complex labels y_j, bounds $|y_j - \Lambda_q(c+\rho\zeta_j)| \le \eta_j$, and outward rectangles enclosing the true roots $\zeta_j = e^{2\pi ij/m}$, where $c = 1/2$, $\rho = 6/5$. The classical primitive theta/Mellin identity and accepted growth bounds $M_q = 4, 10, 6$ on $|s-c| \le r = 12/5$ remain the analytic inputs. This extension accepts an exact rational error request $\varepsilon > 0$ on the fixed compact $|s-c| \le R = 7/10$. It changes the finite readout, while retaining the certified sample errors and the 64-node alias budget. It is not an arbitrary-accuracy theta-sampling interface.

Theorem 2 (Executed rational readout with explicit refusal)**.** *For each supported conductor and rational request, the finite-data selection below either returns a normalized rational polynomial P and an executed, exactly replayable gradient-descent trace with*

$$\sup_{|s-c|\le R} |P(s) - \Lambda_q(s)| \le \varepsilon,$$

or refuses the request with the exact minimum base budget over all 64 allowed degrees. Refusal concerns attainability by this particular fixed-data majorant; it is not an impossibility theorem for approximation.

Proof. Let B_n be an outward rectangle enclosing $b_n^* = m^{-1}\sum_j y_j \bar{\zeta}_j^n$, $0 \le n < 64$, computed from the preserved root rectangles. Put $t_n = \mathrm{mid}(B_n)$ and let e_n be the sum of its coordinate halfwidths. These are exact rationals, and $|t_n - b_n^*| \le e_n$. For a selected $d < 64$ execute, using exact rational operations,

$$b_n^0 = 0, \qquad b_n^{k+1} = \tfrac{3}{4} b_n^k + \tfrac{1}{4} t_n, \qquad 0 \le n \le d.$$

This is genuine gradient descent with step $1/4$ for the explicitly specified real Euclidean objective $J(b) = \frac{1}{2}\sum_{n=0}^{d} |b_n - t_n|^2$, whose gradient is $b - t$. All iterates and exact objective values are retained. The output is literally

$$P_k(s) = \sum_{n=0}^{d} b_n^k \left(\frac{s-c}{\rho}\right)^n.$$

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$, $w = (\rho/r)^{64} = 2^{-64}$, $S_d(x) = \sum_{n=0}^{d} x^n$ and $\eta = \max_j \eta_j$. The five budgets are

$$E_{\mathrm{opt}}(k) = \sum_{n=0}^{d} |b_n^k - t_n|_1 u^n, \qquad E_{\mathrm{oracle}} = \eta S_d(u),$$

$$E_{\mathrm{alias}} = M_q \frac{w}{1-w} S_d(v), \qquad E_{\mathrm{Taylor}} = M_q \frac{v^{d+1}}{1-v},$$

$$E_{\mathrm{arithmetic}} = \sum_{n=0}^{d} e_n u^n.$$

Here $|x+iy|_1 = |x| + |y|$. Cauchy's estimate and true-root Fourier orthogonality give the alias and Taylor terms; certified label errors give the oracle term. Triangle inequality applied to $b_n^k - b_n^* = (b_n^k - t_n) + (t_n - b_n^*)$ gives the other two. Thus their exact sum bounds the compact-uniform error.

Define $C_d = E_{\mathrm{oracle}} + E_{\mathrm{alias}} + E_{\mathrm{Taylor}} + E_{\mathrm{arithmetic}}$. Select the smallest $d < 64$ with $C_d < \varepsilon$; if none exists, report $\min_{d<64} C_d$ and refuse. For a selected degree, $E_{\mathrm{opt}}(k) = (3/4)^k \sum_n |t_n|_1 u^n \longrightarrow 0$. Therefore executing updates until $E_{\mathrm{opt}}(k) \le \varepsilon - C_d$ terminates after finitely many steps. The serialized coefficients, budgets, targets and trace use exact rational endpoints; displayed decimal approximations are not proof inputs. □

9 Objective semantics, precision floors and replay

The original true-circle sample-label least-squares objective is

$$\mathcal{L}_y(b) = \frac{1}{2m} \sum_{j=0}^{m-1} \left| \sum_{n=0}^{d} b_n \zeta_j^n - y_j \right|^2 .$$

With $d < m$, Fourier orthogonality gives $A^*A = mI$ and the real Euclidean coordinate gradient $\nabla\mathcal{L}_y(b) = b - b^*$. The executed objective J instead has the exact rational midpoint target t. The difference $\nabla\mathcal{L}_y - \nabla J = t - b^*$ is enclosed coefficientwise by e_n. Consequently this is an actual rational GD run for J and a certified perturbed-gradient implementation for $\mathcal{L}_y$. It is not the exact optimizer trajectory of the ideal-root objective, nor a replay of neural or quantum training. An interval enclosure of an ideal-root GD trajectory would be a different explicitly labelled object.

Each actual update contracts $b^k - t$ by $3/4$, so the exact stored losses satisfy $J(b^{k+1}) = (9/16)J(b^k)$. A replay recomputes every rational coordinate update, loss and weighted residual. The closed form $b^k = (1 - (3/4)^k)t$ is an independent check on the executed trace, not its producer. Exact rational loop coordinates introduce no further midpoint or floating export error. The arithmetic budget covers the difference between the certified Fourier rectangles and their midpoint targets, including the inherited root enclosure and outward interval operations. A subsequent float conversion requires its own exact conversion-error bound.

The normalized polynomial convention is essential: a stored coefficient with index n multiplies $((s - 1/2)/(6/5))^n$, not $(s - 1/2)^n$. The input certificate is pinned by SHA-256, including its source hashes, all 192 sample enclosures, root rectangles and analytic radius components. Producing source is captured at entry, checked unchanged before emission, and copied into a fresh output directory. An existing output is never overwritten. An independent replay should also check the fixed input hash, all dimensions, degree choice, the first stopping iterate, all five exact terms and their literal sum against the requested rational tolerance.

The attainable boundary of this majorant is

$$C_{\min,q} = \min_{0 \le d < 64} C_{q,d}.$$

Every request $\varepsilon > C_{\min,q}$ has strict budget slack for some degree and the proved finite stop rule. At or below that boundary this interface refuses. The degree minimizing C_d need not be 63: Taylor error decreases with degree while the oracle and arithmetic sums increase. In particular $C_d \ge \eta$ and $C_d \ge M_q 2^{-64}/(1 - 2^{-64})$. Increasing the number of optimization steps does not lower these fixed-data budgets. Approximate boundary values on the stated compact are $7.2929 \cdot 10^{-17}$, $1.7370 \cdot 10^{-16}$ and $3.3741 \cdot 10^{-16}$ for $q = 3, 4, 5$; only the exact exported rational minima are used for acceptance. These are certificate floors, not lower bounds on the actual errors.

A full adaptive theta oracle would separately need proof-preserving choices of omitted-n cutoff, time cutoff, quadrature order, whole-panel discard budget, primitive series lengths, interval precision and serialized-bound precision. The present $N = 11$, $T = 128$, 224-bit arithmetic, 60/92/60 primitive series lengths, 2^{-90} discard threshold and 180-bit rounded budgets cannot support every positive request. Increasing a nominal bit count alone leaves the fixed primitive remainders and analytic tails in place. The preserved labels are not recomputed here. The classical identity, finite arithmetic and informal proof are not new Lean declarations. No zero-location, all-conductor, positive arithmetic determinant, quantum-advantage or GRH conclusion follows.

The interactive replay is available at the certified readout laboratory. All decimal graphics are illustrative views of the linked exact records.

10 Requested-error theta samples for three fixed conductors

For the primitive characters $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$, $\chi_5 = (0, 1, -1, -1, 1)$ with parities $a = 1, 1, 0$, retain the classical primitive theta/Mellin identity and root number one already stated above. Put $D = [-1, 2] + i[-4/3, 4/3]$ and $\delta = 3/q$. The following adaptive extension is implemented with independent interval and sample checks; it is an informal analytic theorem, not a Lean result.

Theorem 3 (Finite adaptive sample specification)**.** *For every supported conductor, exact rational $s \in D$, and rational $\eta > 0$, the choices below terminate and give a rational complex midpoint y and exact rational radius $R \le \eta$ with $|y - \Lambda_q(s)| \le R$. The same conclusion holds for a specified true circle node when its root and node enclosures are refined on every precision retry.*

Proof. For $x \ge 0$ define the exact rational bound $E(x) = 2^{-\lfloor x \rfloor}$. Since $e > 2$, $e^{-x} \le E(x)$; this bound is nonincreasing and tends to zero. Let $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$ and $C_0 = 1$, $C_1 = (1 + \delta^{-1})/2$. For $N \ge 1$, $n_0 = N + 1$, $J \ge 1$, $T = 2^J$, use

$$B_n = 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)},$$

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

Here $\delta n_0 \ge 6/5$ ensures a positive denominator; $E(x)$ equals one for $0 \le x < 1$, so this safeguard is essential. On the four panels of each $[2^j, 2^{j+1}]$, write midpoint m, halfwidth h, $\rho = m/2$, $v = h/\rho \le 2/9$, and $B(m) = \max\{(m + \rho)^2, (m - \rho)^{-2}, 1\}$. Use the Cauchy remainder

$$B_Q = \sum_{\text{panels}} \sum_{n=1}^{N} \frac{2n^a \, 2h \, 3B(m) E(\delta n^2 m/2) v^{2K+2}}{(2K + 3)(1 - v^2)}.$$

Choose in turn the first $N \ge 1$, first $J \ge 1$, and first $K \ge 0$ for which their respective budgets are at most $\eta/8$. Gaussian decay, then exponential time decay, then $v \le 2/9$ prove termination. These are the accepted tail and Cauchy arguments with each true exponential separately replaced by its upper bound E.

There are $P = 4JN$ potential pairs. A nonzero-character pair may be omitted only if its entire two-power contribution has bound

$$b = 4hn^a (m + h)^2 E(\delta n^2 (m - h)) \le \eta/(8P).$$

Add every omitted bound to B_{discard}, giving $B_{\text{discard}} \le \eta/8$. Evaluate all other integrated Taylor polynomials by the stated exact coefficient recurrence using outward rectangles. Refine both interval precision and all primitive series remainders until the final rectangle's sum of coordinate halfwidths satisfies $R_{\text{arith}} \le \eta/4$. For this fixed finite expression all input and primitive intervals shrink; continuity and outward arithmetic imply $R_{\text{arith}} \to 0$. Precision doubling therefore terminates, and the sum of the five raw components is at most $3\eta/4$. Choose an integer output scale B such that $5\, 2^{-B} \le \eta/8$ and round each component upward by $\text{up}_B(x) = 2^{-B} \lceil 2^B x \rceil$. The reported radius is their literal exact sum. With $\mathbf{b} = (R_{\text{arith}}, B_n, B_t, B_Q, B_{\text{discard}})$, it obeys

$$|y - \Lambda_q(s)| \le \sum_{i=1}^{5} \text{up}_B(b_i) \le \frac{3\eta}{4} + 5\, 2^{-B} \le \frac{7\eta}{8} < \eta.$$

The output retains exact rational midpoints, rounded components, B, raw interval radius, parameters, input metadata and source hashes. The raw analytic bounds are exactly reconstructible; they need

not be serialized as potentially enormous fractions. The output precision varies with the request and introduces no fixed reporting floor. □

This fixed-conductor sample result uses no hypothesis on zeros. It does not remove a downstream fixed-node Fourier alias floor, supply an all-conductor theorem or a positive arithmetic determinant, or prove GRH.

11 Adaptive sampling parameters and precision termination

The primitive theta/Mellin identity with root number one yields

$$\Lambda_q(s) = \sum_{n\ge 1} \chi_q(n) n^a \int_1^\infty e^{-\pi n^2 t/q} \big(t^{(s+a)/2-1} + t^{(1-s+a)/2-1}\big)\, dt.$$

On $D = [-1,2] + i[-4/3, 4/3]$, both exponents p satisfy $|\Re p| \le 2$ and $|\Im p| \le 2/3$. For exact $x \ge 0$, put $E(x) = 2^{-\lfloor x \rfloor}$. The elementary inequality $e > 2$ gives $e^{-x} \le e^{-\lfloor x \rfloor} \le E(x)$ and $E(x) \le 2e^{-x\log 2}$. Use this solely as a pointwise nonincreasing upper bound; it is not multiplicative and has no asserted exponential integral identity.

Integration by parts gives $\int_1^\infty t^2 e^{-xt}\, dt = e^{-x}Q(x)$ for $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. For $a = 0$ the bound $n^a e^{-\delta n^2/2} \le C_0 = 1$ is immediate. For $a = 1$, $n \le (1+n^2)/2$ and $n^2 e^{-\delta n^2/2} \le 2/(e\delta) \le 1/\delta$ give $C_1 = (1+1/\delta)/2$. For $n = n_0 + j$, the inequality $n^2 \ge n_0^2 + 2n_0 j$ therefore bounds the omitted-n sum by

$$2C_a Q(\delta n_0^2) \frac{e^{-\delta n_0^2/2}}{1 - e^{-\delta n_0}} \le 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)} = B_n.$$

Take $n_0 = N + 1 \ge 2$; then $\delta n_0 \ge 6/5$ and the latter denominator is positive. Substituting $t = Tu$ in each retained-n time tail gives

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

The two tails are disjoint because the omitted-n tail uses the whole integral and the time tail uses only retained n.

For $j = 0, \dots, J-1$ and $\ell = 0, \dots, 3$ take

$$m = 2^j \left(1 + \frac{2\ell+1}{8}\right), \qquad h = \frac{2^j}{8}, \qquad T = 2^J.$$

These panels cover $[1, T]$ and have $h/m \le 1/9$. On $|z - m| = \rho = m/2$, principal log is analytic, $\Re z \ge m/2$, and $|\arg z| \le 1$. For $f(z) = e^{-\pi n^2 z/q} z^p$, the angular multiplier is at most $e^{2/3} \le 3$, whence

$$|f(z)| \le 3B(m) E(\delta n^2 m/2), \qquad B(m) = \max\{(3m/2)^2, (m/2)^{-2}, 1\}.$$

Cauchy's coefficient estimate, cancellation of odd terms, and geometric summation yield the B_Q in Theorem 3. Including zero-character or later discarded terms in that sum is harmless. Gaussian, time, and geometric decay prove termination of the sequential first-success searches with $B_n, B_t, B_Q \le \eta/8$. For fixed q and $L = \max\{1, \log(1/\eta)\}$ their asymptotic scales can be chosen as $N = O(\sqrt{L})$, $T = O(L)$, $J = O(\log L)$, and $K = O(L)$. Polynomial prefactors in N, T, J contribute only $O(\log L)$ to the logarithm of the error estimates; this is a truncation-size statement, not a rational-arithmetic bit-complexity bound.

For real $t \in [m-h, m+h]$, $t \geq 1$ and $t^{\Re p} \leq (m+h)^2$. The bound for a whole pair, including both powers, is therefore $4hn^a(m+h)^2E(\delta n^2(m-h))$. There are $P = 4JN$ potential pairs; summing only bounds at most $\eta/(8P)$ gives $B_{\text{discard}} \leq \eta/8$. No partial recurrence or isolated exponential is discarded. For active pairs retain the coefficient recurrence

$$r_{-1} = 0, \quad r_0 = 1, \quad r_{j+1} = \frac{(h/m)(p - \lambda m - j)r_j - \lambda m(h/m)^2 r_{j-1}}{j+1}, \qquad \lambda = \pi n^2/q.$$

Coefficient comparison in $tf' = (-\lambda t + p)f$ proves that the integrated Taylor polynomial is

$$2he^{p\log m - \lambda m} \sum_{j=0}^{K} \frac{r_{2j}}{2j+1}.$$

Outward arithmetic encloses its exact value, including the signed character weight and both powers.

Use an isolated p-bit dyadic context, with all primitive remainders refined as p increases. Put $\delta_p = 2^{-p-16}$. Each arctangent series separately stops when its next-term bound is at most δ_p, giving total weighted analytic tail width at most $20\delta_p$ in Machin's formula, before outward dyadic rounding. Each reduced logarithm series separately has tail at most δ_p; for $x = 2^k v$, the combined bound is $(1 + |k|)\delta_p$, before rounding. For the complex exponential, after reduction to $A \leq 1/2$, choose the Taylor order so the bound

$$\frac{A^{H+1}}{(H+1)!}\frac{1}{1 - A/(H+2)}.$$

is at most δ_p. Choose H by finite exact search; its denominator is positive. Add a symmetric exponential error enclosure to each coordinate before outward squaring. The fixed guard16 is not itself an acceptance bound: the final complete rectangle radius must satisfy $R_{\text{arith}} \leq \eta/4$. All interval caches must include p and preserve exact input-type guards. The successful p and retry precisions/radii are recorded. Individual series lengths are determined by these, the exact inputs and the pinned source and are reconstructible by replay; they are not explicitly stored.

For fixed N,J,K and exact target point, the expression has finitely many operations and only nonzero fixed integer denominators. Each primitive input is bounded for sufficiently large p; exponential halving counts are consequently bounded. Exact primitive tails and dyadic rounding widths tend to zero. Induction through the finite continuous expression, including finitely many outward squarings, gives $R_{\text{arith}} \to 0$. Doubling p after a failed comparison terminates. The argument also covers true roots and nodes regenerated at every p, since their widths tend to zero. A fixed nondegenerate s rectangle has no corresponding shrinking-input claim.

A growing-order primitive estimate. To justify convergence despite a growing exponential Taylor order, write $\epsilon_p = 2^{-p}$, $\delta_p = 2^{-p-16}$, and let $d(X)$ be the sum of a rectangle's coordinate widths and $m(X)$ its ℓ^1 magnitude bound. The rounded reduced input W satisfies $a = m(W) \leq 1/2 + 2\epsilon_p < 3/5$ and $d(W) \leq 2^{-k}d(Z) + 4\epsilon_p$; the remainder is at most 2^{-H}, so the first accepted order satisfies $H \leq p + 18$. For the computed term boxes, outward multiplication and integer division give $m_j \leq am_{j-1} + 6\epsilon_p$ and $d_j \leq ad_{j-1} + m_{j-1}d(W) + 12\epsilon_p$. Thus $m_j \leq a^j + 15\epsilon_p$ and $\sum_{j=0}^{H} d_j \leq (25/4)d(W) + 75H\epsilon_p$; adding the remainder increases width by at most $4(\delta_p + \epsilon_p)$. The pre-squaring width is therefore $O(d(Z) + p2^{-p})$ and its magnitude is bounded (indeed below 3 for $p \geq 32$). Each outward square has width at most $2m(X)d(X) + 8\epsilon_p$; the bounded number of squarings consequently preserves convergence to zero. Finally, exact rational partial sums and endpoint rounding give Machin width at most $20\delta_p + 40\epsilon_p$, and logarithm width at most $(1 + |k|)(\delta_p + 2\epsilon_p)$ for fixed $x = 2^k v$. These estimates justify the primitive-width step without assuming a fixed Taylor order or asserting a global arithmetic complexity bound.

For serialization, initialize

$$B = \max\{32, \operatorname{bitlen}(\operatorname{den}\eta) - \operatorname{bitlen}(\operatorname{num}\eta) + 8\}$$

and increase B until $5\,2^{-B} \le \eta/8$. Write $\widehat{x} = 2^{-B}\lceil 2^B x\rceil$ for each of the five raw components. Their exact reported sum is at most $3\eta/4 + 5\,2^{-B} \le 7\eta/8$, and every component remains an upper bound. The output records B, the raw interval radius and complete rectangle, N/J/K, discard threshold/pair counts, exact target metadata and source hashes. These reconstruct all raw analytic bounds and pair sets without serializing huge exact tail denominators. For the practical requests below B is87 and114, respectively. Exact endpoints remain rational; the mathematical specification assumes an integer representation sufficient for the requested size. Dynamic rounding removes the old fixed reporting floor. For rational s, validate its exact coordinates in closed D before conversion: an outward dyadic enclosure of $\Im s = 4/3$ extends beyond D, but the analytic bounds concern the true point. Circle nodes satisfy the strict exact bounds $-7/10 \le \Re s \le 17/10$, $|\Im s| \le 6/5$.

The exact parameter-design checker gives

η	q	N	J	T	K	active pairs
10^{-24}	3	12	7	128	19	71
10^{-24}	4	14	8	256	20	65
10^{-24}	5	16	8	256	20	111
10^{-32}	3	14	7	128	25	84
10^{-32}	4	16	8	256	26	72
10^{-32}	5	18	8	256	26	129

with exact rational budgets and no special-function oracle. The arithmetic precision is selected from the actual interval radius, not inferred from this table. This extension retains the existing classical theta identity; it adds no zero-location, positive determinant, all-conductor, Lean or GRH claim. The following end-to-end adaptive readout supplies its own node-count, alias, degree, Fourier and optimization choices.

12 An end-to-end adaptive analytic readout

The new driver changes the sample truncations, interval precision, circle size and degree; it does not reuse the frozen labels to assert a smaller error. Its scope is still the three stated primitive quadratic characters and the fixed compact $|s - 1/2| \le 7/10$.

Theorem 4 (Requested-error construction in the stated finite family)**.** *Given a positive rational ε and $q \in \{3, 4, 5\}$, the specified exact-arithmetic algorithm terminates in the mathematical unbounded-resource model and returns a normalized rational polynomial P with*

$$\sup_{|s-1/2|\le 7/10} |P(s) - \Lambda_q(s)| \le \varepsilon.$$

Its sample enclosures, rational optimizer updates and all error budgets are explicit. This is an approximation theorem, with the stated classical theta identity as an input; it supplies no zero-location or positive determinant conclusion and is not a Lean theorem or a practical runtime bound.

Proof. Put $u = 7/12$, $v = 7/24$. Choose $d \ge 0$ so $10v^{d+1}/(1 - v) \le \varepsilon/8$, and choose $m > d$ so

$$10\frac{2^{-m}}{1 - 2^{-m}}\frac{1}{1 - v} \le \varepsilon/8.$$

The driver rounds m upward to a multiple of eight, which only decreases this bound. Set the requested sample error $\eta = \varepsilon(1-u)/8$. The adaptive sampler gives all m true circle nodes and labels with error at most η. Form their outward Fourier rectangles through degree d, and refine the nodes and finite evaluation arithmetic if their weighted midpoint radii exceed $\varepsilon/8$. This refinement terminates: for these fixed finite d, m, node uncertainty and every finite-expression arithmetic width tend to zero with precision. Merely lifting fixed boxes would not suffice, so the driver regenerates the nodes and samples on this retry.

Choose rational midpoint targets t_n and execute the previously specified quadratic-objective updates until $\sum_{n=0}^{d} |b_n^k - t_n|_1 u^n \le \varepsilon/4$. Their geometric contraction proves finite termination. The oracle, aliasing, Taylor and Fourier-midpoint terms are each at most $\varepsilon/8$, and the optimization term is at most $\varepsilon/4$. Hence the raw five-term sum is at most $3\varepsilon/4$. Choose a serialization precision B with $5\,2^{-B} \le \varepsilon/8$, round each term upward, then add the five reported rationals exactly. The printed bound is at most $7\varepsilon/8 \le \varepsilon$. The polynomial uses $((s-1/2)/(6/5))^n$, and the preceding Fourier/Cauchy proof gives its asserted compact-uniform bound. □

The trace is losslessly encoded by the fixed target vector and a rational scale extracted from each *executed* coefficient state, together with its objective, weighted residual and full-state hash. Every coefficient can be recovered exactly; a separate replay expands the states and checks every update. The producer does not generate its trace from a closed form. The saved objective remains the rational Fourier-midpoint quadratic, with its perturbation from ideal-root least squares explicitly budgeted.

13 Termination, compression and the boundary of the guarantee

The degree and circle-size choices use only rational comparisons. Since $0 < v < 1$, the Taylor test succeeds at finite d; since $2^{-m} \to 0$, the alias test succeeds at finite $m > d$. The common growth bound10 is safe for all three families; the certificate also recomputes the previously proved exact conductor-specific majorants below $4, 10, 6$.

At fixed d, m, η, the analytic truncation parameters are fixed while precision is doubled. The computed sample centers stay bounded: their underlying finite expression is fixed and the true completed-function values are bounded on the sample circle. Its arithmetic widths tend to zero, as do the regenerated true-root enclosures. Fourier computation is a finite composition of additions, products and division by the positive integer m. Its midpoint-radius sum therefore tends to zero. This proves the driver's additional precision retry, without pretending that changing a box's nominal precision shrinks its uncertainty.

The accepted sampler error may remain above its actual arithmetic width because its analytic tails and output bounds are fixed for this η. That causes no problem: the oracle allocation uses the total certified sample radius, whereas the Fourier-midpoint allocation uses only finite coefficient-rectangle width. No analytic sample error is silently moved into the latter term or removed by numerical convergence.

For the stored training trace, the exact loop gives $b^{k+1} = 3b^k/4 + t/4$ from $b^0 = 0$. Inductively every state is a rational scalar multiple of the same target vector. The producer first executes the actual vector updates, then extracts this scalar from a nonzero target coordinate, checks all coordinates, and records a canonical full-vector SHA-256 hash. If the entire target vector is zero, its zero state needs only scale0. The compressed record is thus an exact representation of the executed states, not a floating-point sketch. Hashes are integrity aids; exact vector reconstruction and the independently checked recurrence establish the mathematical claim.

Coefficient export uses exact rational numerators and denominators. A chunked decimal integer encoder/parser handles endpoints beyond Python's usual digit-conversion guard without changing global interpreter settings. No fixed 180-bit serialization floor remains. The decimal graphics are separate approximate projections, not polynomial coefficients to which the exact certificate may be applied without a conversion-error bound.

All producing source bytes are captured, copied into fresh output paths and checked unchanged. Historical fixed producers are pinned and untouched. For the new modules the run also checks the import-time source snapshots, so a stale imported module cannot be attributed to later source bytes. After output creation, a failed evaluation, final provenance check or serialization emits uncertified failure metadata. Invalid requests and stale import-time sources are refused before creating output artifacts. Old outputs are never overwritten. Extreme requests can be computationally impractical: the termination proof gives no machine-time or memory guarantee, and finite test runs establish only their reported cases.

Nothing requires the output polynomials to be real-rooted, to equal an arithmetic Fredholm determinant, or to have a self-adjoint realization. Consequently arbitrary compact approximation of these known functions does not imply GRH. Uniformity in changing conductor, general Hecke/Dedekind/ elliptic/Artin scopes, the independent determinant comparison, and actual analytic/category Lean objects remain separate obligations.

14 An arithmetic-family moment with every original row retained

Let $K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O} = \mathbb{Z}[\omega]$, and write N for the ideal or element norm. Fix a finite-order Hecke character ν and a finite set S of prime ideals containing those above 6 and those dividing its conductor. For $0 \neq u \in \mathcal{O}$, let κ_u be the character of $K(u^{1/6})/K$ defined using arithmetic Frobenius, and put $\psi_u = \mathrm{prim}(\nu\kappa_u)$. Here prim means the underlying primitive Hecke character: it is formed before introducing any divisor variable or evaluating conductor-dependent zero extensions. The rows below are all actual Eisenstein elements, including units and prime powers supported in S.

The external inputs are abelian Kummer reciprocity and character uniqueness, the primitive Hecke functional equation and its usual vertical growth, Stirling's formula, scalar Dedekind-zeta convexity, and Blomer–Goldmakher–Louvel's large sieve for their specified sixth-order family. We use the pinned arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1 and Appendix II; its Section 5 supplies the all-ideal decomposition whose fixed-twist proof is redone here. No statement about all primitive characters of comparable conductor, Artin holomorphy, Lindelöf bounds or zero locations is assumed.

Theorem 5 (The original-family numerator moment)**.** *For every $\varepsilon > 0$, uniformly for $H \geq 1$ and $t \in \mathbb{R}$,*

$$\sum_{0<Nu\leq H} |L(1/2+it,\psi_u)|^2 \ll_{\nu,S,\varepsilon} H^{1+\varepsilon}(1+|t|)^{1+\varepsilon}. \tag{1}$$

For every fixed $0 < \delta < 1/2$, the same estimate holds with $1/2+it$ replaced by $1/2+\delta+it$, with the constant also depending on δ. If the central bound has constant C_0 at an exponent $0 < \varepsilon_0 \leq 1/4$, an admissible off-central constant is

$$200C_0\left(\frac{3/2+\delta}{1/2-\delta}\right)^2. \tag{2}$$

Proof. For the canonical BGL character $\vartheta_{\mathfrak{a}}$, Appendix II gives a radicand $\alpha_{\mathfrak{a}}$ with $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Since $\mathcal{O}_S$ is principal, comparison with any literal u whose outside-S ideal is $\mathfrak{a}$ gives $\kappa_u = \eta\vartheta_{\mathfrak{a}}$ for one of finitely many fixed S-unit Kummer twists η. This is an equality of primitive characters, established outside a finite set and extended by uniqueness. For each outside-S ideal $\mathfrak{a}$ and S-supported integral

ideal $\mathfrak{b}$, at most six elements generate $\mathfrak{a}\mathfrak{b}$. The resulting S-supported multiplicity sum converges at exponent $1+\varepsilon$.

For each fixed $\xi=\nu\eta$, split the canonical parameter as $\mathfrak{a}=\mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6$, where $\mathfrak{a}_0$ is squarefree, $\mathfrak{a}_1$ is squarefull and sixth-power-free, and $(\mathfrak{a}_0,\mathfrak{a}_1)=1$. The primitive conductor is at most a fixed multiple of $N(\mathfrak{a}_0\mathfrak{a}_1)$. The common-gamma approximate functional equation, with $X=H/(N\mathfrak{a}_2)^6$ and $T=1+|t|$, has common maximal length $B=X^{1/2}T(HT)^{\eta}$ for any sufficiently small fixed $\eta>0$. Mellin inversion removes its row-dependent conductor weight. Split a column as $\mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2$, with $\mathfrak{b}_S$ supported in S, $\mathfrak{b}_1$ squarefree and $\mathfrak{b}_2$ squarefull, the last two coprime and prime to S. The original primitive values at $\mathfrak{b}_S$ remain bounded row multipliers. The S-supported and squarefull column weights are summable at $1/2+\eta$. Multiplying squarefree column coefficients by the fixed ξ preserves their squared norm, whatever its finite order. BGL's large sieve therefore bounds the energy for fixed $\mathfrak{a}_1,\mathfrak{a}_2$ by

$$(HT)^{O(\eta)}\{M+B+(MB)^{2/3}\},\qquad M=X/N\mathfrak{a}_1.$$

Squarefull counting and summation give $X+B\sqrt{X}+(XB)^{2/3}\ll(HT)^{O(\eta)}XT$. The $\mathfrak{a}_2$ sum converges at exponent 6. Principal rows are separated before the pole-free functional equation and bounded by scalar zeta convexity. This proves (1).

For the off-central statement divide each $L(s,\psi)$ by $(s+1)^{5/4}$ and, on principal rows, also multiply by $(s-1)/(s+1)$. These are bounded analytic functions on $\operatorname{Re}s>1/2$, continuous on the boundary. Poisson representation and Jensen's inequality transfer the central squared moment. The convolution of $P_\delta(y)=\delta/[\pi(\delta^2+y^2)]$ with $(1+|t|)^{-3/2+\varepsilon_0}$ is at most eleven times that weight when $0<\varepsilon_0\le 1/4$. Restoring the denominator and the principal pole-removal factor gives (2). The extended edition gives the detailed derivation with all primitive local values. □

Separate fixed-unitary extension. The moment theorem also holds for any one fixed unitary Hecke character ν. For precision choose $\nu(\mathfrak{a})=\nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$, with angular parameter $m\in\mathbb{Z}$ and completed gamma factor

$$\gamma_\nu(s)=2(2\pi)^{-(s+|m|/2+i\tau)}\Gamma(s+|m|/2+i\tau).$$

All finite Kummer twists have this same infinity type. Use $T_\nu=1+|t+\tau|+|m|$, comparable to $1+|t|$ with ν-dependent constants. Pure norm rows have $L(s,\psi)=\zeta_K(s+i\tau)$ and are separated before the pole-free functional equation; their pole at $1-i\tau$ is removed by $(s-1+i\tau)/(s+1+i\tau)$. No uniformity in m, τ or the conductor of ν is asserted. This extension of the moment is separate from the following finite-order type-I residue formula.

14.1 The exact masked remainder and the remaining loss

Retain the original zero-extended symbol $\chi_d(u)=(u/d)_6$ on good squarefree ideals. Let μ be the ideal Möbius function and set

$$\beta_U(d)=\sum_{\substack{ab=d\\ Na,Nb\le U}}\mu(a)\mu(b),\quad T(u,d)=S\cup\operatorname{supp}(u)\cup\operatorname{supp}(d).$$

Let $\psi_{u,2} = \mathrm{prim}(\psi_u^2)$, formed independently of squaring any zero sentinel. With $\mathfrak{f}_u$ and $\mathfrak{f}_{u,2}$ the primitive conductors, the full-mask series is

$$F_{u,d}(s) = \frac{L(s,\psi_u)}{L(2s,\psi_{u,2})} C_{T(u,d)}(s), \tag{3}$$
$$C_T(s) = \frac{\prod_{p\in T,\ p\nmid \mathfrak{f}_u}(1-\psi_u(p)(Np)^{-s})}{\prod_{p\in T,\ p\nmid \mathfrak{f}_{u,2}}(1-\psi_{u,2}(p)(Np)^{-2s})}.$$

Initially it equals $\sum_{c\ \text{squarefree},\ (c,T)=1} \psi_u(c)(Nc)^{-s}$ for $\operatorname{Re} s > 1$. Every prime of T remains deleted, including those forgotten by either primitive conductor.

For finite-order ν, $D \geq 1$, $U \geq 1$, $U^2 \leq D$, and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$, put

$$\widehat{W}(s) = \int_0^\infty W(x) x^{s-1}\, dx, \qquad \mathcal{M}_4(W) = \sum_{j=0}^{4} \sup_x |(x\partial_x)^j W(x)|.$$

On $\sigma = 1/2 + \delta$, the exact residue-subtracted type-I remainder is

$$E_u = \frac{D^\delta}{2\pi} \int_{\mathbb{R}} \widehat{W}(\sigma+it) D^{it} \sum_{\substack{d\ \text{squarefree}\\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{(Nd)^{\sigma+it}} F_{u,d}(\sigma+it)\, dt. \tag{4}$$

This definition retains the actual β_U, all phases, d-masks, conductors and the independently primitive square.

Theorem 6 (An improved masked residual estimate)**.** *Under the preceding finite-order hypotheses, for every $\varepsilon > 0$,*

$$\sum_{0<Nu\leq H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 D^{2\delta} U^{2-4\delta} H (HD)^\varepsilon. \tag{5}$$

Proof. The same deleted set in both factors of (3) gives $|F_{u,d}(\sigma+it)| \ll (Nu\,Nd)^\eta |L(\sigma+it,\psi_u)|$: the reciprocal incomplete Euler product at $2\sigma > 1$ is bounded by $\zeta_K(2\sigma)$, and its remaining finite numerator is bounded by a small power of its radical. Apply Minkowski to (4) in the exact row space and use Theorem 5. The actual coefficient sum is at most

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \leq \left(\sum_{Na\leq U} (Na)^{-\sigma+\eta} \right)^2 \ll U^{1-2\delta+2\eta}.$$

Four Mellin integrations by parts bound the remaining height integral by a constant times $\mathcal{M}_4(W)$. Square the result and absorb the small losses using $U^2 \leq D$. □

At $H = D^{3/4}$ and $U = D^a$ with fixed $0 < a \leq 1/2$, the upper bound's excess over H is $2\delta + (2-4\delta)a$, up to small losses; as $\delta \downarrow 0$ it still leaves $2a$. This is a limitation of the upper bound, not an energy lower bound. The extracted principal residue, the original short term at smaller scales and the signed type-II minus type-I covariance are not controlled by (5). The next arithmetic task is to estimate the complete d-weighted moment before its absolute coefficient sum, or establish sufficient signed covariance. No positive arithmetic determinant, zero-location theorem or GRH proof follows. The analytic argument uses the external classical inputs above; finite diagnostics and selected existing Lean local identities do not formalize the Hecke functions, approximate functional equation, Poisson transfer or these moment bounds.

15 Family identification, the approximate functional equation and masks

We prove the analytic statements of Section 14 from the explicit classical inputs stated there. Write $I(S)$ for the nonzero integral ideals prime to S, and $\mathcal{O}_S$ for the ring of S-integers. Constants may depend on the fixed field, S, ν and any declared positive small exponent. No such constant is asserted uniform in the conductor or infinity type of a varying ν.

15.1 A finite comparison with the literal Kummer rows

Use Appendix II of BGL, with $n = 6$. Its fixed local modulus and representatives may be chosen once, since S contains the primes above 6 and $\mathcal{O}_S$ has class number one. For each $\mathfrak{a} \in I(S)$ the construction writes

$$\mathfrak{a} = (x)E\mathfrak{g}^6, \qquad \alpha_{\mathfrak{a}} = xm_E, \qquad (m_E)_S = E,$$

so $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Choose $y \in K^\times$ generating $\mathfrak{g}$ as an $\mathcal{O}_S$-ideal. If $(u)_S = \mathfrak{a}$, then

$$e = \alpha_{\mathfrak{a}} y^6/u \in \mathcal{O}_S^\times, \qquad [\alpha_{\mathfrak{a}}] = [eu] \quad \text{in } K^\times/K^{\times 6}.$$

There is a precision here: Appendix II allows choosing $\mathfrak{g}$ relative to a tested denominator. Fix one admissible global decomposition first. Its radicand gives the canonical character $\vartheta_{\mathfrak{a}}$ on all primes outside the finite set $S \cup \mathrm{supp}(\mathfrak{a}\mathfrak{g})$ where that choice applies. Uniqueness of Hecke characters, equivalently Chebotarev in the finite abelian quotient, then identifies the underlying primitive characters everywhere. With matching arithmetic-Frobenius conventions,

$$\kappa_u = \kappa_{e^{-1}}\vartheta_{\mathfrak{a}}. \tag{6}$$

If the residue-symbol convention is conjugated, conjugate the canonical family and its coefficient sequence; the same large sieve follows.

The group $\mathcal{O}_S^\times/(\mathcal{O}_S^\times)^6$ is finite. Thus (6) uses one of a finite set of fixed Kummer twists η, all unramified outside S and of bounded S-conductor. This is a comparison of primitive characters, not a pointwise multiplication rule for independently zero-extended arrays at bad primes. In particular, a conductor drop must not be replaced by a product of artificial zeros.

Since $\mathcal{O}$ is principal with six units, for fixed $\mathfrak{a} \in I(S)$ and an S-supported integral ideal $\mathfrak{b}$ there are at most six u satisfying $(u) = \mathfrak{a}\mathfrak{b}$. For any nonnegative row quantity, the literal moment is consequently at most

$$6 \sum_{\substack{\mathfrak{b}\ S\text{-supported} \\ N\mathfrak{b} \le H}} \sum_{\eta} \sum_{\substack{\mathfrak{a} \in I(S) \\ N\mathfrak{a} \le H/N\mathfrak{b}}} \left|L(s, \mathrm{prim}(\nu\eta\vartheta_{\mathfrak{a}}))\right|^2. \tag{7}$$

An ideal bound $X^{1+\varepsilon}T^{1+\varepsilon}$ therefore transfers to all literal rows, since

$$\sum_{\mathfrak{b}\ S\text{-supported}} (N\mathfrak{b})^{-1-\varepsilon} = \prod_{p \in S} (1 - (Np)^{-1-\varepsilon})^{-1} < \infty.$$

No quotient of the original unit rows or deletion of S-prime powers has occurred.

15.2 Fixed twists, exact tame conductors and the squarefree sieve

Fix $\xi = \nu\eta$ and write $\rho_{\mathfrak{a}} = \mathrm{prim}(\xi\vartheta_{\mathfrak{a}})$. Its fixed conductor is supported in S before the Kummer twist. On columns prime to S, $|\xi(\mathfrak{b})| = 1$. BGL Theorem 1.3 thus gives, for any fixed $\mathfrak{a}_1$,

$$\sum_{\substack{\mathfrak{a}_0 \in I(S) \text{ squarefree} \\ N\mathfrak{a}_0 \le M}} \left| \sum_{\substack{\mathfrak{b}_1 \in I(S) \text{ squarefree} \\ N\mathfrak{b}_1 \le B}} \lambda_{\mathfrak{b}_1} \xi(\mathfrak{b}_1) \vartheta_{\mathfrak{a}_1}(\mathfrak{b}_1) \vartheta_{\mathfrak{a}_0}(\mathfrak{b}_1) \right|^2 \ll (MB)^{\varepsilon}\{M + B + (MB)^{2/3}\} \sum_{\mathfrak{b}_1} |\lambda_{\mathfrak{b}_1}|^2. \tag{8}$$

The first fixed multiplier preserves the coefficient norm; the second has modulus at most one. Restricting the positive outer row sum is harmless. A coprimality condition with another fixed ideal belongs to λ. In particular, ν need not have order dividing six and is not asserted to belong to BGL's finite ray-class character set.

There is a unique valuation decomposition

$$\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6, \quad \mathfrak{a}_0 \text{ squarefree}, \quad \mathfrak{a}_1 \text{ squarefull and sixth-power-free}, \quad (\mathfrak{a}_0, \mathfrak{a}_1) = 1.$$

Indeed, for $v_p(\mathfrak{a}) = 6q + r$, $0 \le r \le 5$, put the residual prime in $\mathfrak{a}_0$ if $r = 1$, and its exponent r in $\mathfrak{a}_1$ if $2 \le r \le 5$. BGL Lemma 2.1 gives $\vartheta_{\mathfrak{a}\mathfrak{b}^6} = \vartheta_{\mathfrak{a}}$ and $\vartheta_{\mathfrak{a}_0\mathfrak{a}_1} = \vartheta_{\mathfrak{a}_0}\vartheta_{\mathfrak{a}_1}$ on ideals prime to S in this sixth-power-free case. Those are the identities needed here; no arbitrary multiplicativity assertion is substituted.

At a prime outside S, tame Kummer inertia at residual exponent r has order $6/\gcd(6, r)$. It is nontrivial for $r = 1, \dots, 5$ and has conductor exponent one. The fixed ξ is unramified there, so cannot cancel that inertia. At primes in S the local sixth-power classes are finite, hence only finitely many primitive conductor types occur after multiplication by the fixed ξ. Consequently

$$Q_{\rho_{\mathfrak{a}}} = |D_K| N\mathfrak{f}_{\rho_{\mathfrak{a}}} \le C_{\xi,S} N(\mathfrak{a}_0\mathfrak{a}_1), \tag{9}$$

independently of $\mathfrak{a}_2$. The equality of the conductor's outside-S part with $\mathrm{rad}(\mathfrak{a}_0\mathfrak{a}_1)$ does not identify that conductor with the original larger deleted set.

15.3 Removing the row-dependent functional-equation weights

First take finite-order ν. All characters then have trivial infinity type at the complex place. The primitive completion may be normalized as

$$\Lambda(s, \rho) = Q_\rho^{s/2} \gamma_K(s) L(s, \rho), \qquad \gamma_K(s) = 2(2\pi)^{-s}\Gamma(s), \qquad \Lambda(s, \rho) = \epsilon_\rho \Lambda(1 - s, \bar{\rho}), \quad |\epsilon_\rho| = 1.$$

For nonprincipal rows the completed function is entire. Apply this functional equation to the Mellin integral with even auxiliary factor $G(z) = e^{z^2}$ and denominator z, shifting across $z = 0$. At $s = 1/2 + it$ this gives two weighted Dirichlet series, truncated below to polynomials; the dual prefactor has modulus one. Principal rows are excluded from this pole-free step and treated separately below.

For fixed $\mathfrak{a}_2$, let $X = H/(N\mathfrak{a}_2)^6 \ge 1$ and $T = 1 + |t|$. An explicit normalized kernel is

$$V_t(x) = \frac{1}{2\pi i} \int_{(\eta)} e^{z^2} \frac{\Gamma(s + z)}{\Gamma(s)} T^{-z} x^{-z} \frac{dz}{z}, \qquad x = \frac{2\pi N\mathfrak{b}}{\sqrt{Q_\rho} T}, \tag{10}$$

with $\eta > 0$ small. Its factors combine to $(\sqrt{Q_\rho}/(2\pi N\mathfrak{b}))^z$ in the actual completed-L kernel; there is no extra factor of T. The denominator z supplies the residue at zero. On $z = \eta + iy$, Stirling and $|t + y| \le T(1 + |y|)$ give

$$\left|\frac{\Gamma(1/2 + \eta + i(t + y))}{\Gamma(1/2 + it)T^{\eta+iy}}\right| \ll_\eta (1 + |y|)^{C_\eta} e^{\pi|y|/2}.$$

The factor e^{-y^2} makes this an integrable majorant uniformly in t. The analogous bound holds on every fixed positive shifted line and for the dual gamma kernel. Moving far enough to the right gives arbitrary power decay beyond a common total-ideal cutoff

$$B' = X^{1/2}T(HT)^\eta, \tag{11}$$

up to fixed ξ, S, K constants. For any requested $A > 0$, a sufficiently large positive kernel shift, depending also on η, makes the truncation error $O((HT)^{-A})$ uniformly in all rows in this block. Ideal counting $O_K(Y)$ bounds the absolute tails. Such errors are negligible after the finite row sums by taking A larger if needed.

Insert (10) on $\operatorname{Re} z = \eta$. The factor $(\sqrt{Q_\rho}T)^z$ is independent of the column and is a row multiplier of modulus at most $(CXT^2)^{\eta/2}$. Its imaginary part is a phase. The remaining column coefficient is $(N\mathfrak{b})^{-1/2-\eta-it-iy}$, apart from fixed constants. Minkowski or Cauchy with the integrable gamma majorant therefore removes the row-dependent weight at cost $(HT)^{O(\eta)}$ in the squared norm. No independence of the root numbers is assumed or needed.

For each column use the unique decomposition

$$\mathfrak{b} = \mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2, \quad \mathfrak{b}_S \ S\text{-supported}, \quad \mathfrak{b}_1 \text{ squarefree}, \quad \mathfrak{b}_2 \text{ squarefull}, \quad (\mathfrak{b}_1, \mathfrak{b}_2) = 1,$$

where $\mathfrak{b}_1, \mathfrak{b}_2 \in I(S)$; a prime of outside-$S$ valuation one goes in $\mathfrak{b}_1$, and a prime of valuation at least two goes with its full exponent in $\mathfrak{b}_2$. For fixed $\mathfrak{b}_S, \mathfrak{b}_2$ retain $\rho_{\mathfrak{a}}(\mathfrak{b}_S)$ as the original primitive row value. On the good factors the exact canonical identity applies. The remaining factors involving $\mathfrak{b}_2$, including $\vartheta_{\mathfrak{a}_0}(\mathfrak{b}_2)$, are bounded row multipliers. Their moduli are at most one; only then are they discarded in an inequality. In particular the value at $\mathfrak{b}_S$ is not replaced by the product of zero-extended primitive arrays.

For the $\mathfrak{b}_1$ polynomial the coefficient sequence is independent of $\mathfrak{a}_0$; it includes the indicator $(\mathfrak{b}_1, \mathfrak{b}_2) = 1$ and the common cutoff $N\mathfrak{b}_1 \le B'/N(\mathfrak{b}_S\mathfrak{b}_2)$. Padding it by zeros up to B' is legitimate. Its squared norm is at most $\zeta_K(1 + 2\eta)$. Minkowski over $\mathfrak{b}_S, \mathfrak{b}_2$ is legitimate because

$$\sum_{\mathfrak{b}_S, \mathfrak{b}_2} N(\mathfrak{b}_S\mathfrak{b}_2)^{-1/2-\eta} < \infty.$$

The finite Euler product handles $\mathfrak{b}_S$; the squarefull series converges above $1/2$. Apply (8) to each such polynomial and to the conjugate dual family. For fixed $\mathfrak{a}_1, \mathfrak{a}_2$ the resulting energy is

$$(HT)^{O(\eta)}\{M + B' + (MB')^{2/3}\}, \qquad M = X/N\mathfrak{a}_1. \tag{12}$$

The additional condition $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$ simply restricts the positive row sum. The large-sieve small exponent can be chosen sufficiently small along with η and absorbed in the indicated small loss.

15.4 All-ideal summation and principal rows

Every squarefull ideal $\mathfrak{c}$ has a unique expression $\mathfrak{d}^2\mathfrak{e}^3$ with $\mathfrak{e}$ squarefree. This follows primewise by using exponent zero or one in $\mathfrak{e}$ according to the parity of an exponent at least two. Therefore

$$\#\{\mathfrak{c} \text{ squarefull} : N\mathfrak{c} \le X\} \ll_K X^{1/2}, \qquad \sum_{\mathfrak{c} \text{ squarefull}} (N\mathfrak{c})^{-s} < \infty \quad (s > 1/2).$$

For the count, sum $O((X/(N\mathfrak{e})^3)^{1/2})$ and use $\zeta_K(3/2) < \infty$. Summing (12) over $\mathfrak{a}_1$ gives

$$(HT)^{O(\eta)}\{X + B'\sqrt{X} + (XB')^{2/3}\} \ll (HT)^{O(\eta)}XT,$$

since the squarefull series converges at 1 and 2/3, $B' = \sqrt{X}\,T(HT)^{\eta}$, and $T^{2/3} \le T$ for $T \ge 1$. The $\mathfrak{a}_2$ sum is bounded by $H\sum_{\mathfrak{a}_2}(N\mathfrak{a}_2)^{-6} = H\zeta_K(6)$. Choosing all internal losses small for the requested final ε proves the fixed-twist all-ideal central moment.

There are $O_K(H)$ ideal rows in total. On any row where the primitive character is principal its L-function is ζ_K, and ordinary scalar convexity gives $|\zeta_K(1/2+it)|^2 \ll_\varepsilon T^{1+\varepsilon}$. This treats those rows without a pole-free approximate functional equation. Finally (7) proves (1) for every actual nonzero Eisenstein element.

15.5 Poisson transfer with an explicit constant

For a nonprincipal row define

$$f_\psi(s) = \frac{L(s,\psi)}{(s+1)^{5/4}},$$

using the holomorphic logarithm on $\mathrm{Re}(s+1) > 0$. On a principal row additionally multiply by $(s-1)/(s+1)$. Analytic continuation and ordinary degree-two convexity, used separately at each fixed conductor, show that f_ψ is bounded analytic on the right half-plane $\mathrm{Re}\, s > 1/2$, with continuous boundary and decay at infinity. The principal pole is removed before applying Poisson representation to its real and imaginary parts. Jensen gives

$$|f_\psi(1/2+\delta+it)|^2 \le \int_{\mathbb{R}} P_\delta(t-v)|f_\psi(1/2+iv)|^2\,dv.$$

The exact row family is finite for fixed H, so its sum interchanges with the integral without a limiting-family issue. The boundary pole-removal factor has modulus at most one.

Take $0 < \varepsilon_0 \le 1/4$ and $\alpha = 3/2 - \varepsilon_0 \in [5/4, 3/2)$. Since $|3/2 + iv| \ge (1+|v|)/\sqrt{2}$, the central squared moment of the f_ψ is bounded by $3C_0H^{1+\varepsilon_0}w_\alpha(v)$, where $w_\alpha(v) = (1+|v|)^{-\alpha}$. For $|t| \ge 2$, split $|v-t| \le |t|/2$ from its complement. The near contribution to $P_\delta * w_\alpha$ is at most $2^\alpha w_\alpha(t) < 3w_\alpha(t)$. In the far region use $P_\delta(t-v) \le 4\delta/(\pi t^2)$ and $\int w_\alpha = 2/(\alpha-1)$, giving at most

$$\frac{2\delta 3^\alpha}{\pi(\alpha-1)}w_\alpha(t) \le 8w_\alpha(t).$$

Here $(1+|t|)^\alpha/t^2 \le 3^\alpha/4$ for $|t| \ge 2$. For $|t| \le 2$, the convolution is at most one and hence at most $3^\alpha w_\alpha(t) < 6w_\alpha(t)$. Thus $P_\delta * w_\alpha \le 11w_\alpha$ uniformly for $0 < \delta < 1/2$. Restoring the denominator costs less than $6(1+|t|)^{5/2}$, and the inverse principal factor is bounded by $(3/2+\delta)/(1/2-\delta)$. The product $11 \cdot 3 \cdot 6 = 198 < 200$ proves (2). The constant is uniform as $\delta \downarrow 0$, but not as the contour approaches the principal pole line $\mathrm{Re}\, s = 1$. No zero-free hypothesis appears.

15.6 The separately scoped unitary infinity type

Let $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$ be fixed and unitary, with angular parameter m and gamma factor γ_ν stated in Section 14. The external primitive functional equation is explicitly

$$Q_\rho^{s/2}\gamma_\nu(s)L(s,\rho) = \epsilon_\rho Q_\rho^{(1-s)/2}\gamma_{\bar\nu}(1-s)L(1-s,\bar\rho), \qquad |\epsilon_\rho| = 1.$$

A constant norm-twist phase in another completion convention is absorbed in ϵ_ρ. All finite Kummer twists share this infinity type. Replace T in the kernel by $T_\nu = 1 + |t+\tau| + |m|$. The numerator

gamma ratio now has real base $1/2 + |m|/2 > 0$ and vertical parameter $t + \tau$; its conjugate dual has parameter $-(t + \tau)$. Stirling gives the same integrable Gaussian-majorized kernel on each fixed shifted line. Constants can depend on m, τ; T_ν is comparable to T for this one fixed ν. The finite-prime conductor analysis, modulus-one coefficient twist, all-ideal decomposition and all original primitive row values remain unchanged.

A pure norm row has $L(s, \rho) = \zeta_K(s + i\tau)$; exclude it from the pole-free approximate functional equation and use scalar convexity at $t + \tau$. In the Poisson step cancel its actual pole by

$$h_\tau(s) = \frac{s - 1 + i\tau}{s + 1 + i\tau}.$$

This factor is analytic in the relevant right half-plane. On its boundary $|h_\tau| \le 1$, while inside $|h_\tau^{-1}| \le (3/2 + \delta)/(1/2 - \delta)$. Hence the same transfer factor 200 applies to the now ν-dependent central constant. If the opposite norm convention is chosen, replace τ by $-\tau$ in the gamma factor and pole alike. This proves the separate fixed-unitary moment assertion; it does not claim uniformity in a family of infinity types or replace the finite-order principal-residue expression below.

15.7 Keeping the full type-I mask during transfer

For $\sigma = 1/2 + \delta$, define L^T by removing every prime of $T(u, d)$, not just conductor primes. The identity (3) is equivalently

$$F_{u,d}(s) = \frac{L(s, \psi_u)}{L^{T(u,d)}(2s, \psi_{u,2})} \prod_{p \in T(u,d),\ p \nmid \mathfrak{f}_u} (1 - \psi_u(p)(Np)^{-s}).$$

Absolute convergence at $2\sigma > 1$ bounds the reciprocal incomplete product by $\zeta_K(2\sigma)$, without requiring any zero-free theorem in the critical strip. For every fixed $\eta > 0$,

$$\prod_{p \in T} (1 + (Np)^{-\sigma}) \ll_{\sigma,\eta} (N \operatorname{rad} T)^\eta, \qquad N \operatorname{rad} T \le C_S Nu\, Nd.$$

Indeed $1 + q^{-\sigma} \le q^\eta$ beyond a fixed threshold, and only finitely many prime ideals lie below it. Therefore

$$|F_{u,d}(\sigma + it)| \ll_{\nu,S,\delta,\eta} (Nu\, Nd)^\eta |L(\sigma + it, \psi_u)|. \tag{13}$$

This is a bounded inequality after the exact mask identity, not an identity that removes original zeros. For example, with principal ν and $u = \pi^3$ or π^6 at $N\pi = 7$, the primitive local ratio, restoration and full masked factor at $s = 1$ are respectively

$v_\pi(u)$	primitive ratio	restoration	masked factor
3	48/49	49/48	1
6	8/7	7/8	1

The squared primitive character in the first row is unramified at π although the original primitive character is ramified there. Squaring a zero sentinel would give the wrong denominator.

For finite-order ν, the normalized type-I piece is the retained d sum of the smoothed squarefree series at $Y = D/Nd$, divided by $\sqrt{D}$. Its possible pole is only at $s = 1$, on principal rows. Contour shifting with the full mask gives $I_u = P_u + E_u$, with E_u exactly (4) and

$$P_u = \mathbf{1}_{\psi_u = 1} \sqrt{D}\, \widehat{W}(1) \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d) \nu(d) \chi_d(u)}{Nd} \rho_{T(u,d)},$$

$$\rho_T = \frac{\operatorname{res}_{s=1} \zeta_K(s)}{\zeta_K(2)} \prod_{p \in T} (1 + (Np)^{-1})^{-1}.$$

This does not impose positivity on P_u. Since β_U is supported on $Nd \le U^2 \le D$, every inner scale Y is at least one. Four integrations by parts give $|\widehat{W}(\sigma+it)| \ll_\sigma \mathcal{M}_4(W)(1+|t|)^{-4}$, which justifies the shifted integrals and their horizontal limits with the classical Hecke growth estimates.

Apply Minkowski to (4) in ℓ^2 of the actual u rows, then use (13) and the off-central moment. For sufficiently small internal losses this bounds $\|E\|_2$ by

$$CD^{\delta}H^{1/2+O(\eta)}\sum_{d}\frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}}\int_{\mathbb{R}}|\widehat{W}(\sigma+it)|(1+|t|)^{(1+O(\eta))/2}\,dt.$$

All original $\nu(d)\chi_d(u)$ have modulus at most one and remain inside the exact expression before this inequality. The actual truncated factorization gives, without replacing β_U by arbitrary coefficients,

$$\sum_{d}\frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \le \left(\sum_{Na\le U}(Na)^{-\sigma+\eta}\right)^2 \ll_{\delta,\eta} U^{1-2\delta+2\eta}.$$

Choose $\eta < \delta$ if necessary and use the fixed-field ideal count. The height integral is bounded by $C\mathcal{M}_4(W)$. Squaring and absorbing the small losses in $(HD)^{\varepsilon}$, using $U^2 \le D$, proves (5). Thus the formerly conditional numerator moment input in this residual transfer is discharged, while its positive-power cutoff loss is retained explicitly.

15.8 What remains open and what has been verified

The new remainder envelope improves the old scalar-convexity H power $3/2-\delta+\varepsilon$ to $1+\varepsilon$. At fixed U one may choose δ for a requested small-loss exponent; for $U = D^a$ with fixed $a > 0$, the remaining excess tends to $2a$ as $\delta \downarrow 0$. Neither statement controls P_u, the other signed piece, nor their covariance. In the original Vaughan identity the short $2a_U$ term must be restored at smaller scales where its support can meet the annulus. Uniform $\mathcal{M}_4$ bounds do not justify interchanging a profile supremum with the row moment or assuming $U^2 \le D'$ after a scale D' is reduced.

The independent prose audit is recorded in `planning/analytic-oracle/arithmetic-review/adversarial-afm-review-20261104.md`. The source-hashed finite checker tests valuation/support decompositions, squarefull coordinates, height exponents and rational Poisson envelopes. Its finite success does not prove reciprocity, analytic continuation, the large sieve, the approximate functional equation or the moments. No new Lean declarations accompany these analytic theorems. Existing Lean local Euler identities are a separately limited verification layer. The independent positive arithmetic determinant comparison, original signed low-gcd short-row estimate, all required contour uniformities and GRH remain open; Dirichlet, Hecke, Dedekind and general Artin scopes are not identified with one another.

Pinned primary sources and external inputs. The principal analytic family input is Blomer–Goldmakher–Louvel, arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1, Appendix II and Section 5. Our fixed-ν proof changes column coefficients and redoes the functional-equation reduction; it does not claim the published corollary directly states every fixed twist. For comparison, Gao–Zhao, arXiv:2201.01885v2, Lemma 2.9 and Section 3.2, uses the sextic large-sieve/moment interface. Its GRH-conditional central-density theorem is not invoked. Abelian Kummer identification uses the classical reciprocity theorem, with the project's pinned Milne *Class Field Theory*, version 4.03 (6 August 2020), Theorem 0.8 and Chapter I, Section 1, a reference: Milne's notes. The primitive degree-one Hecke continuation and completed functional equation are explicit classical external inputs; the project's pinned Kudla notes on Tate's thesis, Theorem 4.3 and Corollary 4.4, supply a reference.

Their sketch and incomplete displayed epsilon formulas are not used as independently certified constants. Standard Stirling, Mellin inversion, Poisson representation and scalar convexity complete the external analytic dependencies. None is replaced by a finite diagnostic or a GRH-equivalent assumption.

16 Recombining the actual weighted cutoff

Keep the finite-order Eisenstein family and original masks of Section 14. For $U \geq 1$ and a squarefree ideal m, put

$$A_U(m) = \sum_{d|m} \beta_U(d) = \sum_{\substack{ab|m \\ (a,b)=1,\ Na,Nb\leq U}} \mu(a)\mu(b).$$

Every summation variable here is an integral ideal. The last formula uses squarefreeness of m; an unconstrained square of a Möbius polynomial would also include forbidden overlapping supports.

Theorem 7 (Exact recombination and an absolute-half-plane rate)**.** *Write $T_0 = S \cup \mathrm{supp}(u)$ and $J_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,S)=1} \beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s)$. For $\mathrm{Re}\, s > 1$,*

$$J_{u,U}(s) = \sum_{m \text{ squarefree},\ (m,T_0)=1} A_U(m)\psi_u(m)(Nm)^{-s},$$

$$|A_U(m)| \leq 3^{\omega(m)}, \qquad A_U(m) = \mu(m) \quad (Nm \leq U).$$

For every $\kappa > 0$ and $0 < \epsilon < \kappa$, uniformly for $\mathrm{Re}\, s \geq 1 + \kappa$, all original rows and all $U \geq 1$,

$$\left| J_{u,U}(s) - \frac{1}{L^{T_0}(s,\psi_u)} \right| \leq U^{-\kappa+\epsilon}\{\zeta_K(1+\epsilon)^3 + \zeta_K(1+\epsilon)\}.$$

Here L^{T_0} removes every prime of T_0, including conductor-drop primes; the bound is in the absolute-convergence half-plane only.

Proof. On a surviving term $m = dc$, the two squarefree ideals c, d are coprime, and both avoid T_0. The original product $\nu(d)\chi_d(u)\psi_u(c)$ is then $\psi_u(m)$. Grouping by m gives A_U. Each prime of m has three possible positions in $(a, b, m/(ab))$, so the absolute sum is at most $3^{\omega(m)}$. When $Nm \leq U$ all divisors fit both cutoffs; the primewise sum is $1 - 1 - 1 = -1$, giving $\mu(m)$.

The same bound justifies absolute convergence and regrouping for $\mathrm{Re}\, s > 1$, because on squarefree ideals $3^{\omega(m)} = \tau_3(m)$ and $\sum_m \tau_3(m)(Nm)^{-v} = \zeta_K(v)^3$ for $v > 1$, where this latter sum runs over all nonzero integral ideals. Subtract the absolutely convergent Möbius series for $1/L^{T_0}$. Only $Nm > U$ remain. Rankin's inequality with exponent $\eta = \kappa - \epsilon$ bounds their absolute sum by

$$U^{-\eta} \sum_m \{\tau_3(m) + 1\}(Nm)^{-1-\epsilon},$$

which proves the stated constant independently of deleted primes, row multiplicities or character phases. □

For W supported in $[1, 2]$, the same exact regrouping gives the original normalized type-I polynomial

$$I_u = D^{-1/2} \sum_{m \text{ squarefree},\ (m,S)=1} A_U(m)\nu(m)\chi_m(u)W(Nm/D).$$

This formula retains all original zero sentinels. It permits a direct family large sieve before any absolute d sum; its quantitative moment is developed separately below. The displayed absolute-half-plane rate alone is not that critical-strip moment.

17 Euler packaging, coprimality and the analytic boundary

For $p \notin T_0$, set $z_p = \psi_u(p)(Np)^{-s}$ and $t_p = z_p/(1+z_p)$. On $\operatorname{Re} s > 1$, deleting an additional surviving prime divides the squarefree Euler product by $1 + z_p$. Consequently the exact retained expression is

$$J_{u,U}(s) = F_{u,1}(s)B_{u,U}(s), \qquad B_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,T_0)=1} \beta_U(d) \prod_{p \mid d} t_p.$$

Original zeros enforce $(d, u) = 1$ even when a sixth power drops from its primitive conductor. This factorization cannot restore a forbidden d row by replacing a zero sentinel with an unramified primitive value. The equality continues wherever both sides are holomorphic.

Without the cutoffs, $\beta_\infty(d) = \mu(d)2^{\omega(d)}$ on squarefree ideals. Absolute convergence justifies

$$B_{u,\infty} = \prod_{p \notin T_0} (1 - 2t_p) = \prod_{p \notin T_0} \frac{1 - z_p}{1 + z_p}, \qquad F_{u,1}B_{u,\infty} = \prod_{p \notin T_0} (1 - z_p) = \frac{1}{L^{T_0}(s, \psi_u)}.$$

Theorem 7 gives a quantitative rate rather than an unsupported exchange of this limit with a critical-strip contour.

A local counterexample to the naive square. In a one-prime projection at a good Eisenstein prime of norm 7, take the principal local phase and $s = 1$. Then $z = 1/7$, $t = 1/8$ and, for $U \geq 7$, the exact polynomial is $B_U = 1 - 2t = 3/4$. The unconstrained square $(1 - t)^2 = 49/64$ differs by $1/64$: it admits the forbidden pair $a = b = p$. After multiplying by $F = 1 + z = 8/7$ it gives $7/8$ instead of the correct $6/7$. This is an actual local Euler calculation, not an infinite-series or mean-square counterexample. It identifies the exact coprimality condition that a proposed mollifier-square simplification must preserve.

The exact intermediate coefficient band. For a squarefree m with $U < Nm \leq U^2$, one has $A_U(m) = -\mu(m)$. Indeed the unrestricted disjoint-assignment sum is $\mu(m)$. If $Na > U$, every $b \mid m/a$ automatically has $Nb \leq Nm/Na < U$, and $\sum_{b \mid m/a} \mu(b)$ vanishes unless $a = m$. Hence the excluded a assignments contribute exactly $\mu(m)$; the excluded b assignments contribute another $\mu(m)$. They cannot overlap because $NaNb > U^2 \geq Nm$ contradicts $ab \mid m$. Subtraction proves the claim. Together with the small-norm band, this gives $A_U = \mu$ for $Nm \leq U$ and $A_U = -\mu$ for $U < Nm \leq U^2$. The original smoothed columns have $Nm \geq D \geq U^2$, so the new band does not justify replacing their coefficients by $-\mu(m)$: generally they lie beyond it.

Theorem 8 (A stronger cutoff convergence would require zero-freeness)**.** *Fix a nonprincipal primitive ψ_u. If these actual $J_{u,U}$ converge locally uniformly on $\operatorname{Re} s > 1/2$ as $U \longrightarrow \infty$, then $L(s, \psi_u)$ has no zero there.*

Proof. Classical abelian Hecke continuation makes the numerator of $F_{u,1} = L^{T_0}(s, \psi_u)/L^{T_0}(2s, \psi_{u,2})$ holomorphic. Its denominator is nonzero on $\operatorname{Re} s > 1/2$ by the absolute Euler product at $2s$. Each $B_{u,U}$ is a finite sum with nonzero denominators $1 + z_p$ on $\operatorname{Re} s > 0$. Thus every $J_{u,U}$ is holomorphic on the asserted half-plane. A locally uniform limit G is holomorphic there. On $\operatorname{Re} s > 1$, the already proved convergence gives $L^{T_0}(s, \psi_u)G(s) = 1$. The identity theorem extends this equality to the connected half-plane, precluding a zero of L^{T_0}. The finitely deleted factors $1 - \psi_u(p)(Np)^{-s}$ do not vanish when $\operatorname{Re} s > 0$, so the same conclusion holds for the primitive L-function. □

This is an implication, not a proof that the stronger convergence holds, and it does not assert a converse. In particular, a family-averaged energy bound is not silently promoted to locally uniform convergence for each individual character. Principal rows have a pole at 1 and were excluded from this proposition. General Artin holomorphy is not an input.

Reproducible exact diagnostics. The new source `examples/arithmetic_cutoff/exact_recombination.py` uses actual split/inert prime ideals and raw sextic symbols in $\mathbb{Z}[\omega]$, with exact rational arithmetic. It checks the direct masked sum, coefficient regrouping, Euler packaging, both finite full-cutoff limits and original zeros, including third/sixth-power rows. Its overlapping- support negative controls are not tests of an infinite large sieve or of zero location. The analytic convergence argument above uses ordinary ideal Euler products and the explicitly stated Hecke continuation theorem; no new Lean declaration or complete GRH proof is claimed.

18 A cutoff-uniform envelope for the original masked remainder

Fix $K = \mathbb{Q}(\omega)$, $\mathcal{O} = \mathbb{Z}[\omega]$, a finite-order Hecke character ν, and a finite set S containing the primes above 6 and the conductor of ν. All unspecified ideal sums below are over nonzero integral ideals. Let $\chi_m(u) = (u/m)_6$ be the original symbol, zero when $(m,u) > 1$, and put

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \qquad A_U(m) = \sum_{d \mid m} \beta_U(d).$$

The columns d, c, m are squarefree and prime to S where indicated. Retain the primitive characters $\psi_u = \mathrm{prim}(\nu\kappa_u)$, $\psi_{u,2} = \mathrm{prim}(\psi_u^2)$, and the full deleted set $T(u,d) = S \cup \mathrm{supp}(u) \cup \mathrm{supp}(d)$. For $H, D, U \ge 1$, $U^2 \le D$, and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$, the original normalized type-I piece is

$$I_u = D^{-1/2} \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u) W(Nm/D). \tag{14}$$

Its exact principal residue is

$$P_u = \mathbf{1}_{\psi_u=1} \sqrt{D}\, \widehat{W}(1) \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd} \rho_{T(u,d)},$$

$$\rho_T = \frac{\mathrm{res}_{s=1}\, \zeta_K(s)}{\zeta_K(2)} \prod_{p \in T} (1 + (Np)^{-1})^{-1}. \tag{15}$$

Define $E_u = I_u - P_u$; this is the same residue-subtracted remainder as the fully masked Mellin contour, with no cutoff limit or profile supremum.

Theorem 9 (Uniformity in the actual truncated cutoff)**.** *For every $\varepsilon > 0$,*

$$\sum_{0 < Nu \le H} |E_u|^2 \ll_{\nu,S,\varepsilon} \|W\|_\infty^2 (HD)^\varepsilon \{H + DH^{1/3} + (HD)^{2/3}\}. \tag{16}$$

The sum is over every actual nonzero Eisenstein element, including units, S-supported powers and sixth-power rows. The constant is independent of U under $U^2 \le D$. This is an application of the classical sextic large sieve with exact original coefficients, not a zero-location theorem.

Proof. On an absolutely convergent line, expand the restored squarefree series $F_{u,d}$. Its condition $(c, T(u,d)) = 1$ includes $(c,d) = 1$, and the original symbols multiply. Grouping $m = cd$ gives (14) by Mellin inversion. Each prime of m may belong to the first truncated Möbius divisor, to the second, or to neither. Consequently $|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta$, uniformly in U. The length-D coefficient sequence in (14) therefore has squared norm $\ll_\eta \|W\|_\infty^2 D^\eta$.

The all-ideal row norm in Section 3 of Blomer–Goldmakher–Louvel, arXiv:1112.1650v1 gives

$$B_2^1(X,D) \ll_\eta (XD)^\eta \{X + DX^{1/3} + (XD)^{2/3}\}.$$

For completeness, split $a = a_1 a_2^2 \cdots a_5^5 a_6^6$ dyadically; $a_1, \dots, a_5$ are squarefree and pairwise coprime. If their dyadic sizes are A_j, put $R = \prod_{j=1}^6 A_j$ and $M = \max(A_1, A_2)$. Pivot on the first or second variable, using the squarefree powered sieve proved simultaneously in that paper's Section 3.2. The other variables are bounded column multipliers. The block energy is at most

$$(XD)^\eta \{R + DR/M + D^{2/3}R/M^{1/3}\} \sum_m |\lambda_m|^2.$$

The inequalities $R \le X$, $R^3 \le XM^3$ and $R^3 \le X^2 M$ follow from $\prod_j A_j^j \le X$ and the pivot choice. Summing the logarithmically many blocks proves the displayed bound.

Finite S-unit and ray-class partitions convert the original literal symbol into the canonical column value $\vartheta_m(a)$ with fixed modulus-one coefficient multipliers. On noncoprime pairs both values are zero: squarefree m has outside-S conductor exactly m. This orientation preserves sixth-power zeros. At most six elements generate each outside-S ideal times an S-supported ideal; the S-supported sums converge at powers 1, 1/3 and 2/3. Thus (16) holds first with I_u in place of E_u.

A principal ψ_u requires every outside-S valuation of u to be divisible by 6. There are $O_{\nu,S}(H^{1/6})$ such literal rows. Equation (15), $|\rho_T| \ll_K 1$, and the actual coefficient sum

$$\sum_d \frac{|\beta_U(d)|}{Nd} \le \left(\sum_{Na \le U} \frac{1}{Na} \right)^2 \ll \log^2(2U)$$

give $\sum |P_u|^2 \ll DH^{1/6} \|W\|_\infty^2 \log^4(2U)$. This is absorbed by the right side of (16), using $U^2 \le D$ and a smaller internal exponent. Finally $|E_u|^2 \le 2|I_u|^2 + 2|P_u|^2$. The extended proof specifies the canonical-family transfer and the powered-sieve input precisely. □

Combining this with the previously proved fixed-twist AFM transfer yields

$$\sum_{0 < Nu \le H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 (HD)^\varepsilon \min\{D^{2\delta} U^{2-4\delta} H,\ H + DH^{1/3} + (HD)^{2/3}\}. \tag{17}$$

At $H = D^{3/4}$ and $U = D^a$, $0 \le a \le 1/2$, the excess over H is at most $\min\{2\delta + (2-4\delta)a, 1/2\}$, up to arbitrarily small losses. The new envelope improves the old one for $a > 1/4$ as $\delta \downarrow 0$, but still leaves an exponent loss of 1/2. It does not settle the original signed short-row bound, type-II covariance, positive arithmetic determinant comparison or GRH. The principal-residue bound used here is sufficient for this envelope, not the stronger target needed to reassemble the full signed argument.

19 All-ideal reciprocity and the surviving block obstruction

The proof uses no new zero-free input. Its external analytic ingredient is the pinned BGL large sieve and the powered estimates established in its proof; elementary ideal counting and the already restored Mellin identity supply the remaining interfaces. All constants may depend on fixed K, S, ν and declared small exponents, never on U.

19.1 Exact recombination retains the primitive corrections

On $\operatorname{Re} s > 1$, start from the full defining series, with its original zero masks:

$$F_{u,d}(s) = \sum_{\substack{c \text{ squarefree} \\ (c,T(u,d))=1}} \nu(c)\chi_c(u)(Nc)^{-s}.$$

Its primitive ratio involves $L(s,\psi_u)$, the independently primitive $L(2s,\psi_{u,2})$ and every deleted Euler factor. Expanding this *restored* series gives the exact equality

$$\begin{aligned}\sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} &\beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s) \\ &= \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u)(Nm)^{-s}. \end{aligned} \tag{18}$$

Indeed $(c,d)=1$ makes $m=cd$ squarefree, with $d \mid m$. If m meets u, both original symbols retain their zero. Only on the absolutely convergent line is this infinite series used. Mellin inversion there gives the finite smoothed polynomial (14). Nothing evaluates (18) as a Dirichlet series on $\operatorname{Re} s = 1/2+\delta$ or interchanges a cutoff limit with continuation across unknown zeros.

For squarefree m, a factorization $ab = d \mid m$ assigns each prime to a, to b, or to neither. The actual inequalities $Na, Nb \le U$ select a subset of these assignments. Thus

$$|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta.$$

The last divisor bound follows by choosing a fixed threshold so $3 \le (Np)^\eta$ beyond it, then absorbing the finitely many smaller prime ideals in a constant. When $Nm \le U$ all assignments occur, so $A_U(m) = \prod_{p\mid m}(1-1-1) = \mu(m)$; no such simplification is made for the length-D columns in general. With $\lambda_m = D^{-1/2}A_U(m)\nu(m)W(Nm/D)$, ideal counting and a smaller internal η give

$$\sum_m |\lambda_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta, \qquad \operatorname{supp}\lambda \subseteq \{D \le Nm \le 2D\}. \tag{19}$$

19.2 The powered squarefree input and all-ideal pivot

Let ϑ_a be BGL's canonical primitive sixth-order family. Their Section 3 defines B_1^j using squarefree row/column ideals and kernel $\vartheta_a(m)^j$, while B_2^j uses *all* ideal rows and kernel $\vartheta_m(a)^j$. Theorem 1.3 states the squarefree bound for $j=1$; its Section 3.2 proves the estimates simultaneously for $1 \le j \le 5$, because the recursion in Lemma 3.4 uses $j\ell$. In particular the two inputs needed here are

$$B_1^j(M,D) \ll_\eta (MD)^\eta\{M + D + (MD)^{2/3}\}, \qquad j = 1,2. \tag{20}$$

To see the final step in the published recursion explicitly, its $E_{4/3}$ estimate is $M^{4/3} + D + (MD)^{2/3}$. If $M \le D$, its first term is at most $(MD)^{2/3}$; if $M > D$, use reciprocity/duality (3.2) and the estimate with M, D reversed. The finite set of powers is treated together; no inference from a fixed conductor count to an arbitrary primitive-character sieve occurs.

Write each ideal row uniquely as

$$a = a_1 a_2^2 a_3^3 a_4^4 a_5^5 a_6^6,$$

where $a_1, \dots, a_5$ are pairwise-coprime squarefree ideals and a_6 is arbitrary. For valuation $6q+r$, $0 \le r \le 5$, place the prime in a_r if $r > 0$ and its power q in a_6. Dyadically let $A_j \le Na_j < 2A_j$,

with $A_j \ge 1$, and put $R = \prod_{j=1}^{6} A_j$. A nonempty block satisfies $\prod_j A_j^j \le X$ when $Na \le X$. Choose $\ell = 1$ if $A_1 \ge A_2$, and $\ell = 2$ otherwise; put $M = A_\ell$. Keep every other ideal fixed. Evaluation of the single *column* character gives

$$\vartheta_m(a) = \vartheta_m(a_\ell)^\ell \vartheta_m(a/a_\ell^\ell).$$

The second factor is a bounded column multiplier, including its original zero if m meets any fixed factor. It may depend on all the fixed ideals but not on the pivot row. Pairwise-coprimality and the exact row norm restriction merely select a subset of the positive pivot-row sum. Apply (20) to the first factor. There are $O_K(R/M)$ choices of the other ideals by ideal counting. The block energy is bounded by

$$(XD)^\eta \{R + DR/M + D^{2/3}R/M^{1/3}\} \sum_m |\lambda_m|^2. \tag{21}$$

All three monomial bounds used to sum these blocks are elementary. First $R \le \prod_j A_j^j \le X$. If $M = A_1$, then

$$(R/M)^3 = A_2^3 \cdots A_6^3 \le A_1 A_2^2 A_3^3 A_4^4 A_5^5 A_6^6 \le X,$$

since $A_2 \le A_1$ and $j \ge 3$ for the remaining factors. If $M = A_2$, use $A_1 \le A_2$ in the same comparison. Finally,

$$\frac{R^3}{M(\prod_j A_j^j)^2} = \frac{A_1}{M A_2 A_3^3 A_4^5 A_5^7 A_6^9} \le 1.$$

Thus $R/M \le X^{1/3}$ and $R/M^{1/3} \le X^{2/3}$. Insert these in (21) and sum the logarithmically many blocks. This proves the all-ideal estimate used in Theorem 9. It is precisely the mechanism of BGL Lemma 3.4 and (3.6)–(3.9), with the final powered bound inserted. The simpler fixed-squarefull-row argument would leave the weaker $D\sqrt{X}$ term.

19.3 Literal rows, reciprocal orientation and principal count

Let $a = (u)_S$ denote the outside-S ideal. The finite S-unit comparison gives $\kappa_u = \eta_u \vartheta_a$ as primitive characters, with η_u in a finite fixed set. For $(a, m) = 1$, BGL reciprocity therefore gives

$$\chi_m(u) = \eta_u(m) C([a], [m]) \vartheta_m(a). \tag{22}$$

The notation is deliberate: the left side is the project's original $(u/m)_6$, not BGL's canonical character indexed by m. When $(a, m) > 1$, the left side is zero. The right side is also zero, because squarefree m has outside-S primitive conductor exactly m. Equation (22) hence extends to every pair. Using $\vartheta_a(m)$ on noncoprime pairs instead would be false: primitive ϑ_{b^6} has restored primes of b, whereas the original symbol must still vanish there.

Partition rows by η_u and $[a]$, both finite. On each part the factor $\eta_u(m)C([a], [m])$ is a fixed modulus-one column multiplier. A positive row sum over that part is bounded by the full B_2^1 row sum. For each S-supported ideal b and outside-S ideal a, at most six actual elements generate ab. Thus the required literal energy is bounded by the all-ideal norm summed over b, with $X = H/Nb$. The geometric products

$$\sum_{b \ S\text{-supported}} (Nb)^{-q} = \prod_{p \in S} (1 - (Np)^{-q})^{-1} < \infty, \qquad q = 1,\ 1/3,\ 2/3,$$

transfer the three terms without losing any unit or S-power row. Together with (19), this proves the claimed I energy.

If ψ_u is principal, ν is unramified outside S, and tame Kummer inertia at $v_p(u)$ has order $6/\gcd(6, v_p(u))$. It must be trivial, so each such valuation is divisible by 6. Writing $(u) = ba_6^6$ with b supported in S, ideal counting gives

$$\#\{0 < Nu \le H : \psi_u = 1\} \ll_K H^{1/6} \prod_{p \in S} (1 - (Np)^{-1/6})^{-1}.$$

This is only a necessary condition, sufficient for an upper count. The exact residue formula (15) and $|\widehat{W}(1)| \le \|W\|_\infty$ then give the principal energy bound used above. No sign or positivity of that residue is assumed.

19.4 What the envelope still fails to prove

The $DH^{1/3}$ term is still larger than $H(HD)^\varepsilon$ at $H = D^{3/4}$. The monomial comparison itself is saturated by $A_1 = A_2 = X^{1/3}$ and $A_3 = \cdots = A_6 = 1$. This proves sharpness of that *particular comparison*, not an energy lower bound for actual arithmetic coefficients. In particular, no conclusion that the signed A_U cannot cancel is justified. The next load-bearing input must exploit those actual truncated Möbius coefficients inside the mixed moment, or establish the required signed type-II minus type-I covariance. The smaller-scale short term, principal contribution, full arithmetic determinant identification and GRH remain separate obligations.

Exact finite checks use the split Eisenstein primes generated by $3 + \omega$, $4 + \omega$, $5 + 2\omega$, $6 + \omega$, of norms $7, 13, 19, 31$. Their actual residue symbols are computed in $\mathbb{Q}(\zeta_6)$, retaining sixth-power zeros. Coefficient recombination and dyadic inequalities pass; they check finite algebra and bookkeeping, not the external analytic sieve or the contour theorem. No new Lean declarations accompany this argument.

20 High cutoffs, actual bilinear factors and a signed covariance test

Retain the original Eisenstein family, fixed finite-order ν, fixed S, common $W \in C_c^\infty((0,\infty))$ supported in $[1, 2]$, and every raw shared-prime zero. Put

$$C_U(m) = \sum_{\substack{abc=m \\ Na>U,\, Nb>U}} \mu(a)\mu(b), \qquad r_U(m) = \#\{(a, b) : ab = m,\ Na, Nb > U\}.$$

All factors are ideals, squarefree and pairwise coprime; on the active annulus $A_U = C_U - \mu$. A coefficient identity is different from a row-energy bound.

Proposition 10 (The high-cutoff window)**.** *If $D > 1$ and $2D/7 \le U^2 \le D$, then for every surviving squarefree column $D \le Nm \le 2D$,*

$$C_U(m) = \mu(m) r_U(m), \qquad A_U(m) = \mu(m)(r_U(m) - 1), \qquad |A_U(m)| \ge 1.$$

For each fixed nonzero W, the normalized squared coefficient norm satisfies

$$D^{-1} \sum_{\substack{m \text{ sf} \\ (m,S)=1}} |A_U(m) W(Nm/D)|^2 \gg_{W,S} 1 \tag{23}$$

for large D, uniformly in this window. This is not a lower bound for row energy.

Proof. The least norm of a nonunit ideal outside S is at least 7. A long-long term with $c \neq 1$ would have $Nm > 7U^2 \geq 2D$. Thus $ab = m$, including both endpoints. The fixed-point-free swap $(a, b) \mapsto (b, a)$ makes r_U even, so A_U is an odd nonzero integer. The last claim follows by counting squarefree ideals on a fixed interval where $|W|$ is bounded below; the constant and error are proved in the extended proof. □

The exact remaining polynomial is genuinely bilinear, with $(a, b) = 1$, $U < Na, Nb < 2D/U$, $D \leq Na\,Nb \leq 2D$, and $x_{ab}(u) = x_a(u)x_b(u)$, including zeros. Both factors see the same row $u = \varepsilon\alpha\beta^2$. Cauchy followed by the accepted powered BGL bound gives only

$$Q_{B_X}(C_U, C_U) \ll_{\nu,S,\epsilon} (DX)^{\epsilon}\|W\|_{\infty}^2\{\sqrt{D}\,X^2 + DX + D^{5/6}X^{5/3}\}. \tag{24}$$

At $X \asymp D^{1/4}$ the exponents are $1, 5/4, 5/4$; the earlier regrouped bound has $1/2, 5/4, 13/12$ and is stronger. No BT_σ saving is obtained. Multiplying two marginal operator norms does not supply a coupled fourth moment.

A finite test in the original family. There is no automatic nonnegative signed covariance, even when $C_U(m)\mu(m) \geq 0$. Take $\nu = 1$, S the primes above $2, 3$, $D = 400$, $U = 17$, and all 72 literal rows of B_7. Choose the common nonnegative smooth profile constructed in the extended proof, with values 10 at 439/400, 1 at 475/400, and zero at every other integer norm divided by 400. There are exactly four active squarefree ideal columns, of norms $439, 439, 475, 475$. In that order,

$$A = (1, 1, 1, 1), \quad C = (0, 0, 2, 2), \quad \mu = (-1, -1, 1, 1), \qquad \Re G_{B_7} = \begin{pmatrix} 72 & 0 & 0 & 18 \\ 0 & 72 & 18 & 0 \\ 0 & 18 & 72 & 0 \\ 18 & 0 & 0 & 72 \end{pmatrix}.$$

Exact raw-symbol enumeration, independently reproducible from the ideal and residue-field definitions, gives

$$Q(C, \mu) = -\frac{27}{25}, \qquad Q(C, C) = \frac{36}{25}, \quad Q(\mu, \mu) = \frac{864}{25}, \quad Q(A, A) = \frac{954}{25}. \tag{25}$$

The real Gram matrix is positive definite, with eigenvalues $54, 90$; its positivity does not force a cross covariance to be positive. This fixed finite example refutes that shortcut, not the asymptotic BT_σ target or the original signed mean-square estimate.

The two-cutoff contour remains coupled. For $T_0 = S \cup \mathrm{supp}(u)$, $\psi = \psi_u$ and independently primitive $\psi_2 = \mathrm{prim}(\psi^2)$, put $q_p = \psi(p)(Np)^{-s}$, $\alpha_p = (Np)^{-z}$, $\beta_p = (Np)^{-w}$ outside T_0. The actual three-placement Euler factor is $1 + q_p - q_p\alpha_p - q_p\beta_p$. Its exact factorization is

$$\mathcal{F}(s, z, w) = \frac{L^{T_0}(s, \psi)}{L^{T_0}(s + z, \psi)L^{T_0}(s + w, \psi)} \prod_{p \notin T_0} \left(1 - \frac{q_p^2(1 - \alpha_p)(1 - \beta_p)}{(1 - q_p\alpha_p)(1 - q_p\beta_p)}\right). \tag{26}$$

Exact double Perron inversion is valid on positive auxiliary lines $c, d > 1 - \Re s$ for $1/2 < \Re s < 1$, retaining the original principal pole exclusion. These safe lines reproduce the previous $U^{2-4\delta}$ energy loss at $\Re s = 1/2 + \delta$, with the same common-profile derivative and fixed-margin dependencies. The correction product alone does not justify moving those lines through reciprocal-L poles. In fact $\mathcal{F}(s, z, 0) = 1/L^{T_0}(s + z, \psi)$ as a meromorphic identity. No zero-free region, reciprocal moment, analytic saving, new Lean formalization or GRH proof is supplied by this construction.

21 Proof details for the high-window and two-cutoff tests

21.1 The coefficient lower bound and its precise limitation

The element lattice $\mathbb{Z}[\omega] \subset \mathbb{C}$ has covolume $\sqrt{3}/2$. Counting disks with boundary error $O(\sqrt{x})$ and dividing by the six units, using that this ring is a PID, gives

$$N_K(x) = \kappa x + O(\sqrt{x}), \qquad \kappa = \frac{\pi}{3\sqrt{3}}.$$

Finite inclusion-exclusion at S gives $N_{K,S}(y) = \kappa y \prod_{p\in S}(1-(Np)^{-1}) + O_S(\sqrt{y})$. Insert $1_{\mathrm{sf}}(m) = \sum_{d^2\mid m}\mu(d)$. Absolute convergence at 2, the tail $\sum_{Nd>\sqrt{x}}(Nd)^{-2} = O(x^{-1/2})$, and $\sum_{Nd\le\sqrt{x}}(Nd)^{-1} = O(\log(2x))$ yield

$$N_{\mathrm{sf},S}(x) = c_S x + O_S(\sqrt{x}\log(2x)), \qquad c_S = \frac{\kappa}{\zeta_K(2)}\prod_{p\in S}(1+(Np)^{-1})^{-1} > 0.$$

These tail estimates follow by partial summation from the displayed lattice count. For a fixed nonzero W, choose $[a,b] \subset (1,2)$ with $|W| \ge \eta > 0$. Proposition 10 gives, for large D,

$$D^{-1}\sum |A_U(m)W(Nm/D)|^2 \ge \tfrac{1}{2}\eta^2 c_S(b-a).$$

There is no uniform lower constant over all profiles with one specified supremum norm. Nor can a coefficient lower bound be combined with an operator upper bound to produce a row-energy lower bound: no lower singular value has been established on the actual vector.

21.2 What legal factor Cauchy actually proves

For dyadic $Na \asymp A, Nb \asymp B$, retain the exact $(a,b)=1$ mask, hyperbolic support and $W(Na\,Nb/D)$ in

$$T_a(u) = \sum_b \mu(b)\nu(b)W(Na\,Nb/D)x_b(u).$$

Cauchy gives

$$|\sum_a \mu(a)\nu(a)x_a(u)T_a(u)|^2 \ll A\sum_a |T_a(u)|^2.$$

For fixed a, fixed second row factor and fixed unit/ray classes, the b coefficients are independent of the pivot row and have squared norm $\ll B\|W\|_\infty^2$. The prior classical power-one/two BGL input has $\Delta(X,B) = X + B + (XB)^{2/3}$. Summing the $O(A)$ fixed columns and $O(X)$ other row factors gives

$$\mathcal{E}_{A,B} \ll_{\nu,S,\epsilon} (DX)^\epsilon \|W\|_\infty^2 \frac{A^2B}{D} X\Delta(X,B) \ll (DX)^\epsilon\|W\|_\infty^2\{AX^2 + DX + D^{2/3}A^{1/3}X^{5/3}\}.$$

In the high window $A, B \asymp \sqrt{D}$, and only boundedly many dyadic pairs occur. This proves (24); exchanging factors or row pivots leaves its DX term. A product of marginal second-moment bounds supplies no estimate for the two transforms on the same (r,s). Any useful saving must control that coupled fourth moment or the signed actual-coefficient covariance, with all masks retained.

21.3 A smooth profile and the exact covariance computation

Let

$$\phi(t)=\begin{cases}\exp(1-1/(1-t^2)), & |t|<1,\\ 0, & |t|\ge 1,\end{cases}\qquad W_x(y)=x\phi(4(400y-439))+\phi(4(400y-475)).$$

For $x\ge 0$ this is nonnegative and smooth; its disjoint supports lie strictly inside $(1,2)$. For every integer n, $W_x(n/400)$ equals x at $n=439$, 1 at $n=475$ and zero otherwise. Thus the computation uses exact values of a genuine common C_c^∞ profile, rather than claiming a finite rational weight is smooth. At $x=10$, $\|W_x\|_\infty=10$.

The primary ideal generators, in the order used above, are

$$(-23,-18),\quad (-5,18),\quad (10,-15),\quad (25,15),$$

where (a,b) denotes $a+b\omega$. The first two have norm 439. The last two factor as $(-2,3)(-5,0)$ and $(-5,-3)(-5,0)$, with norms $19\cdot 25$. Since $19,25>17$, the long-factor count is 2 there and zero on the prime columns. All squarefree ideals at both selected norms are included. The complete B_7 uses the four good primes of norms $7,7,13,13$ for distinct ordered row factors and all six units. It has 72 distinct rows, maximum norm 2197; each is nonprincipal by its outside-S Kummer valuation 1 or 2 modulo 6, whose nontrivial tame character has conductor exponent 1.

For the raw Gram $G_{ij}=\sum_{u\in B_7} x_{m_i}(u)\overline{x_{m_j}(u)}$, exact finite-field exponentiation and lattice enumeration give the displayed real matrix. The weighted real vectors are $\ell=(0,0,2,2)$, $v=(-x,-x,1,1)$, and $a=(x,x,1,1)$. Therefore

$$Q(C_U,\mu)=D^{-1}\ell^t(\Re G)v=\frac{36-9x}{50}.$$

Its diagonal contribution is $36/50>0$ and its off-diagonal contribution is $-9x/50$. For $x=10$ this is $-27/25$ and gives (25). The identity $(\Re G-72I)^2=18^2I$ proves the real eigenvalues $54,90$; these are not being asserted as the eigenvalues of the full complex Gram. The exact source is `examples/arithmetic_cutoff/signed_covariance.py`. A second test retains raw zeros on the 246 rows $0<Nu\le 89$ whose outside-S part is nonunit. Its covariance at $x=10$ is $-69/50$. This subset is guaranteed nonprincipal; it is not called the entire nonprincipal family. Here $89=\lfloor 400^{3/4}\rfloor$. These are finite counterexamples to automatic covariance positivity, not asymptotic counterexamples to BT_σ or GRH.

21.4 Double Perron without a divergent critical-strip series

For $T_0=S\cup\operatorname{supp}(u)$ retain $F_0(s)=L^{T_0}(s,\psi)/L^{T_0}(2s,\psi_2)$, with the independently primitive square character. Set $t_p=q_p/(1+q_p)$ outside T_0. The finite actual cutoff is

$$J_U(s,u)=F_0(s)B_U(s),\qquad B_U(s)=\sum_{\substack{a,b\ \mathrm{sf},\ (a,b)=1\\ (ab,T_0)=1,\ Na,Nb\le U}}\mu(a)\mu(b)\prod_{p\mid ab}t_p.$$

For $\sigma=\Re s>1/2$, the series

$$\mathcal{G}(s,z,w)=\sum_{\substack{a,b\ \mathrm{sf},\ (a,b)=1\\ (ab,T_0)=1}}\mu(a)\mu(b)\prod_{p\mid ab}t_p\,(Na)^{-z}(Nb)^{-w}=\prod_{p\notin T_0}\left(1-t_p(Np)^{-z}-t_p(Np)^{-w}\right)$$

converges absolutely on $\sigma+\Re z>1$, $\sigma+\Re w>1$, uniformly in all vertical parts at fixed positive margins. This follows from $|t_p|\le (Np)^{-\sigma}/(1-(Np)^{-\sigma})$. There $\mathcal{F}=F_0\mathcal{G}$, without evaluating an infinite c series in the critical strip. Initial absolute convergence also proves (26) by the local identity

$$(1-q)(1+q-q\alpha-q\beta)=(1-q\alpha)(1-q\beta)-q^2(1-\alpha)(1-\beta).$$

The correction product is holomorphic on

$$\sigma+\Re z>0,\quad \sigma+\Re w>0,\quad 2\sigma+\min(\Re z,0)+\min(\Re w,0)>1.$$

On margins $\sigma+\Re z,\sigma+\Re w\ge\eta>0$ and last exponent $\ge 1+\epsilon$, its local deviation from 1 is at most $4(1-2^{-\eta})^{-2}(Np)^{-1-\epsilon}$. This proves normal convergence and a phase/deletion-uniform vertical upper bound. Nonvanishing of the correction is not asserted.

For $1/2<\sigma<1$, positive lines $c,d>1-\sigma$, and $V=\lfloor U\rfloor+1/2$, define

$$\mathcal{I}_{T_1,T_2}=\frac{1}{(2\pi i)^2}\int_{c-iT_1}^{c+iT_1}\int_{d-iT_2}^{d+iT_2}\mathcal{F}(s,z,w)\frac{V^{z+w}}{zw}\,dw\,dz.$$

The half-integer V includes the actual equality cutoff, since $Na\le U$ is exactly $Na<V$. Put $\Delta_V=\inf_{n\in\mathbb{Z}_{\ge1}}|\log(V/n)|\ge 1/(2V+1)$, and let $\mathcal{A}=\sum_{a,b}|\mu(a)\mu(b)\prod_{p\mid ab}t_p|(Na)^{-c}(Nb)^{-d}$ with the same disjoint support. For $c,T>0$ and $v>0$, $v\ne1$, define

$$K_{c,T}(v)=\frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{v^z}{z}\,dz.$$

Finite-height termwise integration of the absolutely convergent $\mathcal{G}$ and the one-variable tail estimate $|K_{c,T}(v)-1_{v>1}|\le 2v^c/(\pi T|\log v|)$ give

$$|\mathcal{I}_{T_1,T_2}-J_U|\le|F_0|\mathcal{A}V^{c+d}\left\{\frac{2}{\pi T_1\Delta_V}+\frac{2}{\pi T_2\Delta_V}+\frac{4}{\pi^2T_1T_2\Delta_V^2}\right\}. \tag{27}$$

Thus $\mathcal{I}_{T_1,T_2}\to J_U$ as both heights tend independently to infinity, away from the original principal pole. This is an oscillatory limit, not an absolutely integrable infinite vertical integral.

Taking $T_1=T_2=V^2$ gives the safe upper bound $|J_U|\ll(Nu)^\eta|L(s,\psi)|V^{c+d}\log^2(2V)$, with constants retaining fixed margins, ν,S, and σ. Choosing c,d close to $1-\sigma$ reproduces $U^{2-2\sigma+\eta}$ in amplitude and $U^{2-4\delta+\eta}$ in energy on $\sigma=1/2+\delta$. Insertion into the original common-profile Mellin argument retains $M_4(W)^2$ and all previously required height/conductor uniformities. No derivative-free improvement follows from this Perron estimate.

The correction converges even on some negative auxiliary lines, but moving there can cross $z=\rho-s$ or $w=\rho-s$, from zeros of the primitive L-function. The exact meromorphic slice $\mathcal{F}(s,z,0)=1/L^{T_0}(s+z,\psi)$ forbids assuming universal cancellation of these poles. At a numerator zero the slice must be understood meromorphically, not by an unjustified pointwise substitution. Sequential shifts have single boundary integrals and crossed-zero residues as well as the kernel double residue; away from s zeros or poles that double residue is $1/L^{T_0}(s,\psi)$. A principal L pole supplies a reciprocal zero, not a reciprocal pole. Uniform control of actual reciprocal integrals/residues or the signed covariance remains a new input. Numerator AFM alone supplies neither.

22 A quantitative covariance budget without a new exponent

Retain the original Eisenstein sextic symbols, fixed finite-order ν, fixed S, one possibly complex common $W\in C_c^\infty$ supported in $[1,2]$, all squarefree columns prime to S, and every raw shared-prime

zero. In the accepted window $2D/7 \le U^2 \le D$, put $r_m = r_U(m)$, $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $v_m = \overline{\lambda_m}$. Thus $C_U = \mu r$ and $A_U = \mu(r-1)$ on the active annulus. For a finite literal row block B, let

$$G_{mn} = \sum_{u\in B} x_m(u)\overline{x_n(u)}, \qquad R = \operatorname{diag}(r_m), \qquad M_j = \sum_m r_m^j |W(Nm/D)|^2 \quad (0 \le j \le 2).$$

The primitive $\psi_u, \psi_{u,2}$ and full deleted sets retain their previous meanings; no conductor-drop value replaces a raw zero.

Proposition 11 (An actual-vector lower budget)**.** *If $0 \le G \le \Lambda_B I$, then*

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B}{2D}\{\sqrt{M_0 M_2} - M_1\}. \tag{28}$$

In this high window, uniformly in U,

$$\sum_{D\le Nm\le 2D} r_m \ll_K D, \qquad \sum_{D\le Nm\le 2D} r_m^2 \ll_K D\log(2D). \tag{29}$$

Consequently

$$\Re Q_B(C_U, \mu) \ge -C_K \|W\|_\infty^2 \Lambda_B \sqrt{\log(2D)}.$$

The vector inequality is sharp among positive contractions with the specified v, Rv; no such sharpness is asserted for the arithmetic Gram. The moment bound counts actual balanced ideal factorizations, rather than assuming independence of the two row factors.

On B_X, inserting the previously accepted powered-BGL norm gives only

$$\Re Q_{B_X}(C_U, \mu) \ge -C_{\nu,S,\epsilon}\|W\|_\infty^2 (DX)^\epsilon \{X^2 + DX + D^{2/3}X^{5/3}\}.$$

At $X \asymp D^{1/4}$ the exponents remain $1/2, 5/4, 13/12$. At $H = D^{3/4}$ this budget still has the old $D^{1/2}$ excess over H; it supplies no BT_σ saving. A negative-part budget is weaker than $2\Re Q(C_U, \mu) \ge Q(C_U, C_U) + Q(\mu, \mu) - R_\sigma$.

For $S_R = \sqrt{R}$, the exact identity

$$\tfrac{1}{2}(RG + GR) = S_R G S_R + \tfrac{1}{2}[S_R, [S_R, G]]$$

localizes the commutator to unequal factor-count levels, with entry $(\sqrt{r_m} - \sqrt{r_n})^2 G_{mn}$. If B retains all six literal units, the actual unit sum further gives $G_{mn} = 0$ unless $Nm \equiv Nn \pmod{36}$. These are exact deletions, not a power saving. Even zero different-level pairings leave the nonnegative within-level energies $\sum_k (k-1)^2 Q_B(\mu \mathbf{1}_{r=k}, \mu \mathbf{1}_{r=k})$. The unequal-product coupled fourth moment and its mixed Möbius term remain unestimated. No new analytic exponent, energy lower bound, Lean formalization or GRH conclusion follows.

23 Proof and arithmetic boundary of the covariance budget

23.1 The positive-contraction calculation

Since G is the exact raw Gram, it is positive semidefinite. If $\Lambda_B > 0$, put $T = 2G/\Lambda_B - I$, a self-adjoint contraction. Then

$$\Re(v^* R G v) = \frac{\Lambda_B}{2}\{M_1 + \Re(v^* R T v)\} \ge \frac{\Lambda_B}{2}\{M_1 - \|Rv\|_2 \|v\|_2\},$$

which is (28); $\Lambda_B = 0$ is immediate. The Hermitian rank-two matrix $\{(Rv)v^* + v(Rv)^*\}/2$ has possible nonzero eigenvalues $(M_1 \pm \sqrt{M_0 M_2})/2$. Its negative spectral projection realizes the lower bound among all $0 \le G \le \Lambda_B I$, a linear-algebra sharpness statement only. If $a \le r_m \le b$ on the weighted support, $a + b > 0$, the pointwise inequality $r_m^2 \le (a+b)r_m - ab$ also gives

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B M_0 (b-a)^2}{8D(a+b)}.$$

For $a = b = 0$ the covariance is zero. The constant follows by maximizing $\sqrt{(a+b)t - ab} - t$ on $[a, b]$.

23.2 The balanced-divisor second moment

Set $T_0 = \sqrt{D}$. The high-window factor restrictions imply

$$\sqrt{2/7}\, T_0 < Na, Nb < \sqrt{14}\, T_0.$$

The elementary ideal count $\#\{a : Na \le Y\} \ll_K Y$ proves the first assertion of (29). For the second, two ordered factorizations $ab = cd = m$ have the unique form

$$a = gh, \quad c = gk, \quad b = k\ell, \quad d = h\ell, \qquad g = (a, c).$$

Squarefreeness makes g, h, k, ℓ pairwise coprime and squarefree. To upper-bound this positive count, enlarge to arbitrary ideals. For $M = \max(Nh, Nk)$ the individual bounds are

$$Ng \le \frac{\sqrt{14}\, T_0}{M}, \qquad N\ell \le \frac{\sqrt{14}\, T_0}{M}, \qquad 1 \le M \le \sqrt{14}\, T_0.$$

On $2^j \le M < 2^{j+1}$ there are $O_K(2^{2j})$ choices of (h, k) and $O_K((T_0/2^j)^2)$ choices of (g, ℓ), with bounded terminal ranges absorbed in the constant. Each dyadic range contributes $O_K(D)$; there are $O(\log(2D))$ ranges. This proves the second moment with no extra logarithm from counting only $NgN\ell$. Restricting to primes outside fixed S decreases the count. Hence

$$M_0, M_1 \ll_K \|W\|_\infty^2 D, \qquad M_2 \ll_K \|W\|_\infty^2 D \log(2D).$$

These constants are independent of the cutoff in the stated window. Only the common supremum norm is used, including for complex W.

23.3 What the commutator actually controls

Write

$$t_{mn} = D^{-1} \Re\{\lambda_m \overline{\lambda_n} G_{mn}\}, \qquad \delta_{mn} = (\sqrt{r_m} - \sqrt{r_n})^2.$$

The exact commutator identity proves

$$\Re Q_B(C_U, \mu) \ge Q_B(\sqrt{r}\,\mu, \sqrt{r}\,\mu) - \tfrac{1}{2} \sum_{m,n} \delta_{mn} |t_{mn}|.$$

The displayed loss retains the actual character and profile phases before taking absolute values; it is at most the phase-blind loss obtained from $|t_{mn}| \le D^{-1} |\lambda_m \lambda_n G_{mn}|$. On the accepted four-column smooth B_7 fixture at amplitude $x \ge 0$, the positive term is $18/25$, this coefficient-adapted loss is $9x/50$, and the lower bound is attained. Purely imaginary entries of the full complex Gram make the phase-blind loss larger. This calibrates the new budget and does not give an asymptotic obstruction.

For the unit deletion let $\zeta_6 = 1+\omega$. Every good prime norm q is $1 \pmod 6$, and direct residue-field exponentiation gives $x_p(\zeta_6) = \zeta_6^{(q-1)/6}$. Multiplying over $p \mid m$ and reducing $Nm = \prod_{p|m}(1+6e_p)$ modulo 36 gives

$$x_m(\zeta_6) = \zeta_6^{(Nm-1)/6 \bmod 6}.$$

Multiplication of all literal rows by ζ_6 preserves their raw zeros and proves the norm-congruence deletion in the core. The vector budget may be applied separately on these six classes. No D-dependent number of classes or conductor saving is produced.

The moment proof above counts the coefficient diagonal $ab = cd$. It has not estimated the unequal-product character quadruples in the coupled row fourth moment. Replacing the inherited DX norm by $D^{1-\sigma}X$ merely because equal-level and different-class entries vanish is the unsupported step. Small negative commutator loss alone also leaves the within-level energies in the core. Principal residues are unchanged; this covariance is used on the nonprincipal blocks where the original residue is zero, not as a new centering prescription.

23.4 The complete near-saturating face

For a six-factor row block let $R_0 = \prod_{j=1}^{6} A_j$, $M = \max(A_1, A_2)$ and $\prod_j A_j^j \le H$. The accepted exact-mask block norm is

$$\Lambda_A \ll_{\nu,S,\epsilon} (HD)^{\epsilon}\{R_0 + DR_0/M + D^{2/3}R_0/M^{1/3}\}.$$

The column moment and covariance proofs apply to every such block. Write $x_j = \log A_j$, $h = \log H$, $c_0 = \sum_j jx_j$ and $r_0 = \sum_j x_j$. Then, for $x_1 \ge x_2$,

$$h - 3(r_0 - \max(x_1, x_2)) = h - c_0 + (x_1 - x_2) + x_4 + 2x_5 + 3x_6;$$

for $x_2 \ge x_1$ replace $x_1 - x_2$ by $2(x_2 - x_1)$. All terms are nonnegative. Thus the middle monomial is saturated on

$$A_1 = A_2, \qquad A_1A_3 = H^{1/3}, \qquad A_4 = A_5 = A_6 = 1,$$

with full capacity. The balanced $A_3 = 1$ case is only an endpoint. The a_3 direction remains nonprincipal by its outside-S valuation $3 \pmod 6$. If $R_0/M \ge H^{1/3}D^{-\eta}$, the total displayed slack is at most $3\eta \log D$. Therefore a fixed power gain must be uniform on this full neighboring region. The proved budget still costs DR_0/M; it gives no global improvement over the accepted envelope.

The independent standard-library diagnostic checks nine ideal moment cases, eight raw Gram cases with 27 000 phase values and 5 184 raw zeros, exact radical commutator/vector budgets, and 227 exponent cases. Finite tables use common norm profiles realizable by smooth disjoint bumps; they are not tests of an infinite analytic estimate. Full derivation, source bindings and preserved checks are under `planning/covariance-budget-20261105/attempt/` and `outputs/checks/2026-11-05-covariance-budget/attempt/`.

24 The fourth-moment conductor and its full raw zero set

Retain the original Eisenstein sextic symbols $x_m(u) = (u/m)_6$, fixed finite-order ν, fixed S, and one possibly complex common smooth W supported in $[1,2]$. Put $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $r_m = r_U(m)$. Columns remain squarefree and prime to S. In $2D/7 \le U^2 \le D$, the actual polynomial is

$$\mathcal{C}_U(u) = \sum_{\substack{(a,b)=1,\ Na,Nb>U \\ D\le Na\,Nb\le 2D}} \mu(a)\mu(b)\nu(ab)W(Nab/D)x_a(u)x_b(u).$$

Its squared row energy has four factor indices a, b, c, d, with $(a, b) = (c, d) = 1$. These internal masks are essential. At each prime,

$$e_p = \mathbf{1}_{p|a} + \mathbf{1}_{p|b} - \mathbf{1}_{p|c} - \mathbf{1}_{p|d} \quad \text{belongs to } \{0, \pm 1\}.$$

Thus, for the two squarefree products $m = ab, n = cd$, the primitive oscillatory conductor outside fixed S is $q = mn/(m, n)^2$. The full raw zero-preserving period is $L = mn/(m, n)$:

$$x_m(u)\overline{x_n(u)} = \mathbf{1}_{(u,(m,n))=1}\theta_{m/(m,n),\, n/(m,n)}(u).$$

No common-prime zero is restored by taking the primitive character.

A generic fourth moment $|\sum_m \lambda_m x_m(u)|^4$ has four individually squarefree columns $m_1, \dots, m_4$ with no such internal coprimality. Its signed local exponent $e_p = \sum_{i=1}^{2} \mathbf{1}_{p|m_i} - \sum_{i=3}^{4} \mathbf{1}_{p|m_i}$ can also be ± 2, giving a nontrivial cubic component. For $L = \mathrm{rad}(m_1 m_2 m_3 m_4)$, the exact row-power interface is

$$Q_j = \prod_{\substack{p|L \\ je_p \not\equiv 0 \ (6)}} p, \qquad Z_j = L/Q_j, \qquad \prod_{i=1}^{2} x_{m_i}(v^j)\overline{\prod_{i=3}^{4} x_{m_i}(v^j)} = \theta_{j,e}(v)\mathbf{1}_{(v,Z_j)=1}.$$

Here $\theta_{j,e}$ is primitive on Q_j, with its own zero extension. At $j = 3$ an exponent ± 2 becomes primitive-principal but its raw zero remains. For the original two-squarefree-product expansion, $Q_j = q$ for all $1 \le j \le 5$, including the quadratic $j = 3$ direction. For the accepted row blocks $Nr_j \asymp A_j$, put $R_0 = \prod_j A_j$, $P = \max(A_1, \dots, A_5)$ and retain $\prod_j A_j^j \le H$. The full A_3 saturation face and all row-factor masks remain.

The small-conductor pair count recovers an already controlled high-gcd sector; it is not a new analytic advance. On the remaining original sector $Nq > P^2$, favorable smooth completion costs $\min(A_j, \sqrt{Nq}) = A_j$ for $1 \le j \le 5$. The literal sharp cutoff is not replaced by this smooth trial. The signed large-conductor fourth-moment and mixed Möbius sums remain open; no improved full exponent, BT_σ, new Lean theorem, zero-location result or GRH conclusion follows.

25 Scope of the raw fourth-moment reduction

Write $c_{a,b} = \mu(a)\mu(b)\nu(ab)W(Nab/D)$ on the exact factor support in the core. Before absolute Cauchy, the normalized energy is

$$\frac{1}{D} \sum_{(a,b),(c,d) \text{ on that support}} c_{a,b}\overline{c_{c,d}} \sum_{u \in B} x_a(u)x_b(u)\overline{x_c(u)x_d(u)}.$$

Squarefreeness of ab and cd forces each positive and negative prime multiplicity to be at most one, proving $e_p \in \{0, \pm 1\}$. For the mixed term against the actual μ polynomial the exponent is $\mathbf{1}_{p|a} + \mathbf{1}_{p|b} - \mathbf{1}_{p|n}$, with the same conclusion and the same full union mask. The common possibly complex W, ν, hyperbolic support and all raw zeros stay in these identities.

In contrast, unrestricted four-column memberships $(1, 1, 0, 0)$ give $e_p = 2$, a cubic component; $(1, 1, 1, 0)$ give $e_p = 1$ although both product supports contain p. Neither the parity symmetric difference of four supports nor the symmetric difference of two nonsquarefree product supports gives the general conductor. For the interface in the core, its local character order is

j	$1, 5$	$2, 4$	3	6
$e_p = \pm 1$	6	3	2	1
$e_p = \pm 2$	3	3	1	1

Nontrivial entries have tame conductor exponent one; trivial entries remain deleted at $p \mid v$. Multiplying the interface over all six row factors, with the fixed unit/S phase, preserves every original zero, including those at sixth-power factors. All inter-factor, squarefree, ray and capacity restrictions remain external restrictions on the same row sum; this identity asserts no independence.

At a residue field of size Q, let χ have order six and ρ be nontrivial additive. The zero-extended complete sum $G_k(t) = \sum_{z \neq 0} \chi(z)^k \rho(tz)$ satisfies

	$t = 0$	$t \neq 0$
$k \not\equiv 0$ (6)	0	$\lvert G_k(t)\rvert = \sqrt{Q}$
$k \equiv 0$ (6)	$Q - 1$	-1.

For $t \neq 0$, substitution gives $G_k(t) = \overline{\chi(t)^k} G_k(1)$; for nontrivial k, expanding the norm square gives $(Q-1) - \sum_{a \neq 0,1} \chi(a)^k = Q$. The proof applies at split and inert primes. By CRT the full period-L transform vanishes if any local frequency on Q_j is zero; otherwise its magnitude is

$$(NQ_j)^{1/2} \prod_{p \mid Z_j} \left| -1 + Np\, \mathbf{1}_{\text{local frequency at } p \text{ is } 0} \right|.$$

Canceled common primes thus supply Ramanujan factors.

Exact deletion, with all remaining restrictions kept in F, gives

$$\sum_r \theta(r) F(r) \mathbf{1}_{(r,Z)=1} = \sum_{d \mid Z} \mu(d) \theta(d) \sum_z \theta(z) F(dz).$$

For a squarefree pivot let E be the radical of its fixed excluded primes, including Z and the relevant other row factors, with conductor primes omitted since θ already vanishes there. The additional squarefree identity is

$$\mu(r)^2 \mathbf{1}_{(r,E)=1} = \sum_{\substack{h^2 \mid r \\ (h,E)=1}} \mu(h) \sum_{\substack{d \mid E \\ d \mid r}} \mu(d).$$

Since $(d,h) = 1$, its lifted row sum has terms $\mu(d)\mu(h)\theta(d)\theta(h)^2 \sum_z \theta(z) F(dh^2 z)$. The scaled F retains the exact window, primary/ray restrictions and other masks. Divisor triangles cost at most $(HD)^\epsilon$; they do not restore a forbidden nonzero.

The previous fixed-radial-profile high-gcd proof (`research/short-row-high-gcd.md`, independently reviewed in `planning/reviews/book/short-row-formal.md`) already bounds its off-diagonal tail by $O_{\Phi,\epsilon}(D^{2+\epsilon}/G_0^2)$ for $Nh \geq G_0$. For annular $m = hr, n = ht$, $Nq \leq Y$ implies $Nh \geq D/\sqrt{Y}$, so the old smooth-scope bound is $O_{\Phi,\epsilon}(D^\epsilon Y)$. The present elementary count $O(D\sqrt{Y})$, factor multiplicities $r_U(m) r_U(n) \ll_\epsilon D^\epsilon$, and $\#B_A = O(R_0)$ give only $O(\|W\|_\infty^2 D^\epsilon R_0 \sqrt{Y})$ on a literal sharp block. This is a weaker conductor-form recovery in its own bookkeeping scope, not a new high-gcd theorem; the smooth estimate is not silently transferred to a selected sharp off-diagonal piece. The full A_3 face and neighborhood slack are also inherited.

Finally, primitive smooth Poisson completion has bound $O_V(\min(A_j, \sqrt{Nq}))$ on the original support. For $Nq > P^2$ and $1 \leq j \leq 5$, every scaled length $A_j/(Nd\,Nh^2)$ is below $\sqrt{Nq}$, so even this favorable trial takes its trivial length term. Its constant uses fixed smooth seminorms and need not stay bounded when a sharp boundary is approached. In the admissible subsector $(m,n) = 1$, $Nq = Nm\,Nn \asymp D^2$. A complete zero mean does not control the nonzero frequencies of the short original masked row window. The unresolved target is their joint signed sum with weights $r_U(m) r_U(n) \lambda_m \overline{\lambda_n}$ and $r_U(m) \lambda_m \overline{\lambda_n}$, before per-conductor absolute estimates. No new global analytic conclusion is claimed.

26 A quadratic pivot on the cube factor

Retain the original zero-extended sextic symbols $\chi_m(u) = (u/m)_6$ over $K = \mathbb{Q}(\omega)$, the fixed S containing all primes above 6 and comparison moduli, and the squarefree good columns $D \le Nm \le 2D$. Write an outside-S row ideal uniquely as $r_1 r_2^2 r_3^3 r_4^4 r_5^5 r_6^6$: only $r_1, \dots, r_5$ are pairwise-coprime squarefree ideals; r_6 is arbitrary and may overlap them. Use primary generators. Fix one literal unit and one S-supported row factor, retaining their original values. Let $A_j \le Nr_j < 2A_j$, $A_j \ge 1$, $R_0 = \prod_j A_j$ and $\prod_j A_j^j \le H$; the actual capacity restriction on rows is retained. Coefficients $b_m \in \mathbb{C}$ are arbitrary and fixed independently of every variable row.

Proposition 12 (The cube-factor block norm)**.** *For every $\varepsilon > 0$, the raw Gram on this actual block obeys*

$$\sum_{u \text{ in the block}} \left| \sum_m b_m \chi_m(u) \right|^2 \ll_{\nu,S,\varepsilon} (HD)^{\varepsilon} \left\{ R_0 + \frac{DR_0}{A_3} \right\} \sum_m |b_m|^2.$$

The constant is uniform in the fixed unit/S-factor, frozen good factors, H, D and the coefficient sequence. There is no rational-norm-squarefree restriction. Consequently, with $M = \max(A_1, A_2)$, the old powered-BGL bound and this bound give

$$\Lambda_A \ll_{\nu,S,\varepsilon} (HD)^{\varepsilon} \left\{ R_0 + \min\left(\frac{DR_0}{M} + \frac{D^{2/3} R_0}{M^{1/3}}, \frac{DR_0}{A_3} \right) \right\}. \tag{30}$$

The external analytic input is Goldmakher–Louvel, arXiv:1112.1642v2, Corollary 1.2 with $n = 6$, already an explicit classical cube-power sieve. The refinement is this application, not a new large-sieve theorem. For the original normalized coefficients $b_m = D^{-1/2} A_U(m) \nu(m) W(Nm/D)$, the accepted divisor bound gives $\sum_m |b_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta$, uniformly in $U \ge 1$ with $U^2 \le D$ (and the common W may be complex). Choosing smaller internal exponents yields the corresponding uniform block energy envelope. General cutoff or factor-count weights remain inside the arbitrary coefficient norm; that norm is not silently replaced by one. Summing S-supported row factors uses the reduced capacity $H/N\mathfrak{b}$ and the established convergent geometric sums.

On the complete old saturated face $A_1 = A_2 = A$, $A_3 = B$, $A_4 = A_5 = A_6 = 1$, $AB = H^{1/3}$, the new bound is $A^2 B + DA^2$ in place of the old middle monomial DAB. For $H = D^{3/4}$ put $A = D^a$, $B = D^{1/4-a}$, $0 \le a \le 1/4$. When $a < 1/8$, its leading exponent is $1 + 2a < 5/4$; the gain requires B/A to grow by a power for a power improvement. For $a \ge 1/8$, the combined bound still has leading exponent $5/4$. Thus the A_3-dominant part improves, while no new global exponent, BT_σ estimate, signed covariance theorem, Lean theorem or GRH conclusion follows.

Inspecting the surviving sector. The quadratic-pivot inspector plots the complete face, the five monomial powers and the remaining target gap. These are analytic bound exponents, not measured operator norms or trained-network predictions. The refined norm can be inserted in the earlier covariance budget; it does not remove its within-level energies.

A necessary near-face condition. Put $\mathcal{M}_A = DR_0 / \max(M, A_3)$. The combined envelope is at most $(HD)^{\varepsilon}\{R_0 + \mathcal{M}_A + D^{2/3} R_0 / M^{1/3}\}$. If $\mathcal{M}_A \ge DH^{1/3} D^{-\eta}$, then the earlier near-face slack is at most $3\eta \log D$ and, additionally, $A_3 \le MD^\eta$. Thus a uniform remaining leading-term estimate may focus on this smaller neighborhood; it must still cover the entire $B \le A$ part of the face. This is a necessary condition on explicit upper-bound monomials, not a lower bound for actual arithmetic energy.

27 Proof and scope of the quadratic pivot

Freeze all factors except r_3, and write the actual literal row $u = Fr_3^3$. Multiplicativity of the original numerator gives the exact zero-preserving identity

$$\chi_m(u) = \chi_m(F)\chi_m(r_3)^3.$$

Put $c_m = b_m\chi_m(F)$, a coefficient sequence independent of r_3 with $\sum |c_m|^2 \le \sum |b_m|^2$. In particular the arbitrary sixth-power factor is the coefficient mask

$$\chi_m(r_6^6) = \mathbf{1}_{(m,r_6)=1},$$

not the constant one. No condition $(r_3, r_6) = 1$ is introduced. The actual capacity and coprimality of r_3 with r_1, r_2, r_4, r_5 only restrict this positive outer row sum; after freezing, it may be enlarged to all squarefree good pivot ideals in the dyad. The variable collision mask (m, r_3) remains in the original symbol.

Apply the previously proved literal comparison (22) to the primary generator of r_3 itself:

$$\chi_m(r_3)^3 = \eta_{r_3}(m)^3 C([r_3], [m])^3 \vartheta_m(r_3)^3.$$

The twist η_{r_3} ranges over a fixed finite set of S-unit Kummer characters, and $[r_3]$ over the fixed finite reciprocity group. Partition pivot rows by $(\eta_{r_3}, [r_3])$. On each part the first two factors are one fixed modulus-one column multiplier. Thus its coefficient sequence is independent of the pivot row and has squared norm at most $\sum |b_m|^2$. This finite partition, rather than an assertion that arbitrary row-dependent phases may be absorbed, resolves the fixed-S phase issue.

Goldmakher–Louvel Definition 1 and Remark 3 identify the cubic powers of a sixth-order Hecke family as a quadratic Hecke family; their Corollary 1.2 gives directly

$$\sum_{\substack{a \text{ sf,good} \\ Na \le 2A_3}} \left| \sum_{\substack{m \text{ sf,good} \\ Nm \le 2D}} z_m \vartheta_m(a)^3 \right|^2 \ll_{K,S,\eta} (A_3 D)^\eta (A_3 + D) \sum_m |z_m|^2.$$

This is the required column-indexed orientation and allows arbitrary complex coefficients and squarefree ideals of nonsquarefree rational norm. The canonical family has primitive trivial-infinity-type characters, a fixed finite reciprocity group, and the same-class primitive-product property required by Definition 1. At each $p \mid m$ outside S, the local tame sextic character has order six, so its cube has order two and conductor exponent one. Thus its outside-S primitive conductor is exactly m; the S-part has only finitely many fixed conductor types. Accordingly both sides of the comparison are zero at $(m, a) > 1$. No conductor-drop value replaces the raw zero, and no global formula for the full conductor is required. The fixed ν stays in coefficients.

Enlarge the positive row sum separately on each finite part and apply this sieve. Ideal counting gives $O_K(A_j)$ possibilities for each frozen r_j, including arbitrary r_6, hence $O_K(R_0/A_3)$ frozen tuples. Summing gives $(A_3D)^\eta(R_0 + DR_0/A_3)\sum |b_m|^2$. Since $A_3 \le H$, adjust η to obtain Proposition 12; literal unit multiplicity is at most six. Together with the prior block bound this proves (30). It does not change the actual coefficient normalization or principal residues.

On the face in the core, the old bound has exponents $a + 1/4$, $5/4$, $11/12 + 2a/3$, whereas the quadratic bound has $a + 1/4$ and $1 + 2a$. Its improvement for $a < 1/8$ leaves the balanced point and $a \ge 1/8$ with the old $5/4$ leading exponent. Even the improved endpoint $a = 0$ costs D, above the target $D^{3/4}$. The pinned GL PDF has SHA256 `5cc5f03388d3996b4333f07be51f989f`

60779fe172f8bd24c7aa6ddad64dd304. This proof invokes that classical result; finite exponent checks do not reprove it, and no new formalization is claimed.

For the last condition, use $R_0/M \le H^{1/3}$ from the exact capacity inequalities. The hypothesis implies $R_0/A_3 \ge H^{1/3}D^{-\eta}$, so $A_3/M = (R_0/M)/(R_0/A_3) \le D^{\eta}$. It also implies $R_0/M \ge H^{1/3}D^{-\eta}$, which is exactly the previous slack hypothesis. Finally $\min(a+b,c) \le \min(a,c)+b$ for nonnegative a,b,c proves the separated envelope, with no replacement of a signed moment by a positive lower bound.

28 Joint completion before conductor absolute values

In the original C–C expansion write $m = hr, n = ht$ with h,r,t pairwise coprime squarefree, $q = rt \ne 1$, $L = hq$, $Q = Nq$, $\ell = NL$. Set $\theta = x_r\overline{x_t}$ and $\xi_q = x_q^3$. The original internal masks give $\theta^3 = \xi_q$ and primitive θ^j on q for $1 \le j \le 5$, while $\mu(hr)\mu(ht) = \mu(q)$. Fix generators α of h, β of q, choosing $\alpha = 1$ for $h = 1$. With $\delta = 1 + 2\omega$, put $e_\beta(z) = \exp(2\pi i \operatorname{Tr}(z/(\delta\beta)))$, $\tau_\beta(\eta) = \sum_{z \bmod q} \eta(z)e_\beta(z)$, and $c_h(k) = \prod_{p\mid h}(Np\mathbf{1}_{k\in p} - 1)$, $c_1 = 1$. For $f_j(v) = \theta(v)^j\mathbf{1}_{(v,h)=1}$, $1 \le j \le 5$, $f_6(v) = \mathbf{1}_{(v,L)=1}$ and $G_j(k) = \sum_{v \bmod L} f_j(v)e_{\alpha\beta}(kv)$, joint completion gives

$$\prod_{j=1}^{6} G_j(k_j) = Q^2\xi_q(-\alpha)\tau_\beta(\xi_q)\overline{\theta\Big(\prod_{j=1}^{5} k_j^j\Big)} \prod_{j=1}^{5} c_h(k_j)c_L(k_6). \tag{31}$$

This holds when the first five frequencies are q-units; otherwise the left side is zero. The mixed Gauss phase is orientation-independent, with every common-prime zero retained. The exact row Fourier weight still contains all squarefree, ray, inter-factor, capacity and profile restrictions, including the full A_3 face. Its weighted joint conductor/frequency estimate remains open. This identity and its finite Parseval budget give no improved global exponent or GRH conclusion.

29 The surviving joint identity and its missing estimate

CRT gives $G_j(k) = \theta(\alpha)^j\tau_\beta(\theta^j)\overline{\theta(k)^j}c_h(k)$ for q-unit k, and $G_6 = c_L$. Pairing Gauss powers $1,5$ and $2,4$ yields $\prod_{j=1}^5 \tau_\beta(\theta^j) = Q^2\xi_q(-1)\tau_\beta(\xi_q)$; the CRT factors multiply to $\theta(\alpha)^{15} = \xi_q(\alpha)$. This proves (31), with all additive-generator phases explicit.

On a fixed literal class write $u = u_0\prod_{j=1}^6 v_j^j$ and let $F(v)$ be its exact finite row weight. Only $v_1,\dots,v_5$ are squarefree and pairwise coprime; v_6 is arbitrary with its permitted overlaps. Retain every original mask and the sharp capacity boundary in F. Define $\widehat{F}_L(k) = \sum_v F(v)e_{\alpha\beta}(-\sum_j k_jv_j)$ and, with $t = q/r$, put

$$\begin{aligned} a(r) &= r_U(hr)r_U(ht)\nu(r)\overline{\nu(t)} \\ &\qquad \times W(Nh\,Nr/D)\overline{W(Nh\,Nt/D)}\theta_{r,t}(u_0), \\ A(K) &= \sum_{r\mid q} a(r)\overline{\theta_{r,q/r}(K)}, \\ B(K) &= \sum_{\substack{k \bmod L,\ (k_1\cdots k_5,q)=1 \\ \prod_{j=1}^5 k_j^j = K \bmod q}} \widehat{F}_L(k)\prod_{j=1}^{5} c_h(k_j)c_L(k_6). \end{aligned}$$

Set $a(r) = 0$ at excluded orientations; the actual factor counts and common possibly complex W

remain. Fourier inversion gives exactly

$$E_{\mathcal{T}} = \frac{1}{D} \sum_{(h,q)\in\mathcal{T}} \mu(q) \frac{Q^2 \xi_q(-\alpha)\tau_\beta(\xi_q)}{\ell^6} \sum_{K\in(\mathcal{O}/q)^\times} B(K)A(K). \tag{32}$$

Sum the original u_0 classes and blocks. This retains low gcd, including $h = 1$, before conductor absolute values. For C–μ replace the second $r_U(ht)$ by 1. A column-orientation-dependent restriction cannot be hidden in a common F when applying the next identity.

Distinct orientations have nontrivial cubic quotient, so

$$\sum_K |A(K)|^2 = \varphi(q) \sum_{r|q} |a(r)|^2, \qquad |E_{h,q}| \le \frac{Q^{5/2}}{D\ell^6} \left(\varphi(q) \sum_{r|q} |a(r)|^2 \right)^{1/2} \|B\|_2.$$

The product-frequency fibers have size $\varphi(q)^4$ under uniform averaging. The actual B is not uniform: all orientations agree on the coherent set $x_p(K)^2 = 1$ for every $p \mid q$, of size $\varphi(q)/3^{\omega(q)}$. Centered representatives have $|k_j| \le \sqrt{\ell}$. Restricting to any frequency subset Ω has error bounded by the same displayed budget with $\|B - B_\Omega\|_2$; no small lower-height tail is proved.

The full face $A_1 = A_2 = A$, $AA_3 = H^{1/3}$, $A_4 = A_5 = A_6 = 1$ remains. At its pure A_3 endpoint, inversion in the fixed factors leaves $(\sum_r a(r)) \sum_v F_3(v)\xi_q(v)\mathbf{1}_{(v,h)=1}$: the actual row character there is orientation-independent. Literal length-one Fourier factors can undo complete phase balance. A quantitative signed estimate for the actual h, q, K sum is still required. Exact split/inert Gauss, full-mask CRT and orientation tests pass 991 assertions; receipt: `outputs/checks/2026-11-05-joint-dual/research/joint-gauss-check-v1.json`. These diagnostics do not prove an incomplete-moment saving or GRH.

30 A radical-conductor refinement of the raw envelope

Retain the original literal symbols $\chi_m(u) = (u/m)_6$, extended by zero, the fixed good set S, and the squarefree ideal columns $D \le Nm \le 2D$. On an actual dyadic row block write

$$u = u_0 r_1 r_2^2 r_3^3 r_4^4 r_5^5 r_6^6, \qquad A_j \le N r_j < 2A_j, \qquad \prod_{j=1}^{6} A_j^j \le H.$$

Only $r_1, \dots, r_5$ are pairwise-coprime squarefree ideals; r_6 is arbitrary and may overlap them. Fix primary generators, a literal unit and an S-supported factor in u_0. Put $R = \prod_j A_j$, $M = \max(A_1, A_2)$ and $Q = R/A_6$. For arbitrary complex coefficients b_m, independent of every variable row, define

$$\mathcal{E}_A(b) = \sum_{u \text{ in the actual block}} \left| \sum_m b_m \chi_m(u) \right|^2, \qquad \Lambda_A = \sup_{b\ne 0} \frac{\mathcal{E}_A(b)}{\sum_m |b_m|^2}.$$

Proposition 13 (An ordinary conductor comparison)**.** *For every $\varepsilon > 0$,*

$$\Lambda_A \ll_{S,\varepsilon} (HD)^\varepsilon A_6 \{D + Q^2\}. \tag{33}$$

Consequently the original residue-subtracted remainder satisfies, uniformly for $H, D, U \ge 1$ and $U^2 \le D$,

$$\sum_{0<Nu\le H} |E_u|^2 \ll_{\nu,S,\varepsilon} \|W\|_\infty^2 (HD)^\varepsilon \{H + DH^{1/6} + (DH)^{2/3}\}. \tag{34}$$

Here ν, S are fixed and $W \in C_c^\infty((0,\infty))$ is one common, possibly complex profile supported in $[1,2]$; there is no profile supremum.

The input to (33) is the classical ordinary sieve for *all* primitive characters of conductor norm at most Q, with constant $D+Q^2$, not a higher-order-family theorem for arbitrary conductor labels. The following section gives its standard Gauss-sum and two-dimensional lattice proof, including prime-power conductors at S. For fixed r_6, the first five factors have primitive outside-S conductor $q = r_1 \cdots r_5$. The literal primitive character agrees with the raw symbol even at collisions with q; the sixth-power factor instead remains the coefficient mask $\mathbf{1}_{(m,r_6)=1}$. At most $5^{\omega(q)} \ll_\eta (Nq)^\eta$ tuples give the same primitive character. Applying the ordinary sieve and then counting r_6 proves (33). No rational-norm-squarefree restriction is imposed.

To obtain (34), combine this with the accepted powered-BGL block bound from Section 18. With $S_A = R/M$ and $T_A = D^{2/3}R/M^{1/3}$, capacity gives $R \le H$, $A_6 \le H^{1/6}$, $RS_A \le H$ and $T_A \le (DH)^{2/3}$. Hence

$$\begin{aligned}\min(R + DS_A + T_A, DA_6 + R^2/A_6) &\le R + T_A + DA_6 + \min(DS_A, H^2/S_A^2)\\ &\le H + DH^{1/6} + 2(DH)^{2/3}.\end{aligned}$$

Dyadic and fixed-S summation, the uniform actual coefficient norm, and the principal-residue estimate give the assertion for $E_u = I_u - P_u$.

At $H = D^{3/4}$ the new envelope has exponent $7/6$, improving the old $5/4$ by $1/12$; its excess above the target H is still $D^{5/12}$. On the *entire* old face $A_1 = A_2 = A$, $A_3 = B$, $A_4 = A_5 = A_6 = 1$, $AB = H^{1/3}$, one has $Q = A^2B = AH^{1/3} \le H^{2/3}$, so (33) costs $D^{1+\varepsilon}$ there. This includes $B \le A$. The new combined upper bounds reach $7/6$ at $(\log_D A_1, \log_D A_2) = (5/12, 1/6)$ with other coordinates zero; the pure point $\log_D A_1 = 3/4$ also retains the old BGL cross term. These are upper-envelope witnesses, not arithmetic energy lower bounds or counterexamples to signed cancellation.

The simpler proposed replacement of the radicand ab^2 by its radical ab inside BGL remains unjustified: after the existing fixed reciprocity phases, the mixed reciprocity quotient is $\vartheta_b(a')/\vartheta_a(b')$, which varies outside S. The present application does not require that replacement. No target signed moment, type-II covariance estimate, positive arithmetic determinant comparison, zero-location result, new Lean theorem or GRH conclusion follows from (34).

31 Proof of the ordinary-conductor refinement

31.1 Primitive Gauss sums at every conductor

Let $\mathcal{O} = \mathbb{Z}[\omega]$, $\delta = 1 + 2\omega$, and choose one generator f of each nonzero ideal $\mathfrak{f}$. Write $e_f(x) = \exp(2\pi i \operatorname{Tr}(x/(\delta f)))$, $U(\mathfrak{f}) = (\mathcal{O}/\mathfrak{f})^\times$, and $\varphi(\mathfrak{f}) = |U(\mathfrak{f})|$. This additive character is primitive, since the different is (δ). For a primitive multiplicative character ρ modulo $\mathfrak{f}$, extended by zero at nonunits, set $\tau_f(\rho) = \sum_{x \bmod \mathfrak{f}} \rho(x) e_f(x)$. We verify the full-modulus identity

$$|\tau_f(\rho)|^2 = N\mathfrak{f}, \qquad \sum_{x \bmod \mathfrak{f}} \rho(x) e_f(zx) = \overline{\rho(z)}\tau_f(\rho) \quad (z \in \mathcal{O}). \tag{35}$$

No squarefree-conductor hypothesis is used, so this includes wild prime powers in the fixed S-part.

The Ramanujan sum is

$$c_{\mathfrak{f}}(y) = \sum_{x \in U(\mathfrak{f})} e_f(xy) = \sum_{\mathfrak{d} \mid (\mathfrak{f}, (y))} N\mathfrak{d}\, \mu(\mathfrak{f}/\mathfrak{d}).$$

Indeed insert $\mathbf{1}_{(x,\mathfrak{f})=1} = \sum_{\mathfrak{b}\mid(x,\mathfrak{f})} \mu(\mathfrak{b})$; the complete additive sum with $\mathfrak{b} \mid x$ is $N(\mathfrak{f}/\mathfrak{b})$ when $\mathfrak{f}/\mathfrak{b} \mid (y)$ and zero otherwise. Changing variables $x = ay$ in the Gauss square now gives

$$\begin{aligned}|\tau_f(\rho)|^2 &= \sum_{a\in U(\mathfrak{f})} \rho(a) c_{\mathfrak{f}}(a-1) \\ &= \sum_{\mathfrak{d}\mid\mathfrak{f}} N\mathfrak{d}\,\mu(\mathfrak{f}/\mathfrak{d}) \sum_{\substack{a\in U(\mathfrak{f}) \\ a\equiv 1 \pmod{\mathfrak{d}}}} \rho(a) = N\mathfrak{f}.\end{aligned}$$

For every proper $\mathfrak{d} \mid \mathfrak{f}$, the reduction $U(\mathfrak{f}) \to U(\mathfrak{d})$ is surjective by local lifting and CRT. Primitivity says ρ is nontrivial on its kernel, so that inner sum vanishes. For $\mathfrak{d} = \mathfrak{f}$ it equals one. For unit z, substitution proves the second identity in (35). For all frequencies together, finite additive Parseval gives

$$\sum_{z \bmod \mathfrak{f}} \left| \sum_x \rho(x) e_f(zx) \right|^2 = N\mathfrak{f}\,\varphi(\mathfrak{f}).$$

The unit frequencies already exhaust this quantity by the first identity; every nonunit-frequency sum is therefore zero. This proves the asserted zero extension. For $\mathfrak{f} = \mathcal{O}$ there is one residue class and one character; take $\tau_1 = 1$ and the single frequency zero.

31.2 The classical ordinary sieve in these lattice conventions

For $D, Q \geq 1$, coefficients z_α supported on any subset of $\{\alpha \in \mathcal{O} : |\alpha|^2 \leq C_0 D\}$ satisfy

$$\sum_{N\mathfrak{f}\leq Q} \sum_{\rho \text{ primitive mod } \mathfrak{f}} \left| \sum_\alpha z_\alpha \rho(\alpha) \right|^2 \ll_{\mathcal{O},C_0} (D + Q^2) \sum_\alpha |z_\alpha|^2. \tag{36}$$

Here the primitive characters may be restricted to the unit-trivial subfamily. We give the standard proof to make the conductor comparison explicit.

First the reduced fractions x/f, with $x \in U(\mathfrak{f})$ and $N\mathfrak{f} \leq Q$, are distinct modulo $\mathcal{O}$. Their denominator ideal is intrinsic: $\{a \in \mathcal{O} : a(x/f) \in \mathcal{O}\} = \mathfrak{f}$. Thus equality modulo $\mathcal{O}$ forces the same denominator ideal and residue numerator. For different fractions and every $k \in \mathcal{O}$,

$$\left| \frac{x}{f} - \frac{y}{g} - k \right| = \frac{|xg - yf - kfg|}{|fg|} \geq \frac{1}{Q},$$

because the numerator is a nonzero Eisenstein integer and $|fg| = (N\mathfrak{f}N\mathfrak{g})^{1/2} \leq Q$. The fixed codifferent factor $1/\delta$ changes only the lattice constants. The modulus-one frequency zero obeys the same separation from all others.

For completeness, a ρ_0-separated set t_r in a fixed two-dimensional torus obeys the additive bound

$$\sum_r \left| \sum_{|\alpha|\ll\sqrt{D}} z_\alpha \exp(2\pi i \operatorname{Tr}(\alpha t_r)) \right|^2 \ll (D + \rho_0^{-2}) \sum_\alpha |z_\alpha|^2. \tag{37}$$

Use fixed linear coordinates to write the lattice as $\mathbb{Z}^2$ and its additive pairing as $e(n \cdot t)$. The support lies in $|n_i| \leq T$ with $T \asymp \max(1, \sqrt{D})$. To prove the dual inequality, majorize this box indicator by $e^{2\pi} \exp(-\pi\|n\|^2/T^2)$, expand the square, and apply Gaussian Poisson summation. Up to fixed constants the resulting kernel is

$$T^2 \sum_{\ell\in\mathbb{Z}^2} \exp(-cT^2\|t_r - t_s + \ell\|^2).$$

Take ρ_0 at most a fixed torus constant. The lifted points are $c'\rho_0$-separated, so disjoint disks bound their count in a disk of radius L by $O(1+(L/\rho_0)^2)$. Gaussian annular summation bounds every absolute kernel row sum by $O(T^2+\rho_0^{-2})$. Schur's inequality and finite-dimensional duality prove (37), with constants independent of the number of frequencies.

Put $\mathcal{S}(x/f)=\sum_\alpha z_\alpha e_f(\alpha x)$. By (35),

$$\rho(\alpha)=\tau_f(\overline{\rho})^{-1}\sum_{x\in U(\mathfrak{f})}\overline{\rho(x)}e_f(\alpha x).$$

After taking squared absolute values, enlarging the primitive-character sum to all characters of $U(\mathfrak{f})$ and using orthogonality yields

$$\sum_{\rho \text{ primitive mod } \mathfrak{f}}\left|\sum_\alpha z_\alpha\rho(\alpha)\right|^2\le\frac{\varphi(\mathfrak{f})}{N\mathfrak{f}}\sum_{x\in U(\mathfrak{f})}|\mathcal{S}(x/f)|^2\le\sum_{x\in U(\mathfrak{f})}|\mathcal{S}(x/f)|^2.$$

Apply (37) with $\rho_0\gg Q^{-1}$ to prove (36).

Finally $\mathcal{O}$ has class number one. Choose one generator α_m of each ideal column, with $|\alpha_m|^2=Nm$; these are an injective subset of the lattice disk. A primitive trivial-infinity-type Hecke character is a unit-trivial primitive residue character on $\mathcal{O}/\mathfrak{f}$, and its ideal value equals its value on α_m. Descent of the residue character to a proper divisor would preserve unit-triviality and contradict primitive Hecke conductor. Thus (36) applies to arbitrary ideal coefficients of norm at most $2D$, including squarefree ideals of nonsquarefree rational norm.

31.3 Raw zeros, tuple multiplicity and block transfer

Freeze r_6 and the literal unit/S-factor, putting

$$c_m=b_m\chi_m(u_0)\mathbf{1}_{(m,r_6)=1},\qquad\sum_m|c_m|^2\le\sum_m|b_m|^2.$$

This is independent of every first-five variable. In particular the arbitrary overlapping sixth-power factor is not replaced by one. Let $v=\prod_{j=1}^5 r_j^j$ denote the chosen literal product. Every outside-S valuation is between one and five. BGL, arXiv:1112.1650v1, Lemma 2.1, together with the established finite S-unit comparison in (22), gives its primitive literal Kummer character κ_v of trivial infinity type, with

$$q=r_1\cdots r_5\mid\operatorname{cond}(\kappa_v)\mid\mathfrak{c}_S q,\qquad\kappa_v(m)=\chi_m(v)\quad((m,S)=1).$$

The equality includes collisions with q: both sides are zero. The fixed $\mathfrak{c}_S$ includes all wild conductor types and literal comparison twists. No row-dependent phase enters the coefficients.

A fixed primitive character determines its outside-S conductor q. Each prime of q has at most five assignments to the pairwise-coprime $r_1,\dots,r_5$, so tuple multiplicity is at most $5^{\omega(q)}\ll_\eta(Nq)^\eta$. All conductors in the dyadic block have norm at most C_SQ. Enlarge the positive row sum, apply (36), and count $O(A_6)$ possibilities for the frozen arbitrary r_6. This proves (33); the actual capacity, squarefree, ray and inter-factor restrictions only shrink the positive sum. Equivalently $\mathcal{E}_A(b)\le\Lambda_A\sum_m|b_m|^2$; there is no coefficient norm factor in the definition of Λ_A.

31.4 Uniform coefficient and principal-residue assembly

The other block estimate is exactly the accepted powered-BGL input (21), whose squarefree powers are justified in BGL Section 3.2. The capacity comparisons and elementary minimum in Section 30 give

$$\Lambda_A \ll (HD)^{\varepsilon}\{H + DH^{1/6} + (DH)^{2/3}\}.$$

Sum logarithmically many blocks with smaller internal exponents. An S-supported literal factor $\mathfrak{b}$ reduces capacity to $H/N\mathfrak{b}$; the required sums over $\mathfrak{b}$ converge at powers 1, 1/6, 2/3. Literal units have multiplicity at most six. This proves the raw bound with arbitrary fixed b_m.

For the actual coefficients use (19): $|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta$ gives

$$b_m = D^{-1/2}A_U(m)\nu(m)W(Nm/D), \qquad \sum_m |b_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta,$$

uniformly for $U \ge 1$, $U^2 \le D$. The common W is not optimized separately for different rows. If the primitive ψ_u is principal, ν is unramified outside S, so tame Kummer inertia forces every outside-S valuation of u to be divisible by six. Ideal counting and the convergent S-supported sum at power 1/6 give $O_{\nu,S}(H^{1/6})$ such literal rows. In the exact formula (15), $|\rho_T| \ll_K 1$, $|\widehat{W}(1)| \le \|W\|_\infty$, and

$$\sum_d \frac{|\beta_U(d)|}{Nd} \le \left(\sum_{Na\le U}\frac{1}{Na}\right)^2 \ll \log^2(2U).$$

Thus $|P_u| \ll \sqrt{D}\|W\|_\infty \log^2(2U)$ on principal rows, and

$$\sum_{0<Nu\le H} |P_u|^2 \ll DH^{1/6}\|W\|_\infty^2 \log^4(2U).$$

This fits the new envelope. The exact identity $E_u = I_u - P_u$ and $|E_u|^2 \le 2|I_u|^2 + 2|P_u|^2$, with smaller internal exponents, prove (34). No cutoff limit or profile supremum occurs.

The ordinary comparison is an application refinement. A direct BGL radical relabelling would additionally require reciprocity of its mixed family, not just its conductor count: for pairwise-coprime a, b, a', b', the quotient after the original fixed ray phases is $\vartheta_b(a')/\vartheta_a(b')$, which varies outside S. GL, arXiv:1112.1642v2, Definition 1(2), is therefore not supplied by that shortcut. This does not rule out a different mixed-family theorem; it is simply unneeded for the proved weaker ordinary comparison. The remaining exponent gap and every independent signed, spectral, formal and GRH obligation stated in Section 30 remain open.

32 Learning the actual signed arithmetic correlations

The unsigned refinement in Section 30 leaves a signed problem. Here is a separate, executed finite learning objective on the original correlations, with a certificate for every coordinate and observable. It is not a uniform estimate for an expanding family. For ordered distinct squarefree columns $m = hr, n = ht$, put $q = rt$ and

$$\Gamma_i = \sum_{u\in\mathcal{R}} \chi_r(u)\overline{\chi_t(u)}\mathbf{1}_{(u,h)=1}, \qquad i = (h, q, r).$$

The symbols vanish at every collision with q; the common-prime mask is retained separately. With principal ν, actual complementary divisor counts $r_U(m)$, and one common complex profile $W_m = W(Nm/D)$, define

$$d_i^{CC} = r_U(m)r_U(n)W_m\overline{W_n}, \qquad d_i^{C\mu} = r_U(m)W_m\overline{W_n}, \qquad \lambda_i = 1 + |d_i^{CC}|^2 + |d_i^{C\mu}|^2.$$

The exact diagonal Δ_X and these offdiagonal coordinates give

$$E_X = \Delta_X + D^{-1} \sum_i \mu(q_i) d_i^X \Gamma_i, \qquad X \in \{CC, C\mu\}. \tag{38}$$

Thus $E_{CC} = \sum_{u \in \mathcal{R}} |C_u|^2$ and $E_{C\mu} = \sum_{u \in \mathcal{R}} C_u \overline{M_u}$ for the actual normalized row sums. Neither the Möbius sign nor a complex conjugate is discarded.

Proposition 14 (A finite supervised certificate)**.** *Let g_i be complex readout coordinates and $L(g) = \sum_i \lambda_i |g_i - \Gamma_i|^2$. For $K_X = \sum_i |d_i^X|^2/\lambda_i$ and $\lambda_{\min} = \min_i \lambda_i$, the substituted readouts satisfy*

$$|\widehat{E}_X - E_X|^2 \le D^{-2} K_X L(g),$$
$$\max_i |g_i - \Gamma_i|^2 \le L(g)/\lambda_{\min}.$$

The executed metric-gradient update $g^{(t+1)} = (g^{(t)} + \Gamma)/2$ gives $L(g^{(t)}) = 4^{-t} L(g^{(0)})$. Therefore every positive rational requested tolerance on these three squared bounds is attained after finitely many updates on this fixed exact instance, with unrestricted resources.

The implementation's resource guard refuses requests beyond its update budget; termination in this proposition is not a practical runtime guarantee. The labels Γ_i are computed exactly before training, so no speedup over exact label evaluation or generalization to unseen conductors is claimed.

The supplied instance uses $D = 6000, U = 60, H = 625$, S above $2, 3$, principal ν, and all 15 good squarefree ideal columns at activated norms $7507, 7519, 8281, 10507$ (respectively $2, 4, 1, 8$ ideals). One explicitly specified common smooth four-bump profile is zero at every other integer norm. Both the 2262 literal rows and the separately labelled 2148 guaranteed nonprincipal subset are evaluated; the latter is not asserted to be the whole nonprincipal family. There are 95 offdiagonal (h, q) blocks, 65 with $h = 1$, and 210 ordered learned coordinates. Every row case retains 4530 raw column zeros. Real and complex profiles give four requested-10^{-12} certificates, using $53, 55, 53, 55$ metric updates respectively: 216 executed updates, with every full rational state saved. A separate 200 executed ordinary coordinate updates are budget-limited and carry no requested-error claim. The standalone replay is linked from the book and project page. No analytic saving, positive determinant, zero-location assertion, upstream quantum training, new Lean declaration or GRH conclusion follows.

33 Finite phase features, gradients and exact verification

For each block let $G_q = (\mathbb{Z}/6\mathbb{Z})^{\omega(q)}$. A literal admissible row determines local phases e_p at every prime of q; rows with a q-collision or a common-prime collision with h are deleted before forming the exact sparse histogram $n(e)$. Orientation r has

$$\theta_r(e) = \zeta_6^{\sum_{p|q} s_p e_p}, \qquad s_p = 1 \ (p \mid r), \quad s_p = -1 \ (p \mid t), \qquad \Gamma_r = \sum_e n(e) \theta_r(e).$$

Different active orientations have distinct exponent vectors modulo 6, so their characters are orthonormal for the uniform measure on G_q. The model and target projection are

$$F_g(e) = \sum_{r \in J_q} g_r \overline{\theta_r(e)}, \qquad F_\Gamma(e) = \sum_{r \in J_q} \Gamma_r \overline{\theta_r(e)}.$$

Indeed F_Γ is the orthogonal projection of $|G_q| n$ onto this character subspace, and $\mathbb{E}_{G_q} F_g \theta_r = g_r$. The positive spectral metric assigns weight λ_r to this orthonormal basis. Its squared error is exactly

$\sum_r \lambda_r |g_r - \Gamma_r|^2$. Thus the run evaluates the full finite feature-projection loss by Parseval; it does not claim to have executed a sampled or stochastic training loss. This projection need not be a nonnegative density. A Hermitian readout matrix need not be positive semidefinite.

Gradient and contraction. Use ζ_6 coordinates $g = a + b\zeta_6$, so $|g|^2 = a^2 + ab + b^2$ and $M = \left(\begin{smallmatrix} 1 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right)$. For the objective $L/2$, the ordinary real-coordinate gradient in block i is $\lambda_i M(x_i - x_i^*)$. In the block metric $\lambda_i M$, the gradient is $x_i - x_i^*$. A metric step of size $1/2$ therefore halves every complex error and quarters L. Conjugate initial coordinates and this step preserve the Hermitian reverse orientations. The ordinary trajectory treats all ordered coordinates independently; asymmetric weights need not preserve that symmetry. For ordinary steps $\eta = 2/(3\lambda_{\max})$, the Hessian eigenvalues lie between $\lambda_{\min}/2$ and $3\lambda_{\max}/2$, whence

$$L_{t+1} \le \left(1 - \frac{\lambda_{\min}}{3\lambda_{\max}}\right)^2 L_t.$$

This last bound is conservative; the saved 50 ordinary steps per case are not asserted to attain the requested tolerance.

Proof of Proposition 14. Subtract (38) with the same exact diagonal. Weighted Cauchy–Schwarz, applied only to the error, gives

$$\left|\sum_i \mu(q_i) d_i^X (g_i - \Gamma_i)\right|^2 \le \left(\sum_i \frac{|d_i^X|^2}{\lambda_i}\right)\left(\sum_i \lambda_i |g_i - \Gamma_i|^2\right).$$

Here $|\mu(q_i)| = 1$ because q_i is squarefree; the predicted sum itself keeps its signs and phases. Each summand in the second factor is nonnegative, which proves the coordinate bound. The exact affine update gives $g^{(t)} - \Gamma = 2^{-t}(g^{(0)} - \Gamma)$, proving contraction. For finite rational positive λ_i, K_X, L_0 and $\tau > 0$, some integer t makes all three bounds at most τ^2. The zero-error case stops immediately. No varying-family constants enter this finite assertion. □

The activated original family. Let

$$\phi(t) = \begin{cases} \exp(1 - 1/(1-t^2)), & |t| < 1, \\ 0, & |t| \ge 1, \end{cases} \qquad W(y) = \sum_{N \in \{7507,7519,8281,10507\}} w_N \phi(4(6000y - N)).$$

These disjoint bumps are supported in $(1, 2)$ and $\phi(0) = 1$. For the real case $w_N = 1$; for the complex case $w_N = 1 + \zeta_6(4N - 30000)/6000$. At integer norms $W(n/6000) = w_n$ at the four activated norms and is zero elsewhere. Hence all active original squarefree columns are present, rather than an undeclared column submatrix. No exponential evaluation is needed at these exact nodes. The profile is a fixed finite demonstration and does not establish a profile-uniform analytic estimate. The counts r_U are $0, 2, 4, 2$ at the four norms. The guaranteed nonprincipal subset removes precisely those literal rows whose outside-S norm is 1; no classification of the omitted 114 rows as a whole is asserted. Every nonunit outside-S factor has a prime exponent from 1 to 3, since its prime norm is at least 7 and $7^4 > 625$; its nonzero residue modulo 6 gives a nontrivial local character. All six units and all original S-power rows belong to the separately saved full literal case.

The seed 20261105 selects deterministic rational conjugate initial coordinates by SHA256 of the ordered column identifiers. There is no random sampling. Exact $\mathbb{Q}(\zeta_6)$ arithmetic stores every metric and ordinary state. Training accepts only when both observable certificates and the

uniform-coordinate certificate are at most τ^2, preventing a scalar cancellation from masquerading as accurate correlation learning. The real full-row targets are $E_{CC} = 2393/125$ and $E_{C\mu} = 431/50$; the guaranteed-subset targets are $4293/250$ and $771/100$. Complex-profile targets and the complete phase histograms are saved alongside. Each case has 64 nonzero negative-$\mu(q)$ C–C target orientations. Replacing all signs by plus is an explicit failing control, not an estimator.

Reproduction and scope. The complete run is reproducible.

The source is `examples/arithmetic_learning/learn.py`; its README states the exact smooth profile, CLI, resource guard and loss convention. The accepted immutable output is `outputs/experiments/2026-11-05-arithmetic-learning-v2/`. All source/helper hashes, exact input bytes and artifact hashes are bound before and after execution and checked on replay. Existing outputs, nonpositive or binary-float tolerances, and stale source/input/case bytes are refused. The preserved v1 logging failure is not the accepted run. The focused verifier independently reconstructs 31668 local symbols and 450 Gram targets, then checks the full saved metric trajectory. Independent mathematical review and exact code checks are recorded separately; finite arithmetic checks are not a proof of an asymptotic signed estimate. Full exact training labels are supervision, so this is a certified finite reference architecture, without an efficiency, generalization, PSD, zero-location or GRH claim. Analytic and categorical Lean remain unfinished.

34 Why the common profile must stay fixed

The unsigned envelope of Section 30 and the unresolved signed estimate have different profile quantifiers. We give an obstruction to an H-sized bound uniform through the sup norm alone. It uses the *actual* coefficients and nonprincipal rows, rather than replacing the columns by arbitrary weights. It does not contradict an estimate for one fixed smooth profile or one with adequate derivative dependence.

Fix $K = \mathbb{Q}(\omega)$, $\omega^2 + \omega + 1 = 0$, principal $\nu = 1$, and a finite good set S containing the primes above $2, 3$. Write $\zeta = 1 + \omega$, $\delta = 1 + 2\omega$, so ζ has order six and $\delta^2 = -3$. Retain the literal raw symbols and normalization

$$I_u(W) = D^{-1/2} \sum_{m \text{ squarefree } (m,S)=1} A_U(m)\chi_m(u)W(Nm/D), \qquad E_u = I_u - P_u.$$

Here and below squarefree means *ideal* squarefree. For $U^2 = D$ the accepted high-window identity gives $A_U(m) = \mu(m)(r_U(m) - 1)$ on $D < Nm < 2D$, with $r_U(m)$ even and $|A_U(m)| \geq 1$. The multiset of prime-ideal norms of a squarefree ideal is determined by its rational norm. Thus $A_U(m) = A_N$ depends only on $N = Nm$, including divisor-norm multiplicities. Moreover

$$\eta(m) := \chi_m(\zeta) = \zeta^{(Nm-1)/6}, \qquad (m, S) = 1. \tag{39}$$

The primitive character η has conductor $(2)^2(\delta)^3$, of norm 432; its powers $1, 5$ have conductor norm 432, powers $2, 4$ norm 27, and power 3 norm 16. All five powers are nonprincipal. These row conductors do not remove the raw zero at any prime dividing a sixth factor.

Put $\phi(t) = \exp(1 - 1/(1 - t^2))$ for $|t| < 1$ and zero otherwise. For integer D set $w_N = \operatorname{sgn}(A_N)\overline{\zeta^{(N-1)/6}}$ when a good squarefree ideal of norm N exists and $D < N < 2D$, and zero otherwise. Define one common, complex profile

$$W_D(y) = \sum_{D<N<2D} w_N \phi(4(Dy - N)). \tag{40}$$

Its disjoint supports lie strictly inside $(1,2)$, its sup norm is one for large D, and $\|W_D^{(j)}\|_\infty \ll_j D^j$. In particular the endpoints have value zero. This is a profile varying with scale, independent of the row, not a separate choice for each ideal or each row.

Proposition 15 (A moving-profile lower bound)**.** *For every fixed $0<\theta<1$, take $H=D^\theta$, $U=\sqrt{D}$, and $X=H^{1/6}$. As integer $D\to\infty$,*

$$\sum_{0<Nu\le H} |E_u(W_D)|^2 \ge (c_S+o(1))DH^{1/6}, \qquad c_S>0. \tag{41}$$

This lower bound already comes from literal rows $u_v=\zeta\beta_v^6$, where v is an arbitrary good ideal of norm at most X and β_v is a generator. These rows have nonprincipal primitive character η, so $P_{u_v}=0$. Their exact values are

$$E_{u_v}(W_D) = D^{-1/2} \sum_{\substack{m \text{ squarefree, } (m,Sv)=1\\ D<Nm<2D}} |A_U(m)|. \tag{42}$$

Every collision with v remains deleted, even when the primitive conductor drops to 432. No squarefree condition is imposed on v.

At $H=D^{3/4}$ the lower exponent is $9/8$, whereas the current upper exponent is $7/6$. Thus this does not establish sharpness of $7/6$ or a new upper saving. For $0<\theta<2/3$, the term $DH^{1/6}$ dominates the current unsigned envelope; it is necessary up to subpower losses in this *varying-profile* class, even on nonprincipal rows.

For $k\ge 0$ put $M_k(W)=\sum_{j=0}^k \|(y\partial_y)^j W\|_\infty$. If, for every $\varepsilon>0$, a bound uniform over all common profiles with constants independent of W, D were to assert

$$\sum_{0<Nu\le H} |E_u(W)|^2 \ll_\varepsilon M_k(W)^\alpha (HD)^\varepsilon H, \qquad \alpha\ge 0,$$

then necessarily $k\alpha \ge 1-5\theta/6$. At $\theta=3/4$ this is $k\alpha\ge 3/8$. For a single fixed ε the necessary condition is only $k\alpha+(1+\theta)\varepsilon \ge 1-5\theta/6$. These are necessary cost comparisons, not sufficient estimates. A fixed-W result, a class with bounded seminorms, or the existing $M_4(W)^2$ dependence is not refuted.

The same rows expose a concrete centering error in the joint completion: removing the orientation mean of the actual correlation can delete a nonzero signed contribution on *nonprincipal* rows. The extended proof retains the literal unit phase, gives an exact four-orientation example, and explains why orientation projection alone supplies no saving. The standalone profile and coherent-mean inspector shows the complete finite norm groups, arbitrary sixth-factor masks and exact residual quantities. Its finite tables are diagnostics of this construction, not asymptotic estimates, training or GRH evidence.

35 Proof and joint-mean boundary of the moving-profile construction

35.1 Norm groups and primitive conductors

For a split rational prime p, a squarefree ideal contains zero, one or both conjugate primes. Its rational norm exponent is respectively $0,1,2$; at exponent one there are two choices, and at exponent two its divisor norms include two copies of p. For an inert prime, the exponent is zero or two, and

the nontrivial choice is the single prime ideal of norm p^2. Thus the rational norm determines μ, the divisor-norm multiset and r_U, even though it need not determine the ideal. For example norm 49 has squarefree ideal divisor norms $1, 7, 7, 49$, whereas the inert prime of norm 25 has divisor norms $1, 25$. Additional fixed S deletions remove some ideals without changing this constancy among those remaining. The high-window formula and its parity are the already proved facts of Section 20, not a new coefficient estimate.

At a good prime ideal $\mathfrak{p}$ of norm q, Euler's definition gives $\chi_{\mathfrak{p}}(\zeta) = \zeta^{(q-1)/6}$. All such norms are 1 modulo 6; for $a = 1 + 6r, b = 1 + 6s$ one has $(ab - 1)/6 = r + s + 6rs$. Multiplicativity proves (39). Write $\chi_4(N) = (-1)^{(N-1)/2}$ for odd N. On good ideals,

$$\eta = \chi_4(N)\omega^{(N-1)/3},$$
$$\eta^2 = \omega^{2(N-1)/3} = \omega^{-(N-1)/3}, \qquad \eta^3 = \chi_4(N).$$

The inverse orientation in the middle expression is essential.

For completeness the local conductors are exact, not merely upper bounds. The quadratic norm character factors modulo $(2)^2 = (4)$. Since $\mathrm{Tr}(\delta\gamma) \in 3\mathbb{Z}$ for every $\gamma \in \mathcal{O}$, expanding the norm gives $N(\alpha + \delta^3\beta) \equiv N\alpha \pmod 9$. The cubic norm character therefore factors modulo δ^3. The coprime local witnesses

$$\alpha_2 = 13 + 6\omega, \quad N\alpha_2 = 127, \quad \alpha_2 - 1 = 2\omega\delta^3,$$
$$\alpha_3 = -11, \quad N\alpha_3 = 121, \quad \alpha_3 - 1 = 4\delta^2.$$

show minimality. The first is 1 modulo 2, is $1 + 2\omega$ modulo 4, and is 1 modulo δ^3; its η value is -1. The second is 1 modulo 4 and $-2 = 1 + \delta^2$ modulo δ^3, but not 1 modulo δ^3; its η value is ω and its square is ω^2. Thus the quadratic local exponent is two and the cubic one is three. The two moduli are coprime, giving the stated norms $432, 27, 16$ for the nontrivial powers. Global units have norm one, so these residue characters are unit trivial and descend to ideal Hecke characters in this class-number-one field. A norm-seven prime has value ζ, proving order six. A larger finite S does not principalize the character: CRT gives representatives of the same nontrivial ray classes avoiding the extra excluded primes. Primitive conductor reduction never restores raw primes excluded by S or the sixth factor.

35.2 Exact alignment, with arbitrary overlapping sixth factors

The bumps in (40) have radius $1/(4D)$ and center N/D. Distinct supports are disjoint. At an active integer node their value is exactly w_N, and all endpoints are zero. Differentiation gives $\|W_D^{(j)}\|_\infty \le (4D)^j \|\phi^{(j)}\|_\infty$. Expanding $(y\partial_y)^j$ into ordinary derivatives and using $1 < y < 2$ gives $M_k(W_D) \ll_k D^k$.

Every nonzero ideal v has a generator β_v. The literal element $u_v = \zeta\beta_v^6$ is independent of that choice, since all units have sixth power one; distinct ideals yield distinct rows. Its norm is $(Nv)^6$. For squarefree good m, exact raw multiplicativity says

$$\chi_m(u_v) = \eta(m)\mathbf{1}_{(m,v)=1}.$$

Repeated primes of v are allowed and its collisions with m remain zero. The primitive row character is always η, so the actual principal indicator in (15) is zero when $\nu = 1$. Substituting the nodal profile proves (42) with no factor six, unit averaging or residue replacement.

35.3 The coprime-triple density and the selected $h = 1$ block

Put $\kappa = \pi/(3\sqrt{3})$. Lattice counting, with six unit generators per ideal, gives

$$A_K(Y) := \#\{a : Na \le Y\} = \kappa Y + O(\sqrt{Y} + 1), \qquad A_K(Y) \le CY \quad (Y \ge 1).$$

This is the existing elementary ideal count. Let $T(D, X)$ count ordered triples (m, n, v) with m, n squarefree good, $D < Nm, Nn < 2D$, $Nv \le X$, v good and arbitrary, and the three ideals pairwise coprime. For $D, X \to \infty$, $X \le D$, we claim

$$\begin{aligned} T(D, X) &= (\kappa^3 \mathcal{C}_S + o(1))D^2 X, \\ \mathcal{C}_S &= \prod_{\mathfrak{p} \in S} (1 - t_{\mathfrak{p}})^3 \prod_{\mathfrak{p} \notin S} (1 - 5t_{\mathfrak{p}}^2 + 6t_{\mathfrak{p}}^3 - 2t_{\mathfrak{p}}^4) > 0, \\ t_{\mathfrak{p}} &= (N\mathfrak{p})^{-1}. \end{aligned} \tag{43}$$

Here a product over S means its prime ideals. To prove this, first impose the local restrictions at a fixed finite set of primes containing S. They are finite Boolean combinations of divisibility by $\mathfrak{p}$ and $\mathfrak{p}^2$ in m, n, and by $\mathfrak{p}$ in v. Finite inclusion-exclusion and the count for fixed ideal multiples give the product of the local densities, with relative error tending to zero. At a good prime the allowed cases are no positive valuation, a single valuation one in m or n, or an arbitrary positive valuation in v. The factor is therefore

$$(1 - t)^3 \left(1 + 2t + \frac{t}{1 - t}\right) = (1 - t)^2(1 + 2t - 2t^2) = 1 - 5t^2 + 6t^3 - 2t^4.$$

At S all three valuations must vanish, giving $(1 - t)^3$. The geometric series for v is essential; v is not squarefree.

An omitted failure belongs to one of five events: $\mathfrak{p}^2 \mid m$, $\mathfrak{p}^2 \mid n$, $\mathfrak{p} \mid (m, n)$, $\mathfrak{p} \mid (m, v)$ or $\mathfrak{p} \mid (n, v)$. Divisibility counts and $A_K(Y) \le CY$ bound each by $O(D^2X/(N\mathfrak{p})^2)$, uniformly; if a quotient scale is below one its count is zero. Thus the tail after norm P is

$$O\left(D^2 X \sum_{N\mathfrak{p} > P} (N\mathfrak{p})^{-2}\right) = o_P(D^2 X).$$

The series converges by the ideal count (or $\zeta_K(2) < \infty$). Take the large-scale limit first at fixed P, then $P \to \infty$. Each local factor is positive and differs from one by $O(t^2)$, proving (43). Strict annulus endpoints have no effect on the main term by the same lattice count. The secondary range $X/2 \le Nv < X$ has half this limiting volume, though it may be empty at small X.

On the selected literal unit class every expanded pair term in (42) is nonnegative. Keeping only $(m, n) = 1$ gives its exact $h = 1$ A–A contribution

$$\mathcal{E}^{AA}_{\text{selected},h=1} = D^{-1} \sum_{v \text{ good } Nv \le X} \sum_{\substack{m,n \text{ squarefree,good } D < Nm, Nn < 2D \\ (m,n)=(m,v)=(n,v)=1}} |A_U(m) A_U(n)| \ge D^{-1} T(D, X).$$

Consequently, with $X = H^{1/6}$,

$$\sum_{0 < Nu \le H} |E_u|^2 \ge \sum_{Nv \le X,\ v \text{ good}} |E_{u_v}|^2 \ge \mathcal{E}^{AA}_{\text{selected},h=1} \ge (\kappa^3 \mathcal{C}_S + o(1))DX.$$

The first inequality is positivity of the *outer squared row sum*. The selected $h = 1$ block does not lower-bound the signed $h = 1$ sum over other literal rows or a unit-complete family. No positive-semidefinite claim about a gcd-truncated Gram matrix is needed. This is A–A, not a positive-density assertion for C–C long-long weights, which may vanish. This proves Proposition 15 with $c_S = \kappa^3 \mathcal{C}_S$. The derivative tradeoff follows by comparing powers in $DH^{1/6} \ll D^{k\alpha + (1+\theta)\varepsilon} H$ and then letting $\varepsilon \downarrow 0$ only when the proposed bound holds for every positive ε.

35.4 Projection preserves the coupled problem, not a saving

Use the exact joint notation of Section 29. For fixed h, q with $q \neq 1$ (including $h = 1$), let $n = \varphi(q)$, $\ell = N(hq)$, $Q = Nq$, and

$$c = \frac{Q^2 \xi_q(-\alpha)\tau_\beta(\xi_q)}{\ell^6},$$
$$\Gamma_r^F = c \sum_K B_F(K)\overline{\theta_r(K)}, \qquad A(K) = \sum_{r \in J} a_r \overline{\theta_r(K)}.$$

The active orientations J include all original restrictions, and the literal unit phase belongs to a_r. The distinct θ_r are orthonormal for $n^{-1}\sum_K$. If P projects onto their span, then

$$PB_F = (cn)^{-1} \sum_{r \in J} \Gamma_r^F \theta_r, \qquad \sum_K |PB_F(K)|^2 = (|c|^2 n)^{-1} \sum_r |\Gamma_r^F|^2.$$

Replacing B_F by PB_F in the full bilinear pairing is exact. Cauchy–Schwarz is precisely $|c\sum_K B_F A| \leq \|a\|_2 \|\Gamma^F\|_2$. The conductor and group-size factors cancel. Projection rank alone gives no arithmetic saving for this particular coupled vector.

For $s = \omega(q)$ let $C = \{K : x_{\mathfrak{p}}(K)^2 = 1 \text{ for all } \mathfrak{p} \mid q\}$. All orientations agree on C. If M_C denotes multiplication by its indicator, the matrix in the orthonormal orientation basis is

$$PM_C P = 3^{-s}\mathbf{1}\mathbf{1}^*, \qquad \lambda = |J|/3^s, \qquad \|(1-P)M_C P\|^2 = \lambda(1-\lambda).$$

Thus coherent restriction does not commute with projection. For example

$$\left| \sum_C (B_F - PB_F)A \right| \leq \sqrt{\lambda(1-\lambda)} \|(1-P)B_F\|_2 \|A\|_2,$$

where all norms in this inequality are counting ℓ^2 norms. Normalized norms would require an additional factor n on the right. Counting a coherent subset alone is not a lower bound for its actual signed pairing.

35.5 A nonzero orientation mean on actual nonprincipal rows

For the selected rows let V be any finite set of good ideals, put the first five row factors equal to one, and retain the sixth-factor mask. For each h, q define $Z_{h,q} = \#\{v \in V : (v, hq) = 1\}$. After removing the separately displayed literal unit phase, the exact correlation is

$$\Gamma_r^F = Z_{h,q} = c \sum_K B_V(K)\overline{\theta_r(K)} \quad \text{for every active orientation } r.$$

To pass from the published C–C identity to A–A replace each r_U by $r_U - 1$. At $h = 1$, $m = r, n = q/r$, the coefficient is

$$a_r^{AA} = (r_U(m) - 1)(r_U(n) - 1) W_m \overline{W_n} \theta_{m,n}(\zeta), \qquad \mu(q) a_r^{AA} = |A_U(m) A_U(n)|.$$

This retains $\mu(q)$, the common complex profile and its conjugation, all deleted common primes, and the literal ζ phase. If $Z_{1,q} > 0$ and at least one active interior A–A coefficient is nonzero, the actual pairing is positive, whereas $\sum_r |\Gamma_r^F - \text{mean}\,\Gamma^F|^2 = 0$. An orientation-variance-only replacement with no mean term is therefore false. The mean is not a principal residue: these rows have character η, with nonprincipal powers. This does not refute a properly formulated fixed-profile signed estimate.

A fresh exact example uses $D = 4000$, $U = 63$ and chosen prime ideals of norms $7, 13, 79, 67, 97$. Their product has four active orientations: the pairs 6097/7663, 7189/6499 and both reversals. Here $r_U = 2$, and A_U is -1 on the three-prime columns and $+1$ on the two-prime columns. The four good ideals with $10 \le Nv \le 20$ have norms $13, 13, 19, 19$. Exactly one collides with the chosen 13-prime, so $Z = 3$ for all four orientations. The exact selected $h = 1$ contribution is $12/4000 = 3/1000$ despite zero orientation variance. Dropping the literal ζ phase incorrectly changes the coefficient sum from 4 to -1, giving $-3/4000$. The coherent fraction is $4/243$ and the squared leakage is $956/59049$. The residue group and infinite Fourier tail are not estimated by this finite check. The fixed row conductor 432 must be distinguished from the pair-column conductor $q = mn$, whose norm here and in the asymptotic annulus is comparable to D^2.

The exact implementation at `examples/profile_uniformity/` keeps complete norm groups and arbitrary sixth factors, including distinct ideals of norm 49 with different masks. The saved six-scale examples, exact joint fixture and their independent checks certify finite identities only. Full-union masks, coherent means, fixed-profile uniformity and the remaining mixed/pure upper-envelope neighborhoods still require a new signed estimate. No determinant comparison, zero location, analytic or categorical Lean theorem, quantum result or GRH proof is obtained.

36 A bound for the signed coprime slice and an exact character twist

The coprime-column contribution can be negative, so it needs its own bound. Fix the original ν, S, one common possibly complex smooth W supported in $[1, 2]$, $D > 1$, $U \ge 1$ and $U^2 \le D$. Consider an actual block

$$u = u_0 v_1 v_2^2, \quad A_j \le N v_j < 2A_j \ (j = 1, 2), \quad Nu \le H, \qquad R = A_1 A_2, \quad M = \max(A_1, A_2).$$

The good ideals v_1, v_2 are squarefree and coprime, with their literal primary/ray representatives; all other good factors are the unit ideal. Let F be the actual indicator, including capacity and every row mask. In the remaining neighborhoods the first factor is nonunit, so the primitive row character is nonprincipal and $E_u = I_u$ there. For $X, Y \in \{A_U, C_U, \mu\}$ define the actual signed slice

$$S_1^{XY} = \frac{1}{D} \sum_u F(u) \sum_{\substack{m,n \text{ squarefree,good} \\ (m,n)=1}} X(m) Y(n) \nu(m) \overline{\nu(n)} \chi_m(u) \overline{\chi_n(u)} W(Nm/D) \overline{W(Nn/D)}. \quad (44)$$

All columns in this formula are the original annular columns, including raw zero sentinels. No positive-semidefinite claim about the coprime truncation is made.

Proposition 16 (A bound for the actual signed $h = 1$ slice)**.** *Uniformly in the original cutoff and this actual block,*

$$|S_1^{XY}| \ll_{\nu,S,\epsilon} \|W\|_\infty^2 (HD)^\epsilon \min\left\{ R + \frac{DR}{M} + \frac{D^{2/3} R}{M^{1/3}},\ D + R^2 \right\}. \quad (45)$$

The exact identity underlying the estimate is

$$\begin{aligned} D S_1^{XY} &= \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \mu(d) \sum_{\substack{u \text{ in the actual block} \\ (d,u)=1}} T_d^X(u) \overline{T_d^Y(u)}, \\ T_d^X(u) &= \sum_{\substack{a \text{ squarefree} \\ (a,dS)=1}} X(da) \nu(a) \chi_a(u) W(Nd\,Na/D). \end{aligned} \quad (46)$$

In particular $X(da)$ is not replaced by $X(d)X(a)$, and the mask $(d,u)=1$ remains.

The proof applies the accepted block sieves inside each positive row norm in (46), then explicitly loses the d signs in an absolute triangle. At $H=D^{3/4}$ the two remaining power-scale neighborhoods $A_1 \asymp D^{5/12}, A_2 \asymp D^{1/6}$ and $A_1 \asymp D^{3/4}, A_2 \asymp 1$, with feasible capacity constants, both retain power $7/6$. The desired $3/4$ still needs a signed saving of $5/12$. This proves a separate signed-slice bound; it does not improve the published global envelope or supply the missing cancellation.

There is also an exact arithmetic description of the two-pivot phase. Put $\delta = 1+2\omega$ and, at each good prime ideal p, set $j(p) = J_p(\chi_p, \chi_p^2)$. Extend j multiplicatively on squarefree columns. Let $\Psi(a)$ be the unique generator of an ideal prime to 2δ satisfying $\Psi(a) \equiv 1 \pmod{2\delta}$.

Proposition 17 (The actual Jacobi column twist)**.** *The map Ψ is an algebraic Hecke character with exact finite conductor (2δ), of norm 12, and principal convention $\Psi((\alpha)) = \alpha$ for $\alpha \equiv 1 \pmod{2\delta}$ (infinity type $(1,0)$). For good squarefree columns,*

$$(j(p)) = p, \quad -j(p) = \Psi(p), \qquad \mu(m)j(m) = \Psi(m). \tag{47}$$

Define $\vartheta(m) = \Psi(m)/\sqrt{Nm}$ and $\rho(m) = j(m)/\sqrt{Nm} = \mu(m)\vartheta(m)$ on those columns. In the high window the actual gauged coefficients are $b_A = (r_U - 1)\vartheta\nu W$, $b_C = r_U\vartheta\nu W$ and $b_\mu = \vartheta\nu W$.

This does not replace the original row kernel. Its compensating inverse gauge remains, so all ordinary coefficient/operator norms are unchanged. The new character has nonzero infinity type and is not silently covered by a theorem stated only for finite-order twists. The identities in (47) involving μ must not be extended to arbitrary nonsquarefree sixth factors.

A fresh fixed-profile example takes $\nu = 1$, S the primes above $2, 3$, literal unit one and primary row generators 1 modulo 3. It lies in an actual $A_6 = 1$ block at $D = 60000$, $U = 244$, $H = 3833$, $A_1 = 37$, $A_2 = 7$. All 20 rows of the stated literal class are nonprincipal. A selected q has exactly two legal column orientations, of norms $97051/96091$ and its reversal. Their correlations in the ω basis are $(-4,-2)$ and $(-2,2)$, with mean -3. Their selected contribution is $S_{1,q}^{AA} = -1/10000$; deleting the mean gives zero. The centered squared norm is 6, not zero, and the proposed variance-only Cauchy bound would incorrectly require $36 \le 12$. This refutes that centering rule in a fixed-profile row block, not the asymptotic target. Other q contributions remain. The coprime-kernel inspector shows the actual rows, raw zeros, two-column Gram matrix and compensated gauge; no training or arithmetic spectrum is inferred from that finite view.

37 Proof of the signed-slice budget and its surviving analytic burden

37.1 Gcd inversion before an absolute bound

Insert $\mathbf{1}_{(m,n)=1} = \sum_{d \mid m,\ d \mid n} \mu(d)$ in (44), and write $m = da, n = db$. Squarefreeness forces $(a,d) = (b,d) = 1$. The common factors give $\nu(d)\overline{\nu(d)} = 1$ and $\chi_d(u)\overline{\chi_d(u)} = \mathbf{1}_{(d,u)=1}$. This proves (46), retaining the original coefficient $X(da)$ and the full row-zero mask. For $X = Y$ each inner squared row sum is nonnegative; its $\mu(d)$-weighted sum need not be.

Put $L = D/Nd$. Whenever a column survives, $L \ge 1/2$. The actual divisor-assignment bound and ideal counting give

$$\sum_{\substack{a \text{ squarefree} \\ (a,dS)=1}} |X(da)\nu(a)W(Nd\,Na/D)|^2 \ll_\eta D^\eta \|W\|_\infty^2 L.$$

Indeed $N(da) \le 2D$, $|X(da)| \ll_\eta D^\eta$ uniformly in d, U, and there are $O(L)$ ideals of norm at most $2L$ for $L \ge 1/2$. This includes C_U by $C_U = A_U + \mu$ on the active annulus. There is no multiplicativity assertion for $X(da)$. For $L \ge 1$, the powered block bound and the ordinary-conductor comparison of Section 30, followed by Cauchy for the cross term, imply

$$|G_d^{XY}| \ll (HD)^\eta \|W\|_\infty^2 L \min\left\{R + \frac{LR}{M} + \frac{L^{2/3}R}{M^{1/3}},\ L + R^2\right\}, \tag{48}$$

where G_d^{XY} denotes the inner row sum in (46). The raw (d, u) restriction is only enlarged inside these positive norms; u_0 is a fixed column multiplier of modulus at most one. For $1/2 \le L < 1$, only the unit column can survive. The direct row count is $O(R)$, and the right side of (48) is at least a constant times R since $R \ge 1$ and $L \ge 1/2$. Thus this endpoint does not require a sieve with an unstated subunit length.

Now take the absolute triangle in d explicitly. The two estimates may be summed separately and their minimum taken. The elementary ideal sums

$$\sum_{Nd \le 2D} \frac{1}{Nd} \ll \log(2D), \qquad \sum_d (Nd)^{-2} \ll 1, \qquad \sum_d (Nd)^{-5/3} \ll 1$$

turn the first estimate into $R\log(2D) + DR/M + D^{2/3}R/M^{1/3}$ after division by D; the second gives $D + R^2\log(2D)$. Absorb the logarithms with smaller internal exponents to obtain (45). All raw zeros, literal/ray/squarefree/capacity restrictions were retained before the stated positive enlargements. No bound for an indefinite submatrix was inferred from positivity of a larger Gram matrix.

At the mixed power-scale point the powered exponents are $7/12, 7/6, 10/9$, and the ordinary exponents are $1, 7/6$. At the pure point they are $3/4, 1, 7/6$ versus $1, 3/2$. These are upper-bound exponents with feasible dyadic constants, not actual energy lower bounds. A sufficient $h = 1$ target for the A–A combination is

$$\left|\sum_d \mu(d) G_d^{AA}\right| \ll_{\nu,S,W,\epsilon} DH(HD)^\epsilon. \tag{49}$$

At $H = D^{3/4}$ this asks for power $7/4$ before division by D, while (45) gives $13/6$, leaving $5/12$. This is sufficient for this slice, not a necessary bound for every separate C–C/C–μ term if their signed assembly cancels further. The quantities G_d depend on prime supports, $X(da)$ and row deletions; they are not an independent smooth function of Nd to which a scalar Möbius partial-sum estimate can simply be applied.

37.2 The correctly oriented Jacobi generator

At a good prime ideal p, let $n = Np$ and $\chi = \chi_p$ be the literal positive sextic residue character. In $\mathcal{O}/p$, its value reduces to $x^{(n-1)/6}$. For $j(p) = J(\chi, \chi^2)$, the elementary Gauss–Jacobi identity and the three nontrivial Gauss norms give $|j(p)|^2 = n$. Reduction in that *same* residue field gives

$$j(p) \equiv \sum_x x^{(n-1)/6}(1-x)^{(n-1)/3} = 0 \pmod p.$$

Every exponent after expansion lies strictly between 0 and $n-1$, so the finite-field power sums vanish. Hence $(j(p)) = p$. This includes an inert prime of norm p_0^2, and inverting the literal character conjugates the ideal at a split prime.

This raw sum equals the old $J(\chi^2,\chi^3)$: the quotient of the two Gauss–Jacobi expressions is $G(\chi)G(\chi^5)/G(\chi^3)^2=1$. It differs from the theta-curve Frobenius normalization by its explicitly stated factor $-\chi^3(-1)$; that sign is not imported silently. No new Gauss or Jacobi norm theorem is claimed.

Let $f=(2\delta)$. The six units map bijectively onto $(\mathcal{O}/f)^\times$: the components at 2 and δ have orders three and two, with complementary kernels. Since $\mathcal{O}$ is a PID, every ideal prime to f has a unique generator $\Psi(a)\equiv 1 \pmod f$. Uniqueness gives multiplicativity. If $u_f(\alpha)$ is the unit having the residue of α modulo f, then

$$\Psi((\alpha))=\alpha/u_f(\alpha), \qquad |\Psi(a)|^2=Na.$$

This explicitly defines the finite part and the principal infinity-type convention. The finite conductor divides f. A proper divisor omits a component: the unit -1 is 1 modulo 2, and ω is 1 modulo δ. A smaller defining modulus would require the unit ideal to have value -1 or ω. Thus the finite conductor is exactly f.

To identify the actual $j(p)$, now reduce in the *character-value* ring, a different operation from reduction modulo p above. Modulo δ, a sixth root v satisfies $v=v^3$ and $v^2=1$. The zero branches at $x=0,1$ are kept, so $j(p)\equiv\sum_{x\neq 0,1}\chi^3(x)=-1 \pmod\delta$. Modulo 2, $v=v^4$, and $J(\chi,\chi^2)\equiv J(\chi^4,\chi^2)=-\chi^4(-1)=-1$. The inverse-character Jacobi identity here follows by the bijection $x/(1-x)$ onto the nonzero field elements other than -1. Consequently $-j(p)\equiv 1 \pmod f$, proving $-j(p)=\Psi(p)$. Multiplication proves (47) on squarefree columns. For nonsquarefree ideals μ vanishes, so that last identity must not be extended to arbitrary sixth factors. This construction establishes the fixed character directly; it assumes no elliptic modularity, spectral comparison, or distribution of prime arguments.

37.3 The actual two-factor completion, not a new row kernel

Freeze the literal u_0,v_3,v_4,v_5,v_6 and put $Z=u_0v_3^3v_4^4v_5^5v_6^6$. The last factor is arbitrary, with its allowed overlaps. For $h=1$, write $q=mn\neq 1$, $Q=Nq$, $\theta=\chi_m\overline{\chi_n}$ and $\xi_q=\theta^3$. For $q=(\beta)$ use the original inverse-different pairing $e_\beta(z)=\exp(2\pi i\,\mathrm{Tr}(z/(\delta\beta)))$. Let $F(v_1,v_2)$ be the exact common finite weight, retaining the first-five coprimality, primary/ray and capacity restrictions. It must be independent of the column orientation. Define

$$\widehat{F}_q(k_1,k_2)=\sum_{v_1,v_2}F(v_1,v_2)e_\beta(-k_1v_1-k_2v_2),$$
$$T_{q,F}(\theta)=\sum_{k_1,k_2\in(\mathcal{O}/q)^\times}\widehat{F}_q(k_1,k_2)\overline{\theta(k_1k_2^2)}.$$

Periodization sums every actual representative; no injectivity assumption is needed. The primitive powers θ,θ^2 have zero Gauss sums at nonunit frequencies, including at inert primes. Exact inversion gives

$$\sum F\theta(v_1)\theta(v_2)^2=Q^{-2}\tau_\beta(\theta)\tau_\beta(\theta^2)T_{q,F}(\theta)=Q^{-2}J_q(\theta,\theta^2)\tau_\beta(\xi_q)T_{q,F}(\theta).$$

Jacobi sums CRT without additive-generator multipliers, so $J_q=j(m)\overline{j(n)}=\sqrt{Q}\,\rho(m)\overline{\rho(n)}$. The quadratic central phase is the already recorded period-four law

$$\gamma(\beta)=\tau_\beta(\xi_q)/\sqrt{Q}=\tfrac12\{1+i^{-b}+i^a+i^{b-a}\}, \qquad \beta=a+b\omega.$$

For composite squarefree q, identifying $\tau_\beta(\xi_q)$ with $\sum_x e_\beta(x^2)$ requires CRT expansion of the local root counts; $1+\xi_q$ is not the root-count function on a composite ring. The known quadratic Gaussian evaluation is reused, not retested as a new phase theorem. Its pairing is exactly the displayed e_β.

With $b_X = \rho X \nu W$, the actual two-pivot identity is

$$DS_1^{XY} = \sum_q \frac{\gamma(\beta_q)}{Q} \sum_{mn=q} b_X(m)\overline{b_Y(n)}\theta_{m,n}(Z)T_{q,F}(\theta_{m,n}). \tag{50}$$

The two Gauss magnitudes, equivalently the central Gauss and Jacobi magnitudes, give exactly $1/Q$. There is no extra $\xi_q(-1)$ in this two-factor identity; the full six-factor formula of Section 28 retains that sign. If β changes to $u\beta$, both γ and T acquire the same quadratic sign $\xi_q(u)$, which cancels in their product. If Z shares any prime with q, $\theta(Z) = 0$, including at a sixth-power collision. No dropped primitive-conductor prime is restored.

Multiplying columns by ρ changes the kernel by its compensating inverse unitary gauge. It preserves operator norms and eigenvalues even for an indefinite coprime-truncated matrix. A fixed-character coefficient estimate cannot replace a bound for the actual q-dependent $T_{q,F}$. At the pure endpoint $v_2 = c$ with $(c, q) = 1$, exact summation gives

$$T_{q,F}(\theta) = \theta(c)^2\tau_\beta(\overline{\theta^2}) \sum_{k_1 \in (\mathcal{O}/q)^\times} \widehat{F}_1(k_1)\overline{\theta(k_1)}.$$

Since $\theta^2(-1) = 1$, the extra Gauss pair multiplies to Q, returning the original one-pivot scalar $\tau_\beta(\theta)/Q$. A nonunit c gives the original zero. Thus a putative central-phase gain cannot be retained after discarding a literal length-one Fourier factor.

37.4 A central reciprocal remains after the gauge

On squarefree columns put $B_U = \mu A_U$ and $\vartheta = \Psi/\sqrt{N}$. The original ternary assignment gives exactly

$$B_U(m) = \sum_{\substack{abc=m \\ Na,Nb \le U}} \mu(c), \qquad \rho(m)A_U(m) = \vartheta(m)B_U(m).$$

All factors are squarefree and pairwise coprime. Fix a nonzero numerator z and keep the full deleted set $T = S \cup \operatorname{supp}(z)$. Write Ξ for the unitary primitive character of $\vartheta\nu\kappa_z$, with these raw deletions retained. Its nonzero infinity type prevents a silent reuse of a finite-order-only moment theorem. For $\Re s > 1$,

$$J_U^{\sharp,T}(s) = \sum_{m \text{ squarefree},\ (m,T)=1} B_U(m)\Xi(m)(Nm)^{-s} = N_U^T(s)/L^T(s,\Xi),$$

$$N_U^T(s) = \sum_{\substack{a,b \text{ squarefree} \\ (a,b)=1,\ (ab,T)=1 \\ Na,Nb \le U}} \frac{\Xi(ab)}{(Nab)^s} \prod_{p \mid ab} (1 - \Xi(p)(Np)^{-s})^{-1}.$$

Fixing a, b leaves the Möbius Euler product in c; this proves the identity by absolute convergence. The numerator is finite and analytic for $\Re s > 0$. At $U = 1$ it is one, so the series is exactly $1/L^T(s,\Xi)$. That boundary identity is not a high-window counterexample. It shows why cancellation of central reciprocal poles cannot simply be inferred from the column-character identification.

For $x_p = \Xi(p)(Np)^{-s}$, $\alpha_p = (Np)^{-v}$ and $\beta_p = (Np)^{-w}$, the gauged three-placement Euler factor is $1 + x_p(\alpha_p + \beta_p - 1)$, with the opposite signs from the ungauged factor in Section 20. In the initial absolute domain it gives

$$\prod_{p \notin T} \{1 + x_p(\alpha_p + \beta_p - 1)\} = \frac{L^T(s+v,\Xi)L^T(s+w,\Xi)}{L^T(s,\Xi)} \prod_{p \notin T} H_p,$$

where

$$H_p = \frac{\{1 + x_p(\alpha_p + \beta_p - 1)\}(1 - x_p\alpha_p)(1 - x_p\beta_p)}{1 - x_p}.$$

Expansion gives

$$H_p - 1 = \frac{x_p^2(\alpha_p + \beta_p - \alpha_p^2 - \beta_p^2 - \alpha_p\beta_p)}{1 - x_p} + \frac{x_p^3\alpha_p\beta_p(\alpha_p + \beta_p - 1)}{1 - x_p}.$$

Thus the correction converges normally in the open region $\Re s > 0$ where the real parts of all of

$$2s + v,\ 2s + w,\ 2s + 2v,\ 2s + 2w,\ 2s + v + w,$$
$$3s + 2v + w,\ 3s + v + 2w,\ 3s + v + w$$

exceed one. In particular, for $\Re s \geq 1/2 + \eta$ and $\Re v, \Re w \geq 0$, its terms differ from one by $O_\eta((Np)^{-1-2\eta})$. Deleted primes only remove factors. Finite correction factors may vanish; no zero-free product is asserted. The displayed factorization is first an absolute-domain identity and is used meromorphically only where the functions have been continued. At $v = w = 0$ it simplifies to $L^T(s, \Xi)/L^T(2s, \Xi^2)$, a special slice cancellation which does not remove the general central reciprocal. No finite-height inversion is declared exact without truncation errors; no new contour, reciprocal family moment or analytic saving is proved. Most importantly, this fixed-z coefficient series is not the actual q-dependent joint weight in (50).

37.5 The fixed-profile remaining-sector diagnostic

Use the fixed smoothstep $s(t) = e^{-1/t}/(e^{-1/t} + e^{-1/(1-t)})$ for $0 < t < 1$, extended by zero and one at the two sides, and set $W(y) = s(8(y-1))s(8(2-y))$. It is smooth, supported in $[1, 2]$, and identically one on $[9/8, 15/8]$, independently of D. Take $\nu = 1$ and S the primes above $2, 3$. At $D = 60000, U = 244, H = 3833$, select the actual block $A_1 = 37, A_2 = 7$, literal unit one and primary row generators 1 modulo 3. Complete enumeration gives 44 candidate pairs, with two coprimality and 22 capacity rejections, leaving 20 distinct nonprincipal rows. The first outside-S valuation is one in each row.

Choose $\mathfrak{p}_p = (p, \omega - \lambda_p)$ for

$$(p, \lambda_p) \in \{(37, 10), (43, 6), (61, 13), (307, 17), (313, 98)\}.$$

Their product q has exactly two legal annular column orientations: 97051/96091 and its reversal. Direct small- and long-divisor assignments give $r_U = (0, 2)$, $\mu = (-1, 1)$, $C_U = (0, 2)$ and $A_U = (1, 1)$. Both profile values are exactly one. Twelve orientation/row evaluations are raw zeros. Exact residue-field exponentiation yields

$$\Gamma_1 = -4 - 2\omega, \qquad \Gamma_2 = -2 + 2\omega, \qquad \Gamma_{\mathrm{av}} = -3.$$

Here Γ_{av} is the orientation average. With $a^{AA} = (-1, -1)$ and $\mu(q) = -1$,

$$S_{1,q}^{AA} = -6/60000 = -1/10000, \qquad \sum_r a_r(\Gamma_r - \Gamma_{\mathrm{av}}) = 0.$$

The centered norm is $\sum_r |\Gamma_r - \Gamma_{\mathrm{av}}|^2 = 6$ and $\sum_r |a_r|^2 = 2$, whereas $|\sum_r a_r\Gamma_r|^2 = 36 > 12$. A variance-only Cauchy bound therefore fails even with nonzero variance. It is not a negative full row energy or an asymptotic counterexample.

The corresponding *selected two-column* Gram matrix has diagonal $14, 20$ and the displayed conjugate off-diagonal entries. Its eigenvalues are $17 \pm \sqrt{21}$ and its A quadratic form is 28 before division by D. Removing the diagonal for coprimality gives eigenvalues $\pm 2\sqrt{3}$ and the A form -6. Other original columns and q values have not been suppressed in a full-family claim: this display is a selected-q contribution with its complete stated row block. All finite data and scalar identities have independent exact checks; no finite spectrum or learned label implies a determinant or GRH theorem. The next estimate must retain the signed mean, reciprocal burden and actual coupled q weights, closing the still-open 5/12 gap.

38 The signed divisor mean and a complete-column laboratory

The preceding signed-slice estimate loses the sign of the common ideal divisor. We now retain that variable in an exact analytic reduction and in a complete finite computation. Neither supplies the missing 5/12 saving at $H = D^{3/4}$. The computations below use a new fixed profile; they do not fill in the uncomputed columns of the earlier broad-plateau profile.

Let $\mathcal{C}$ be original squarefree good ideal columns in $[D, 2D]$, and let $\mathcal{R}$ be an actual finite literal row block, with nonnegative multiplicities w_u. All shared-prime zeros of the original $\chi_m(u)$ remain. For arbitrary complex coefficient vectors define the linear-first form

$$DQ_1(x, y) = \sum_{u \in \mathcal{R}} w_u \sum_{\substack{m,n \in \mathcal{C} \\ (m,n)=1}} x_m \overline{y_n} \chi_m(u) \overline{\chi_n(u)}. \tag{51}$$

The original component is obtained with $x_m = X_U(m)\nu(m)W(Nm/D)$ and the corresponding y_m. On the nonprincipal row blocks used below the original residue term vanishes. Equation (51) is a raw coprime form on a generic block; principal rows would still require their separate residue assembly.

For every ideal divisor of an active column put

$$F_d(x; u) = \sum_{m \in \mathcal{C},\ d|m} x_m \chi_m(u), \qquad Z_d(x, y) = \sum_u w_u F_d(x; u) \overline{F_d(y; u)}.$$

The exact identity is

$$DQ_1(x, y) = \sum_d \mu(d) Z_d(x, y). \tag{52}$$

There is no extra $\mu(q)$: the vectors already contain the original coefficients. Full divisors $d = m$ are essential to cancel self-pairs. Divisors and squarefreeness refer to ideals, including distinct ideals with the same rational norm.

Proposition 18 (A weighted divisor-incidence lift)**.** *Suppose $\lambda_d \geq 0$ is an arbitrary-coefficient bound*

$$\sum_u w_u |F_d(x; u)|^2 \leq \lambda_d \sum_{m \in \mathcal{C},\ d|m} |x_m|^2.$$

With $B(m) = \sum_{d|m} \lambda_d$ and $\|x\|_B^2 = \sum_m B(m)|x_m|^2$, one has

$$|DQ_1(x, y)| \leq \|x\|_B \|y\|_B. \tag{53}$$

For a finite literal phase table one may take the exactly computed

$$\lambda_d = E_d := \sum_u w_u \#\{m \in \mathcal{C} : d \mid m,\ \chi_m(u) \neq 0\}.$$

This last choice is a Frobenius certificate, not a new analytic estimate. For $x = t + e$ it gives the finite perturbation bound

$$|Q_1(x,x) - Q_1(t,t)| \le D^{-1}(2\|t\|_B\|e\|_B + \|e\|_B^2). \tag{54}$$

Proof. Insert ideal-divisor inversion in (51). Apply row Cauchy to each Z_d and weighted Cauchy to the resulting sum over d. Then $\sum_d \lambda_d\|x_d\|_2^2 = \|x\|_B^2$. The squared Frobenius norm of the actual F_d matrix is exactly E_d. Expand $Q_1(t+e,t+e) - Q_1(t,t)$ to obtain the last assertion. Positivity of the coprime-truncated matrix was never assumed. □

Using an accepted positive-row sieve bound for λ_d only recovers the old exponent up to ideal-divisor factors. It does not estimate the signed sum before the triangle in (52).

There is also a genuinely signed scalar expansion. Fix squarefree residual ideals a, b, which may overlap, and an actual row u. Keep the full deleted set $\mathcal{T} = S \cup \operatorname{supp}(abu)$ and set

$$\mathcal{Z}_{U;a,b,\mathcal{T}}(s) = \sum_{\substack{d \text{ squarefree} \\ (d,\mathcal{T})=1}} \frac{\mu(d)A_U(da)A_U(db)}{(Nd)^s}, \qquad \Re s > 1.$$

For four positive cutoff variables put $P_p = 1-(Np)^{-v_1}-(Np)^{-v_2}$ and $Q_p = 1-(Np)^{-w_1}-(Np)^{-w_2}$. The common-divisor Euler factor is $1-(Np)^{-s}P_pQ_p$. Its first-order factorization has four numerator and five denominator zeta factors:

$$\prod_{p\notin\mathcal{T}}(1-(Np)^{-s}P_pQ_p) = \frac{\prod_{i=1}^2 \zeta_K^{\mathcal{T}}(s+v_i)\zeta_K^{\mathcal{T}}(s+w_i)}{\zeta_K^{\mathcal{T}}(s)\prod_{i,j=1}^2 \zeta_K^{\mathcal{T}}(s+v_i+w_j)}\,\mathcal{H}_{\mathcal{T}}(s;v,w). \tag{55}$$

The correction is normally convergent at $\Re s \ge \frac12+\eta$, $\Re v_i, \Re w_j \ge 0$, for each fixed $\eta > 0$. This says nothing about zeros or residues of the zeta ratio. Finite-height Perron kernels, with their explicit endpoint/error costs, recover the cutoffs by successive limits.

A direct finite cutoff expansion gives more explicit information:

$$\mathcal{Z}_{U;a,b,\mathcal{T}}(s) = \frac{\mathcal{N}_{U;a,b,\mathcal{T}}(s)}{\zeta_K^{\mathcal{T}}(s)}. \tag{56}$$

The numerator is a finite sum over eight nonzero joint short-divisor states, holomorphic on $\Re s > 0$. Uniformly for $\sigma \ge \sigma_0 > 0$,

$$|\mathcal{N}_{U;a,b,\mathcal{T}}(\sigma+i\tau)| \ll_{K,\sigma_0,\epsilon} (Na\,Nb)^{\epsilon}U^{4\max(1-\sigma,0)+\epsilon}.$$

The costs near $\sigma = 1/2$ persist. The remaining long state has the Möbius sign, but the eight finite short states cannot be omitted. A fixed-residual reciprocal zeta factor also does not prove an uncancelled pole in the complete coherent sum.

Exact complete-column reference. Choose once $\delta = 1/240000$ and $W_\delta(y) = s((y-1)/\delta)s((2-y)/\delta)$, where s is the smooth step from zero to one. For the finite grid $D \le 60000$ every strict integer norm node has $W_\delta = 1$; endpoints vanish. Its derivatives cost $O_k(\delta^{-k})$. The three cases have different declared row blocks, not a common asymptotic scaling experiment. They use principal ν, S above $2, 3$, literal class one and nonprincipal actual rows.

D	columns	divisors	rows	S_1^{AA}
144	41	53	5	$-19/36$
576	164	212	11	$-221/288$
60000	16953	21750	20	$-5119/5000$

In the last case the exact even/odd Möbius masses are 543416 and 604844, giving numerator -61428 before division by D. All 339060 raw column–row values, including 47908 zeros, and all nine cross components are retained. In particular 1361 columns have nonsquarefree rational norm while remaining squarefree as ideals. The signed-divisor inspector exposes the actual states and dimensions. These finite negative slices are neither whole-family asymptotic counterexamples nor a positive arithmetic determinant or GRH result.

39 Proofs and limits of the signed-divisor development

39.1 Actual divisor incidence and a finite error certificate

The identity $\mathbf{1}_{(m,n)=1} = \sum_{d|m,\ d|n} \mu(d)$ is an identity of ideal divisors. Expanding the finite sums proves (52). When $m = da$ is squarefree, $(d, a) = 1$. If $(d, u) = 1$, multiplication by $\chi_d(u)$ is a rowwise unit phase; if $(d, u) > 1$ it is the original zero. For original $x_m = X_U(m)\nu(m)W(Nm/D)$, precisely $F_d(x; u) = \nu(d)\chi_d(u)T_d^X(u)$ on $(d, u) = 1$, and it is zero otherwise. Thus their row norms agree on the actual restricted support. For general vectors the coefficient x_{da} remains unfactored. No identity $X_U(da) = X_U(d)X_U(a)$ is used. Residual a, b may overlap.

Write $\|x_d\|_2^2 = \sum_{d|m} |x_m|^2$. The hypotheses of Proposition 18 give

$$\begin{aligned} |DQ_1(x, y)| &\le \sum_d |Z_d(x, y)| \le \sum_d \lambda_d \|x_d\|_2 \|y_d\|_2 \\ &\le \left(\sum_d \lambda_d \|x_d\|_2^2 \right)^{1/2} \left(\sum_d \lambda_d \|y_d\|_2^2 \right)^{1/2} = \|x\|_B \|y\|_B. \end{aligned}$$

This is a divisor-incidence lift of positive-row bounds to an indefinite coprime form. It is not an inference from positivity of a larger Gram matrix. In particular $B(m) \le \lambda_{\max} \tau_{\text{ideal}}(m)$, with $\tau_{\text{ideal}}(m) = 2^{\omega(m)}$ on squarefree ideals. The subpolynomial divisor bound recovers the existing exponent; this triangle does not supply the desired signed cancellation.

For the original unweighted two-pivot block ($w_u = 1$), the accepted arbitrary-coefficient inputs at $L = D/Nd \ge 1$ allow

$$\lambda_d \ll_\epsilon (HD)^\epsilon \min \left\{ R + \frac{LR}{M} + \frac{L^{2/3}R}{M^{1/3}},\ L + R^2 \right\}, \qquad R = A_1 A_2, \quad M = \max(A_1, A_2).$$

Their constants are classical-input constants, not computed numbers. Any removal of a d-row mask occurs only inside the positive norm. For $1/2 \le L < 1$, $Na \in [L, 2L]$ permits at most the unit ideal; its operator norm is bounded directly by the actual row measure. For an empty subsequence take $\lambda_d = 0$.

For the fully computable choice $\lambda_d = E_d$, the matrix entries are $\sqrt{w_u}\chi_m(u)$ on d-divisible columns. The squared Frobenius norm is precisely the stated raw nonzero-entry count. It is valid for arbitrary coefficient vectors, not just the saved trajectory. No unknown sieve constant enters that finite certificate. For approximations t_x, t_y with errors $e_x = x - t_x, e_y = y - t_y$,

$$|Q_1(x, y) - Q_1(t_x, t_y)| \le D^{-1}(\|e_x\|_B \|y\|_B + \|t_x\|_B \|e_y\|_B).$$

The AA specialization follows by expanding three terms. To verify it without floating square roots, put $T = \|t\|_B^2, E = \|e\|_B^2$ and $L = D|Q_1(x, x) - Q_1(t, t)| - E$. If $L > 0$, the certificate is the equivalent real inequality $L^2 \le 4TE$; if $L \le 0$ it is immediate. In the saved exact $\mathbf{Z}[\omega]$ computations with rational weights, these are integer/rational checks rather than floating-root tests. These are finite coefficient-error statements. They do not prove a varying-family remainder estimate, spectral positivity or a determinant comparison.

39.2 A signed common-divisor Euler product

From here restore ALL original annular squarefree good columns, not an arbitrary subset from the vector lemma, and use the actual unweighted raw row block of Section 36. On the stated nonprincipal blocks it is the corresponding component of the residue-subtracted moment. On a block containing principal rows the raw identity remains valid, but the original principal correction is additional and is not discarded. Expand the exact d identity while keeping its sign. For fixed a, b, u the inner sum is

$$\sum_{\substack{d \text{ squarefree} \\ (d,Sabu)=1}} \mu(d) A_U(da) A_U(db) W(Nd\,Na/D)\overline{W(Nd\,Nb/D)}. \tag{57}$$

The exact outer coefficient is $\nu(a)\overline{\nu(b)}\chi_a(u)\overline{\chi_b(u)}$. The residual a, b are individually squarefree but need not be coprime. If their quotient character loses a common prime, that prime remains in $\mathcal{T} = S \cup \operatorname{supp}(abu)$ and in every original zero. The common d character cancels only to $\mathbf{1}_{(d,u)=1}$.

For $\Re s > 1$ the series $\mathcal{Z}$ is absolutely convergent, with coefficient majorant $3^{\omega(a)+\omega(b)}9^{\omega(d)}$, uniformly in U. Use four positive Perron variables. Write $\mathcal{E}_{\mathcal{T}}(s; v, w) = \prod_{p \notin \mathcal{T}}(1-(Np)^{-s}P_pQ_p)$. The fixed residual prefactor is $P_a(v)Q_b(w) = \prod_{p|a} P_p(v) \prod_{p|b} Q_p(w)$; a common prime of a, b contributes both factors. The common d prime has nine joint cutoff positions and the sign $\mu(p) = -1$, giving local factor $1 - zPQ$ with $z = (Np)^{-s}$, $\alpha_i = (Np)^{-v_i}$ and $\beta_j = (Np)^{-w_j}$. Thus the exact correction in (55) is

$$H_p = \frac{(1 - zPQ)\prod_{i=1}^{2}(1 - z\alpha_i)\prod_{j=1}^{2}(1 - z\beta_j)}{(1 - z)\prod_{i,j=1}^{2}(1 - z\alpha_i\beta_j)}. \tag{58}$$

The linear coefficient of numerator and denominator is $-1 - \sum_{i,j} \alpha_i\beta_j$, fixing all nine zeta signs. At all four auxiliary variables zero the product is $1/\zeta_K^{\mathcal{T}}(s)$ and $H_p = 1$. At $v_2 = w_2 = 0$ it is $1/\zeta_K^{\mathcal{T}}(s + v_1 + w_1)$, again with $H_p = 1$. These are generating-function checks, not finite-cutoff identifications.

For $\Re s \geq 1/2 + \eta$, $\Re v_i, \Re w_j \geq 0$, the degree-five numerator and denominator of (58) have equal constant and linear coefficients. The sums of absolute coefficient majorants in degrees two through five are at most 146 and 26. Since $|PQ| \leq 9$ and $|\alpha_i|, |\beta_j| \leq 1$,

$$|H_p - 1| \leq C_\eta (Np)^{-1-2\eta}, \qquad C_\eta = \frac{172}{(1 - 7^{-1/2-\eta})^5}. \tag{59}$$

Indeed the five denominator factors have modulus at least $1 - |z|$; good prime ideal norms are at least seven. Ideal counting proves normal convergence, uniform in deleted primes and vertical phases, and

$$|\mathcal{H}_{\mathcal{T}}| \leq \exp\left(C_\eta \sum_p (Np)^{-1-2\eta}\right).$$

Local correction factors can vanish; no inverse-correction bound is asserted. Normal convergence controls neither the four numerator nor five reciprocal zeta factors in the critical region. Their poles, zeros and crossed residues cannot be suppressed.

39.3 Finite Perron heights and the exact eight-state numerator

Set $V = \lfloor U \rfloor + 1/2$ and $\Delta_V = \min_{n \geq 1} |\log(V/n)| \geq 1/(2V+1)$. For four finite vertical heights T_j on positive real lines c_j, let $I_T(s)$ be the fourfold integral of $P_aQ_b\mathcal{E}_{\mathcal{T}}(s; v, w)$ against $\prod_j V^{z_j}\, dz_j/(2\pi i z_j)$.

In the initial half-plane $\sigma = \Re s > 1$, finite-height Fubini is justified by the absolute placement majorant. Put

$$A_{\mathrm{abs}}(\sigma, c) = \prod_{p|a}(1 + q^{-c_1} + q^{-c_2}) \prod_{p|b}(1 + q^{-c_3} + q^{-c_4}) \\ \cdot \prod_{p \notin \mathcal{T}} \{1 + q^{-\sigma}(1 + q^{-c_1} + q^{-c_2})(1 + q^{-c_3} + q^{-c_4})\}, \quad q = Np.$$

Then

$$|I_T(s) - \mathcal{Z}_{U;a,b,\mathcal{T}}(s)| \le V^{\sum c_j} A_{\mathrm{abs}}(\sigma, c) \left\{ \prod_{j=1}^{4} \left(1 + \frac{2}{\pi T_j \Delta_V}\right) - 1 \right\}, \tag{60}$$

and $A_{\mathrm{abs}} \le 3^{\omega(a)+\omega(b)} \zeta_K(\sigma)^9$. For one short integer norm n, the indicator is at most $(V/n)^c$, and the truncated Perron error is at most $2(V/n)^c/(\pi T \Delta_V)$. Expand the difference of four products and sum every nonempty error subset to prove (60). Consequently successive Perron limits recover the exact cutoffs. The half-integer gap also gives a height-uniform $C_U(V/n)^c$ bound, allowing dominated convergence in the initial absolutely convergent series. Four infinite Perron integrals are not declared absolutely convergent or interchanged silently with Mellin inversion.

There is an entirely finite alternative. Partition $a = a_0 a_1 a_2$ and $b = b_0 b_1 b_2$, internally coprime in each column. Cross-column overlaps remain allowed. A common d prime occupies one state $(i, j) \in \{0, 1, 2\}^2$. Separate its $(0, 0)$ part e from eight pairwise coprime squarefree atomic ideals h_{ij} avoiding $\mathcal{T}$, and let h be their product. Define

$$h_i^L = \prod_{j=0}^{2} h_{ij} \ (i = 1, 2), \qquad h_j^R = \prod_{i=0}^{2} h_{ij} \ (j = 1, 2).$$

An atomic h_{ij} with both indices nonzero belongs to one short factor in each column. These four short factors must not be forced globally disjoint. Let $\mathcal{A}_U$ comprise precisely these tuples satisfying $N(a_i h_i^L) \le U$ and $N(b_j h_j^R) \le U$ for $i, j = 1, 2$. The numerator in (56) is exactly

$$\mathcal{N}_{U;a,b,\mathcal{T}}(s) = \sum_{\mathcal{A}_U} \frac{\mu(a_1 a_2)\mu(b_1 b_2)\mu(h_{11}h_{12}h_{21}h_{22})}{(Nh)^s} \prod_{p|h}(1 - (Np)^{-s})^{-1}. \tag{61}$$

At a common prime the sign is $(-1)^{1+\mathbf{1}_{i>0}+\mathbf{1}_{j>0}}$. It is negative for $(0, 0)$, positive for the four single-short states, and negative for the four double-short states. Summing the remaining squarefree e, coprime to $\mathcal{T}h$, gives $1/\zeta_K^{\mathcal{T}}(s)$ times the local product in (61). This proves the exact identity, including its sign.

Multiplying the four cutoff inequalities yields $Nh \le U^4$; double states are counted twice in that product. Hence the numerator is finite and holomorphic on $\Re s > 0$. At $U = 1$ it equals one: this boundary is not a high-window counterexample or a global pole obstruction. For $\sigma \ge \sigma_0 > 0$ and every $\epsilon > 0$,

$$|\mathcal{N}_{U;a,b,\mathcal{T}}(\sigma + i\tau)| \ll_{K,\sigma_0,\epsilon} (Na\, Nb)^{\epsilon} U^{4\max(1-\sigma,0)+\epsilon}. \tag{62}$$

There are at most $3^{\omega(a)+\omega(b)}$ residual partitions and $8^{\omega(h)}$ atomic placements at fixed h. The finite local product is $\ll_{\sigma_0,\epsilon} (Nh)^{\epsilon}$. Apply the ideal-divisor bound and ideal counting up to U^4, using smaller internal exponents; a logarithm at the transition is absorbed in U^{ϵ}. Uniformity as $\sigma_0 \to 0$ is not claimed. For $\sigma = 1/2 + \eta$ with $0 < \eta < 1/2$, the old-type cutoff cost is $U^{2-4\eta+\epsilon}$, not a new saving.

39.4 A uniform common-profile Mellin embedding of the actual mean

For residual a, b put $M = \max(Na, Nb)$ and $\varrho = \min(Na, Nb)/M$. At $\varrho \le 1/2$ the product of the two profile values vanishes, including equality because of endpoint vanishing. Otherwise put

$$\Phi_\varrho(t) = W(t)\overline{W(\varrho t)}$$

when $Na \ge Nb$, and use its conjugate for the reversed orientation. It is supported in $[1/\varrho, 2] \subset [1, 2]$. With $M_k(W) = \max_{j \le k} \|(t\, d/dt)^j W\|_\infty$,

$$|\widehat{\Phi}_\varrho(\sigma + i\tau)| \le C_{I,k} M_k(W)^2 (1 + |\tau|)^{-k} \tag{63}$$

uniformly for σ in any fixed compact strip I and all $\varrho \in [1/2, 1]$. Leibniz and log-coordinate integration by parts prove this bound; no ratio-dependent change of W is made. The transform is entire. For $c > 1$, ordinary Mellin inversion after the finite numerator identity gives the exact scalar (57) as

$$\frac{1}{2\pi i} \int_{\Re s = c} \widehat{\Phi}_\varrho(s) (D/M)^s \frac{\mathcal{N}_{U;a,b,\mathcal{T}}(s)}{\zeta_K^{\mathcal{T}}(s)}\, ds. \tag{64}$$

This integral is vertically absolutely convergent in its initial half-plane. The original profile forces $M \le 2D$, so the residual a, b and actual row sums are finite.

Keep the full local deletion factor

$$\frac{1}{\zeta_K^{\mathcal{T}}(s)} = \frac{1}{\zeta_K(s)} \prod_{p \in \mathcal{T}} (1 - (Np)^{-s})^{-1}.$$

At fixed $\sigma_0 > 0$ it costs $\ll_{S,\sigma_0,\epsilon} (N \operatorname{rad}(abu))^\epsilon$, hence at most $(HD^2)^\epsilon$ on actual support. This estimates the finite mask dependence, not the central reciprocal. Multiplying (64) by every original outer coefficient and summing the actual rows and residuals gives

$$DS_1^{AA} = \frac{1}{2\pi i} \int_{\Re s = c} \frac{D^s \mathcal{A}_{D,H,U,W}(s)}{\zeta_K(s)}\, ds. \tag{65}$$

Explicitly, with each profile orientation retained,

$$\mathcal{A}_{D,H,U,W}(s) = \sum_{u \in \mathcal{R}} \sum_{\substack{a,b \text{ squarefree,good} \\ Na, Nb \le 2D,\ \varrho > 1/2}} \nu(a)\overline{\nu(b)}\chi_a(u)\overline{\chi_b(u)}\, M^{-s}\, \widehat{\Phi}_\varrho(s) \cdot \mathcal{N}_{U;a,b,\mathcal{T}}(s) \prod_{p \in \mathcal{T}} (1 - (Np)^{-s})^{-1}.$$

This is a *finite coherent sum*, not an independent marginal moment or a positive sum. All raw zeros remain in its outer coefficients. For fixed finite D, H, U it is holomorphic on $\Re s > 0$ and rapidly decreasing on fixed vertical strips there, with constants depending on those finite parameters. No uniform target estimate for it is supplied. This is an exact signed embedding, not a zero-location theorem.

39.5 A nonzero short-state correction and the unproved step

An actual annular control uses good prime ideals of norms $7, 13, 19, 31, 43$ with primary generators $(1, 3), (4, 3), (-2, 3), (1, 6), (7, 6)$ in the $1, \omega$ basis. Take $D = 1500, U = 25$, $d = p_7$, $a = p_{13}p_{19}$, $b = p_{13}p_{31}$ and the nonprincipal row $u = p_{43}$ in the pure block $A_1 = 37, A_2 = 1, H = 241$, literal

class one. The full raw supports are disjoint from u, and $N(da) = 1729, N(db) = 2821$ lie in the annulus. A single fixed smooth profile with plateau $[1.1, 1.9]$ equals one at both nodes. Direct assignments give $A_U(a) = A_U(b) = -1$, $A_U(da) = 1$, $A_U(db) = -1$. The actual signed d coefficient is $+1$. The $(0,0)$ state alone would give -1; states $(1,0)$ and $(2,0)$ each contribute $+1$. Dropping all short states therefore misses a nonzero $e = 1$ correction. This illustrates the new eight-state identity, not a new discovery of coefficient nonmultiplicativity.

The original pair has gcd p_7p_{13}. Its contributions across $d = 1, p_7, p_{13}, p_7p_{13}$ cancel exactly. Thus a fixed-residual reciprocal does not prove an uncancelled pole or an obstruction in the complete coherent mean. The missing step is to estimate that full mean by $DH(HD)^\epsilon$, retaining fixed-profile/seminorm dependence, raw masks and conductor/ray/capacity restrictions. Scalar Mertens cancellation in the long state, a normal correction product, or an absolute cutoff numerator bound does not prove it. Principal and short assembly, independent positive determinant/zero-location bridges and separate Dirichlet, Hecke, Dedekind, elliptic and Artin obligations remain open. No holomorphy or desired zero location is assumed to close them.

39.6 Exact arithmetic and source-bound computational scope

The evaluator streams each actual row and accumulates original $x_m\chi_m(u)$ into every ideal-divisor bucket $d \mid m$. It then forms all nine cross products for A_U, C_U, μ and sums with $\mu(d)$. Write $n_{\mathrm{col}} = |\mathcal{C}|$, $I = \sum_m \tau_{\mathrm{ideal}}(m)$, K for the distinct divisors and r for the listed rows. This bucket stage costs $O(r(I + K))$ arithmetic operations with a fixed three channels. Its streaming storage is $O(K)$ plus incidences and either a streamed row or separately retained phase table. Character construction, factorization and integer bit lengths are additional costs, not declared free. For coefficients bounded by C, bucket coordinates are $O(n_{\mathrm{col}}C)$, products $O(n_{\mathrm{col}}^2C^2)$ and accumulated values $O(rn_{\mathrm{col}}^2C^2)$ for unit row weights. For compressed integer multiplicities replace r in this bit-size bound by $\sum_u w_u$ and account for weight multiplication.

For $D = 60000$, the actual phase table has shape 20×16953; the incidence has 73420 nonzeros over 21750 divisor buckets, and F_d has $20 \times 21750 \times 3$ complex states. There are 271688516 ordered coprime column pairs. The naive all-pair row loop would visit 5748084180 pairs and was *not executed* at this size. This is an operation count, not an unexecuted timing benchmark. Independent direct pair evaluation covers the two complete small families; the large family is checked linearly from literal field arithmetic and divisor-poset inversion.

The exact large results also include

$$S_1^{CC} = \frac{661}{7500}, \qquad S_1^{C\mu} = \frac{13931}{30000}, \qquad S_1^{\mu\mu} = -\frac{229}{1250}.$$

For the ordered vectors (A_U, C_U, μ) the full pairing table is

$$\frac{1}{30000}\begin{pmatrix} -30714 & -11287 & 19427 \\ -11287 & 2644 & 13931 \\ 19427 & 13931 & -5496 \end{pmatrix}.$$

This is the table of three original coefficient vectors, not a spectral carrier for GRH. The conjugate μC entry and the AA identity retain their actual linear-first convention. All three saved computations use source v1. A separate v2 helper fixes its unused $N = U = 1$ high-window assertion; every one of the 17158 executed columns has $N > U$, so no accepted row/table or source snapshot was silently replaced. The fixed-profile flat-node implementation refuses requests above its stated $D \le 60000$ certificate. Neither these three different finite blocks nor their measured runtimes imply an asymptotic saving.

40 A target-sized budget for rough-prime masks

The actual signed divisor mean admits a useful quantitative simplification: the rough-prime coprimality mask can be restored with an explicit correction bound. The remaining main term still retains the Möbius sign and the coherent row mean; it is not estimated by this result.

Work over $\mathbb{Z}[\omega]$. Fix a finite S containing the primes above $2, 3$, a unitary multiplicative ν on good ideals, and one common complex profile W supported on $[1, 2]$, smooth and zero at both endpoints. Put $B = \|W\|_\infty$ and assume

$$1 \le U, \qquad U^2 \le D, \qquad 2D \le 7U^2, \qquad H \ge 1. \tag{66}$$

Let $\mathcal{R}$ be the actual finite family of nonzero literal rows, with original weights $w_u \ge 0$, $1 \le Nu \le H$, and $R_w = \sum_u w_u$. All original internal squarefree/coprime conditions, unit and S-ray classes, capacity and sixth-factor masks remain in this family. Write S_1^{AA} for the raw form (51) with $x_m = A_U(m)\nu(m)W(Nm/D)$, where

$$A_U(g) = \sum_{\substack{xy \mid g,\ (x,y)=1 \\ Nx \le U,\ Ny \le U}} \mu(x)\mu(y). \tag{67}$$

Here the columns in (51) are the COMPLETE original good squarefree annulus $[D, 2D]$, with the common W as its norm cutoff and the original full-good-ideal/ν, S formulation. No selected-column indicator or additional column-ray filter is allowed in this reduction; all actual row unit/ray/capacity filters remain. Principal-row residues, if present, require their separate assembly. For the intended $A_1 > 1$ blocks the defining conductor of ν is supported in S, and the rows are nonprincipal.

Let f range over good squarefree U-smooth ideals, put $F = Nf$, and set

$$T_f(t; u) = \sum_{\substack{a \text{ good squarefree} \\ (a,f)=1}} A_U(fa)\nu(a)\chi_a(u)W(FtNa/D). \tag{68}$$

Here $\chi_a(u) = 0$ on $(a, u) > 1$. For a prime ideal p, let $T_f^{(p)}(t; u)$ impose the additional condition $(a, p) = 1$. The smooth factor f is distinct from the original column-pair gcd sector $h = 1$. Define the following *unnormalized* expressions:

$$K_0 = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f)|T_f(1; u)|^2, \tag{69}$$

$$K_T = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f) \sum_{\substack{p \notin S,\ Np > U \\ p \nmid u}} |T_f^{(p)}(Np; u)|^2, \tag{70}$$

$$K_S = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f) \sum_{p \notin S,\ Np > U} |T_f(Np; u)|^2. \tag{71}$$

Throughout, p denotes a prime ideal. None of these expressions is declared positive, since the $\mu(f)$ signs remain.

The exact decomposition of the common ideal divisor $d = fe$ has at most one rough prime when the residual pair is nonunit. The unit residual pair also permits two distinct rough primes, necessarily with $f = 1$. Its exact correction is

$$C_2 = \sum_{\substack{\{p,q\} \text{ unordered, } p \ne q \\ p,q \notin S,\ Np, Nq > U}} |W(NpNq/D)|^2 \sum_{\substack{u \in \mathcal{R} \\ (u,pq)=1}} w_u. \tag{72}$$

Conjugate primes of equal rational norm are included. One has

$$DS_1^{AA} = K_0 - K_T + C_2. \tag{73}$$

In K_S, terms with $p \mid a$ or $p \mid u$ are explicitly auxiliary: fpa can fail to be squarefree. They use $A_U(fa)$ as written, not a new original column or a substitution for $A_U(fpa)$.

Theorem 19 (Explicit rough-prime mask budget)**.** *Put* $\kappa_U(u) = \#\{p \notin S : p \mid u,\ Np > U\}$ *and* $\Delta = (K_S - K_T)/D$. *Under* (66)*,*

$$|\Delta| \le B^2 \left(229376 \sum_u w_u \kappa_U(u) + 2048 R_w \right), \tag{74}$$

$$0 \le C_2/D \le 1792 B^2 R_w. \tag{75}$$

Consequently

$$S_1^{AA} = \frac{K_0 - K_S}{D} + E, \tag{76}$$

$$|E| \le B^2 \left(229376 \sum_u w_u \kappa_U(u) + 3840 R_w \right). \tag{77}$$

If $H \le D^{3/4}$*, then every row has* $\kappa_U(u) \le 1$*, and in particular*

$$|E| \le 233216 B^2 R_w \le 2^{18} B^2 R_w. \tag{78}$$

Only the rough-prime deletion has been restored; the smooth-divisor mask, literal row family and original coefficients remain. The explicit bound uses the exact restricted identity

$$A_U(g) = \mu(g)(2\mathbf{1}_{Ng \le U} - 1) \quad (Ng \le 7U,\ g \text{ good squarefree}).$$

Every active term of $T_f(Np; u)$ has $N(fa) < 7U$, hence $|A_U(fa)| = 1$ there. This is not a shrinkage or proxy for the whole original coefficient family. Only absolute correction bounds use it; the signed coefficients in the clean main remain unchanged.

For general H, one has $\kappa_U(u) \le 2\log(2H)/\log(2U)$, also at $U = 1$. For an actual two-pivot block with $R_w \ll A_1 A_2 \le H$ and $H \le D^{3/4}$, (78) gives an $O(B^2 H)$ error with no divisor-power or profile-derivative loss. The remaining target problem for this raw A–A slice is the clean signed main $(K_0 - K_S)/D$. It has not been bounded here; the 7/6 envelope and missing 5/12 improvement for the complete original remainder are unchanged. The argument neither deletes a coherent mean nor treats the coprime-truncated form as positive.

Three exact finite controls. These use principal ν, S above $2, 3$, literal unit class one and complete original good squarefree annuli. The first two row variables are squarefree, good and coprime and may include the unit ideal; all other factors equal one.

D	U	H	A_1	A_2	columns	rows
200	10	53	13	1	52	5
512	16	107	19	1	139	7
900	30	384	7	7	250	2

Put $\eta_0(t) = \exp(-1/t)$ for $t > 0$ and $\eta_0(t) = 0$ for $t \le 0$, and define

$$s(t) = \frac{\eta_0(t)}{\eta_0(t) + \eta_0(1-t)}.$$

Choose once $\delta = 1/4096$ and $W_\delta(t) = s((t-1)/\delta)s((2-t)/\delta)$. For integer $D \le 1024$, strict integer norm nodes have weight one and endpoints have weight zero; its derivatives cost $O_k(4096^k)$. The following numbers are RAW, before division by D:

D	K_0	K_S	K_T	E_{row}	E_{res}	C_2	DS_1^{AA}
200	339	235	209	20	6	21	151
512	366	781	711	48	22	89	−256
900	436	410	366	0	44	28	98

In every case $K_S - K_T = E_{\mathrm{row}} + E_{\mathrm{res}}$ and $DS_1^{AA} = K_0 - K_T + C_2$. The displayed nonnegative values of $E_{\mathrm{row}}, E_{\mathrm{res}}$ are finite observations; no general nonnegativity is assumed for either signed correction. Only C_2 is nonnegative by definition. The first two capacities are $\lfloor D^{3/4} \rfloor$. The third $H = 384$ is a declared noncritical fixture capacity, not $\lfloor 900^{3/4} \rfloor$; both of its rows have $\kappa_U = 0$. All ideal and raw-zero masks, unit residual states and explicitly auxiliary non-squarefree p^2 terms are retained. The exact control summary records the scopes and exact fractions after one division by D. These are finite identities, not a proof or numerical fit of the analytic bounds, and do not estimate the signed clean main.

41 Proof of the rough-prime mask budget

41.1 The common ideal divisor and the exact rough part

Here (51) uses the complete original good squarefree annulus and common W, without an additional selected-column or column-ray indicator. The actual row filters are unchanged. The identity $\mathbf{1}_{(m,n)=1} = \sum_{d \mid m,\ d \mid n} \mu(d)$ in (51) is an identity of ideal divisors. Write $m = da, n = db$. The residual ideals a, b are squarefree but may overlap; squarefreeness of the original columns requires $(d, ab) = 1$. On $(d, u) = 1$ the original common phases satisfy

$$|\nu(d)|^2 \chi_d(u)\overline{\chi_d(u)} = 1.$$

On $(d, u) > 1$ the product is the original raw zero. Hence the full deleted set is $\mathcal{T} = S \cup \mathrm{supp}(abu)$, including residual primes where $\chi_a\overline{\chi_b}$ might become imprimitive. With

$$F_{a,b}(x) = W(xNa/D)\overline{W(xNb/D)}, \qquad M = \max(Na, Nb), \quad \alpha = \min(Na, Nb),$$

the inner signed sum is

$$\sum_{\substack{d \ \mathrm{SF} \\ (d,\mathcal{T})=1}} \mu(d) A_U(da) A_U(db) F_{a,b}(Nd). \tag{79}$$

If its profile is nonzero, then

$$D/\alpha < Nd < 2D/M. \tag{80}$$

All endpoints have zero profile value. In particular the profile is identically zero if $\alpha \le M/2$. No independent marginal row family or changed profile has been substituted.

Factor $d = fe$ uniquely, with f squarefree U-smooth and e squarefree with every prime norm greater than U. Whenever ez is squarefree,

$$A_U(ez) = A_U(z). \tag{81}$$

Indeed neither short divisor in (67) can contain a prime of e; deleting e preserves the two cutoffs and both Möbius signs in a bijection of the original assignments. There is no factor $\mu(e)$ in (81). This invariance is specific to A_U; it does not hold unchanged for the other coefficient channels. The values $A_U(fa), A_U(fb)$ are not factorized.

If $M > 1$, goodness gives $M \geq 7$. From (66) and (80), $Nd \leq 2D/M \leq U^2$. Thus e is either 1 or one prime ideal, and (79) is exactly

$$\sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,\mathcal{T})=1}} \mu(f) A_U(fa) A_U(fb) \left(F_{a,b}(Nf) - \sum_{\substack{p \notin \mathcal{T} \\ Np > U}} F_{a,b}(NfNp) \right). \tag{82}$$

The rough part here belongs to the *common divisor* d; the full columns can have additional rough primes in a or b.

If $M = 1$, then $a = b = 1$. Three rough prime ideals would have norm product strictly greater than $7U^2 \geq 2D$: two contribute more than U^2, and the third good prime has norm at least 7. Two rough primes are possible, but then a nonunit f would similarly give $Nf\,Ne > 7U^2 \geq 2D$. Hence that case has $f = 1$ only. Its sign is $\mu(pq) = 1$ and its coefficient is $A_U(pq) = A_U(1) = 1$. Summing this case over the unchanged literal rows gives (72); summing all other cases of (82) over a, b gives the completed squares in (69)–(70). This proves (73). The unit ideal among residuals and conjugate split primes must both be retained. The condition $f = 1$ is not a termwise assertion that the original pair has gcd 1: the Möbius expansion cancels noncoprime original pairs across its divisor sectors.

41.2 An exact disjoint error identity

We restore only $(p, abuS) = 1$ to $p \notin S$. In particular the conditions $(f, abuS) = 1$ and every raw $\chi_a(u)$ remain in the two residual sums. The full deleted union has the disjoint partition

$$\mathbf{1}_{p|abu} = \mathbf{1}_{p|u} + \mathbf{1}_{p\nmid u}\mathbf{1}_{p|a \text{ or } p|b}. \tag{83}$$

Therefore $K_S - K_T = E_{\text{row}} + E_{\text{res}}$, where the errors are *unnormalized*:

$$E_{\text{row}} = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f) \sum_{\substack{p \notin S,\ Np > U \\ p|u}} |T_f(Np; u)|^2, \tag{84}$$

$$E_{\text{res}} = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f) \sum_{\substack{p \notin S,\ Np > U \\ p\nmid u}} \left(|T_f(Np; u)|^2 - |T_f^{(p)}(Np; u)|^2 \right). \tag{85}$$

The two errors are signed; their bounds below use absolute values, and neither is assumed nonnegative. At $p \mid u$, all residuals divisible by p already have raw character zero. This is consistent with the full square in (84); it does not make that square zero. These formulas keep the complete row/residual union without double counting.

Suppose a nonzero omitted residual has $p \mid a$, and write $a = pc$. Its profile requires $Nf\,(Np)^2 Nc < 2D$. If fc is nonunit, then

$$Nf\,(Np)^2 Nc > 7U^2 \geq 2D,$$

a contradiction. Thus necessarily $f = c = 1$ and $a = p$. This is a support statement about the explicitly auxiliary extension; the formal column $fpa = p^2$ is not an original squarefree column. With $P = Np$, set

$$z_p(u) = \nu(p)\chi_p(u) W(P^2/D), \qquad T_p(u) = T_1(P; u). \tag{86}$$

Here $A_U(p) = 1$, since $P > U$. The difference $T_f - T_f^{(p)}$ vanishes unless $f = 1$, when it equals z_p. Consequently

$$E_{\mathrm{res}} = \sum_u w_u \sum_{\substack{p \notin S,\ Np > U \\ p \nmid u}} \left(2\,\mathrm{Re}(T_p(u)\overline{z_p(u)}) - |z_p(u)|^2 \right). \tag{87}$$

The subtraction $-|z_p|^2$ is the exact intersection $p \mid a$ and $p \mid b$; omitting it would count that intersection twice. Complex profiles and all remaining literal phases are allowed. If $z_p \neq 0$, then

$$\sqrt{D} < P < \sqrt{2D}. \tag{88}$$

The condition $P > U$ is automatic here from $U^2 \leq D$.

41.3 Elementary ideal and coefficient estimates

We give all counting inputs; no prime distribution theorem or Mertens estimate is needed. The Eisenstein norm is Euclidean. After rounding both real coordinates of a complex quotient in the basis $1, \omega$, the error $s + t\omega$ has $|s|, |t| \leq 1/2$ and norm at most $3/4 < 1$. A least-norm nonzero element of any ideal therefore generates it by Euclidean division. Multiplication by $a + b\omega$ has determinant $a^2 - ab + b^2$, giving the principal ideal norm. Since $a^2 - ab + b^2 \geq (a^2 + b^2)/2$, counting all possible generators, with harmless overcounting by units, proves for $X \geq 1$ that

$$\#\{a : Na \leq X\} \leq (2\sqrt{2X} + 1)^2 < 16X. \tag{89}$$

An annulus $L < Na < 2L$ therefore contains at most $32L$ ideals when $2L > 1$. If $2L \leq 1$ its profile contribution is zero, including the unit endpoint when $2L = 1$. This proves the needed uniform bound even when $1/2 < L < 1$; the unit ideal must not be dropped in that range.

There is an exact coefficient identity on the restricted support:

$$A_U(g) = \mu(g)\big(2\mathbf{1}_{Ng \leq U} - 1\big) \quad (g \text{ good squarefree},\ Ng \leq 7U). \tag{90}$$

Every proper divisor of a nonunit good g has norm at most $Ng/7 \leq U$. If $Ng \leq U$, all ternary assignments are allowed and their sum is $\prod_{p \mid g}(1 - 1 - 1) = \mu(g)$. If $Ng > U$, only the assignments $(x, y) = (g, 1), (1, g)$ are omitted; each has sign $\mu(g)$, leaving $-\mu(g)$. The unit ideal gives $A_U(1) = 1$ and is included. No approximation or multiplicativity of the cutoff coefficient is used.

In an active term of (68), $P = Np > U$ gives

$$N(fa) < 2D/P < 2D/U \leq 7U.$$

Therefore $|A_U(fa)| = 1$ exactly there. This statement concerns only the rough-prime support, not the entire original coefficient family. With $L = D/(FP)$, the ideal count and $|\chi_a(u)| \leq 1$ now give

$$|T_f(P; u)| \leq 32B\,\frac{D}{FP}. \tag{91}$$

The same statement holds if the sum is empty, and covers $L < 1$ by the preceding endpoint discussion. A dyadic use of (89) gives

$$\sum_f \frac{1}{(Nf)^2} \leq \sum_{j \geq 0} 16\,2^{j+1} 2^{-2j} = 64, \tag{92}$$

where the left side may even include all ideals. The smooth squarefree restriction only decreases this bound.

41.4 The normalized row-mask budget

Apply triangle only to (84), use (91), and then (92). The raw square contributes D^2/P^2; dividing by D once gives

$$\frac{|E_{\mathrm{row}}|}{D} \le 65536B^2 \sum_u w_u \sum_{\substack{p\mid u,\ p\notin S\\ Np>U}} \frac{D}{(Np)^2}. \tag{93}$$

By (66), $D/(Np)^2 \le D/U^2 \le 7/2$. Let $\kappa_U(u)$ count the distinct good rough prime ideals dividing u. Each $P = Np$ is at least 7 and greater than U, so $P^2 > 7U \ge 2U$. The complete ideal norm of the literal row therefore gives

$$\kappa_U(u) \le \frac{2\log(Nu)}{\log(2U)} \le \frac{2\log(2H)}{\log(2U)}. \tag{94}$$

The denominator is positive also at $U = 1$; unit rows have $\kappa_U = 0$. All sixth-factor divisors remain in this bound, and no rational-norm gcd is substituted for an ideal mask. Using $D/(Np)^2 \le 7/2$ in (93) gives the sharper form

$$\frac{|E_{\mathrm{row}}|}{D} \le 229376B^2 \sum_u w_u \kappa_U(u). \tag{95}$$

The logarithmic bound (94) is available for arbitrary H.

41.5 The residual intersection budget

Use (87), $|z_p| \le B$, and (91) at $f = 1$. After the single division by D,

$$\frac{|E_{\mathrm{res}}|}{D} \le B^2 R_w \sum_{\sqrt{D}<Np<\sqrt{2D}} \left(\frac{64}{Np} + \frac{1}{D}\right). \tag{96}$$

Here absolute values bound only the correction; the exact identity (87) retains its original phases. Since $D \ge 1$ and $Np \le \sqrt{2D} \le 2D$, one has $1/D \le 2/(Np)$. Also

$$\sum_{\sqrt{D}<Np<\sqrt{2D}} \frac{1}{Np} \le 16\sqrt{2} \tag{97}$$

by (89). Thus

$$|E_{\mathrm{res}}|/D \le 1056\sqrt{2}\, B^2 R_w \le 2048B^2 R_w.$$

Together with (95), this proves (74).

41.6 The exact unit-residual correction and the clean main

In (72), each contributing pair has $P, Q > U$ and $PQ \le 2D$. Hence both prime norms are less than $2D/U$. There are at most $32D/U$ such prime ideals by (89), so the number of unordered distinct pairs is at most $512D^2/U^2$. Only inside this *nonnegative* expression enlarge the masked row count to R_w and the profile to B^2. Thus

$$0 \le \frac{C_2}{D} \le 512\frac{D}{U^2} B^2 R_w \le 1792B^2 R_w,$$

proving (75). Conjugate primes of equal rational norm remain included; no rational-squarefree restriction or new diagonal exclusion is imposed. This is a bookkeeping bound for the exact

correction, not an oscillatory saving or a lower bound for the full signed slice. Other residual sectors can cancel it.

Now (73) gives (76) with $E = \Delta + C_2/D$, and the explicit bounds give (77). Suppose $H \le D^{3/4}$. If two distinct good rough primes divided a row, then $H \ge Nu \ge (Np)(Nq) \ge 49$, but also

$$H \ge Nu \ge (Np)(Nq) > U^2 \ge 2D/7.$$

Combining with $H \le D^{3/4}$ gives $D^{1/4} < 7/2$ and hence $H < (7/2)^3 = 343/8 < 49$, a contradiction. Thus $\kappa_U(u) \le 1$ at every scale in this capacity range. Substitution in (77) gives $|E| \le 233216B^2R_w \le 2^{18}B^2R_w$, proving (78). This is an exact endpoint argument, not an asymptotic approximation.

For a two-pivot block, counting each of its two dyadic ideal variables by (89) gives $R_w \ll A_1A_2$ when the original fixed unit classes and multiplicities are bounded. Internal coprimality and capacity reduce that count; grouping literal fibers preserves total multiplicity. Under $A_1A_2 \le H \le D^{3/4}$ the correction is therefore $O(B^2H)$. The exact local coefficient identity and these correction bounds do not estimate the remaining signed main $(K_0 - K_S)/D$. Its original $A_U(fa)$, coherent rows, common fixed profile and Möbius mean remain the analytic obligation; no new global exponent follows.

41.7 Finite control scope

The tables in the shared core retain complete original squarefree ideal columns and raw character zeros. Their residual-prime extension is a separate auxiliary sum, including the non-squarefree p^2 terms needed for (87); those are not original columns. The exact intersection subtraction and unit residual states with $1/2 < L < 1$ are checked separately. The common profile has fixed $\delta = 1/4096$ and large fixed derivative constants. The capacities and row blocks differ between cases, and the $H = 384$ case is outside the critical capacity range. These finite equalities neither determine the analytic constants nor imply an asymptotic saving.

41.8 A coherent prime-substitution budget, and its remaining cost

This subsection requires $U \ge 7$. It does not extend the prime-density argument to $U = 1$, although the coefficient and mask lemmas above hold for all $U \ge 1$. For real $t \ge U$, put

$$G_f(t) = \sum_{\substack{u \in \mathcal{R} \\ (f,u)=1}} w_u |T_f(t;u)|^2, \qquad M_1(W) = \max\{B, \|xW'(x)\|_\infty\}.$$

The active residuals satisfy $Na \le 2D/(FU)$, so their union is finite; G_f is smooth and vanishes for $t \ge 2D/F$. Since the rough prime in K_S is unmasked, it enters only this radial common profile. Let

$$E_S(t) = \pi_K^S(t) - \mathrm{Li}(t), \qquad \delta_S(U) = \sup_{t \ge U} \frac{|E_S(t)|}{t}, \qquad \mathcal{J} = \sum_f \mu(f) \int_U^\infty \frac{G_f(t)}{\log t}\, dt,$$

where π_K^S counts prime *ideals* outside the fixed finite S with multiplicity, including inert prime norms and distinct conjugate ideals. Stieltjes partial summation gives the exact coherent identity

$$\begin{aligned} K_S &= \mathcal{J} + \mathcal{E}_{\mathrm{PIT}}, \\ \mathcal{E}_{\mathrm{PIT}} &= \sum_f \mu(f) \left(-G_f(U)E_S(U) - \int_U^\infty E_S(t) G_f'(t)\, dt \right). \end{aligned} \tag{98}$$

The lower term uses $\pi_K^S(U)$, so the condition $Np > U$ is exact even when a prime norm equals U. A conjugate pair contributes a jump of two. This equality precedes any absolute sum over f.

Let $\lambda(L)$ be a TRUE arbitrary-coefficient squared norm bound for the specified positive row operator on good squarefree ideals of norm in $[L, 2L]$, including the actual weights. For $1/2 \le L < 1$, set $\lambda_*(L) = R_w$ by the direct unit-column bound; for $L \ge 1$, use $\lambda_*(L) = \lambda(L)$. No columns are active for $L < 1/2$. With $L = D/(Ft)$, differentiate only the common W: multiplication by $t\,d/dt$ replaces $W(x)$ by $xW'(x)$. The exact restricted identity (90), ideal counting and weighted row Cauchy give

$$G_f(t) + |tG_f'(t)| \ll M_1(W)^2 \lambda_*(L) L.$$

Any enlargement of a common f-row mask here occurs only inside these positive norm estimates. Substitution in (98), with $L_f = D/(FU)$, proves

$$\frac{|\mathcal{E}_{\mathrm{PIT}}|}{D} \ll M_1(W)^2 \delta_S(U) \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ F < 2D/U}} \frac{1}{F} \left(\lambda_*(L_f) + \int_{1/2}^{L_f} \lambda_*(L)\, \frac{dL}{L} \right). \tag{99}$$

The original Möbius sign remains in K_S and $\mathcal{J}$ themselves; the absolute f sum appears only in this error majorant.

Specialize now ONLY to the established positive input for the actual $A_6 = 1$ unweighted two-pivot block, or bounded natural literal weights for which the same input has been established. Put

$$R = A_1 A_2, \qquad M_{\mathrm{row}} = \max(A_1, A_2), \qquad A_1 A_2^2 \le H.$$

At $L \ge 1$ that input is

$$\lambda(L) \ll_\epsilon (HD)^\epsilon \min \left\{ R + \frac{LR}{M_{\mathrm{row}}} + \frac{L^{2/3} R}{M_{\mathrm{row}}^{1/3}},\ L + R^2 \right\}. \tag{100}$$

It is not asserted for arbitrary nonnegative weights, enlarged primitive conductor families or arbitrary sixth-factor enlargements. Using its first branch in (99) gives

$$\frac{|\mathcal{E}_{\mathrm{PIT}}|}{D} \ll_\epsilon (HD)^\epsilon M_1(W)^2 \delta_S(U) \mathcal{B}_{\mathrm{op}}, \tag{101}$$

$$\mathcal{B}_{\mathrm{op}} = R \log^2(2D) + \frac{D}{U} \frac{R}{M_{\mathrm{row}}} + \left(\frac{D}{U} \right)^{2/3} \frac{R}{M_{\mathrm{row}}^{1/3}}. \tag{102}$$

The logarithmic term uses $\sum_{Nf \le X} (Nf)^{-1} \ll \log(2X)$; the other terms use convergent ideal sums of powers 2 and 5/3. The same positive bounds give $|\mathcal{J}|/D \ll_\epsilon (HD)^\epsilon B^2 \mathcal{B}_{\mathrm{op}} / \log U$. These are inherited majorants, not a new signed gain or an estimate of $K_0 - \mathcal{J}$.

Capacity implies

$$\mathcal{B}_{\mathrm{op}} \ll H \log^2(2D) + (D/U) H^{1/3} + (D/U)^{2/3} H^{2/3}.$$

At $H = D^{3/4}$ and $U \asymp \sqrt{D}$, these terms have powers $3/4, 3/4, 5/6$. The last power occurs in this majorant already on pure scales $A_1 \asymp H, A_2 = 1$ with fixed capacity slack. It is not an actual-error lower bound.

For the fixed field, Thorner–Zaman, *Algebra & Number Theory* 13 (2019), 1039–1068, Theorem 1.1, applied to the trivial extension K/K with absolute degree $[K : \mathbb{Q}] = 2$ and absolute discriminant 3, supplies for sufficiently large U a bound

$$\delta_S(U) \ll_{K,S} \exp(-c_K \sqrt{\log U}).$$

The possible exceptional real zero is fixed: retain its secondary $\mathrm{Li}(x^{\beta})$ term, or absorb it only into fixed-field constants and the threshold. Finite S deletions are included. No prime theorem uniform in the growing deleted set or row conductor is used. The DOI, publisher URL and pinned reading-cache hashes are in the public citation record.

A fixed-field subpower prime error does not make the available $D^{5/6+\epsilon}\delta_S(U)$ majorant certify the target $D^{3/4+\epsilon}$ for every $\epsilon > 0$. Actual signed errors might cancel. Even an exact prime density would leave the coherent signed $K_0 - \mathcal{J}$ unestimated. The elementary correction theorem uses only B, whereas the prime-substitution error requires $M_1(W)^2$. The complete original remainder's $7/6$ envelope remains unchanged.

Classical reference NIST Digital Library of Mathematical Functions, Section25.15, Dirichlet L-functions, equations 25.15.5–6; primitive functional equation and finite Gauss sums (snapshot pinned 4 November 2026). The normalization and explicit finite Gauss sums used here are stated above.